

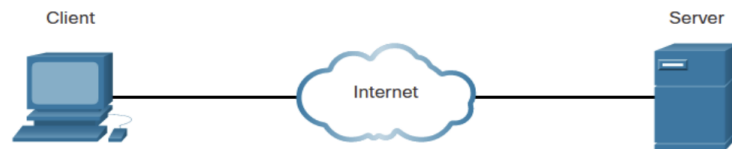
COMP2004 - Computer Networking

Introduction to Networking

Network Components

- Host Roles

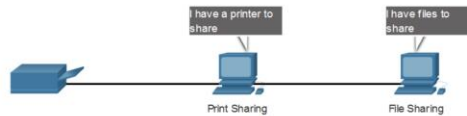
- Servers are computers that info to end servers e.g. email servers, web servers and files servers.
- Clients are computers that send requests to the servers to retrieve information e.g. web page from a web server or email from an email server



Server Type	Description
Email	Email server runs email server software. Clients use client software to access email.
Web	Web server runs web server software. Clients use browser software to access web pages.
File	File server stores corporate and user files. The client devices access these files.

- Peer to Peer

- When a client is also a server. This is only recommended for very small networks.



Advantages	Disadvantages
Easy to set up	No centralized administration
Less complex	Not as secure
Lower cost	Not scalable
Used for simple tasks: transferring files and sharing printers	Slower performance

End Devices

- An end device is where a message originates from or where it is received. Data originates with an end device, flows through the network, and arrives at an end device.

Intermediary Network Devices

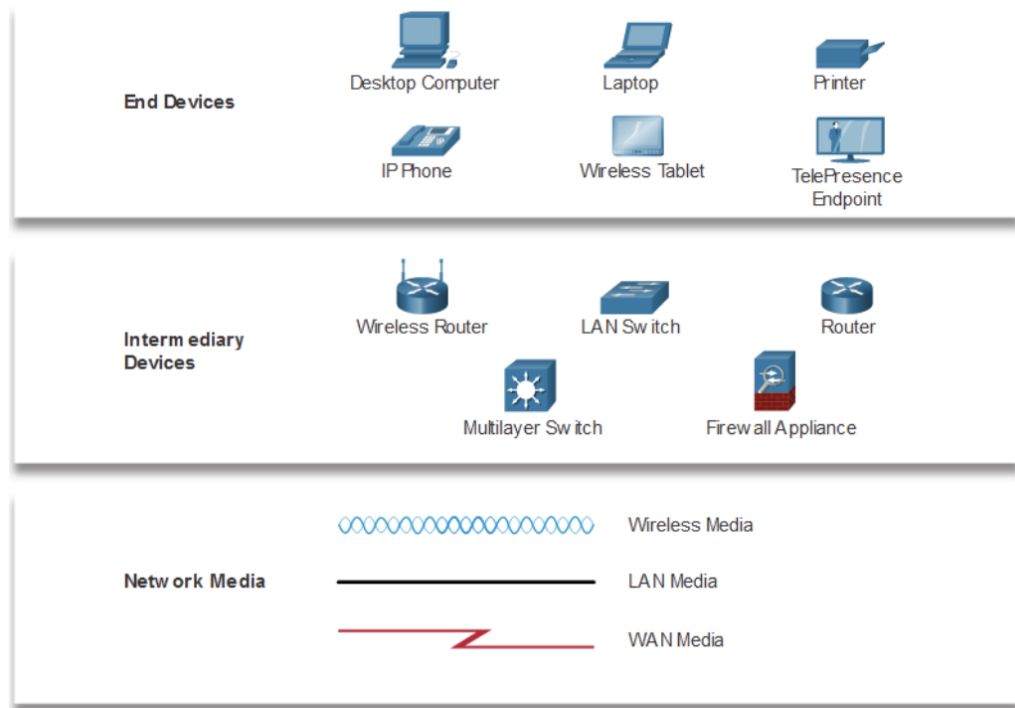
- An intermediary device interconnects end devices. Examples include switches, wireless access points, routers, and firewalls.
- Management of data as it flows through a network is also the role of an intermediary device, including:
 - Regenerate and retransmit data signals.
 - Maintain information about what pathways exist in the network.
 - Notify other devices of errors and communication failures.

Network Media

Media Types	Description	
Metal wires within cables	Uses electrical impulses	Copper
Glass or plastic fibers within cables (fiber-optic cable)	Uses pulses of light.	Fiber-optic
Wireless transmission	Uses modulation of specific frequencies of electromagnetic waves.	Wireless

Network Representations and Topologies

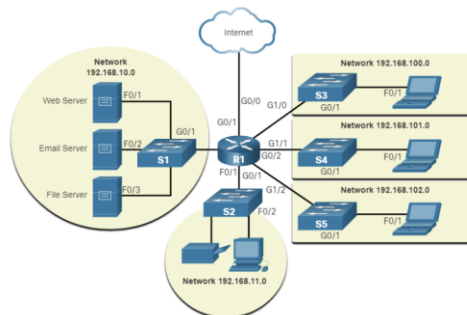
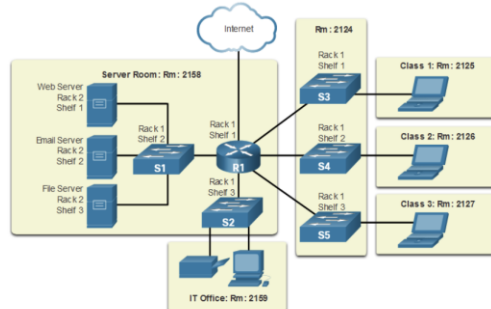
- Network diagrams, often called topology diagrams, use symbols to represent devices within the network. Important terms to know include:
 - Network Interface Card (NIC)
 - Physical Port
 - Interface



• Topology Diagram

Physical topology diagrams illustrate the physical location of intermediary devices and cable installation.

Logical topology diagrams illustrate devices, ports, and the addressing scheme of the network.



• Common Types of Networks

○ Networks of Many Sizes

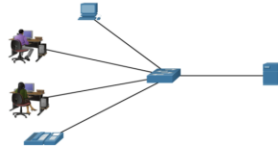
- Small Home Networks – connect a few computers to each other and the Internet
- Small Office/Home Office – enables computer within a home or remote office to connect to a corporate network
- Medium to Large Networks – many locations with hundreds or thousands of interconnected computers
- World Wide Networks – connects hundreds of millions of computers worldwide – such as the internet

○ LANs and WANs

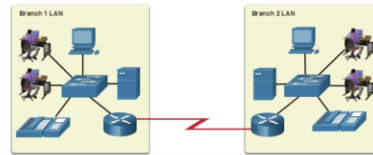
- LAN – Local Area Network

- WAN – Wide Area Network

A LAN is a network infrastructure that spans a small geographical area.



A WAN is a network infrastructure that spans a wide geographical area.



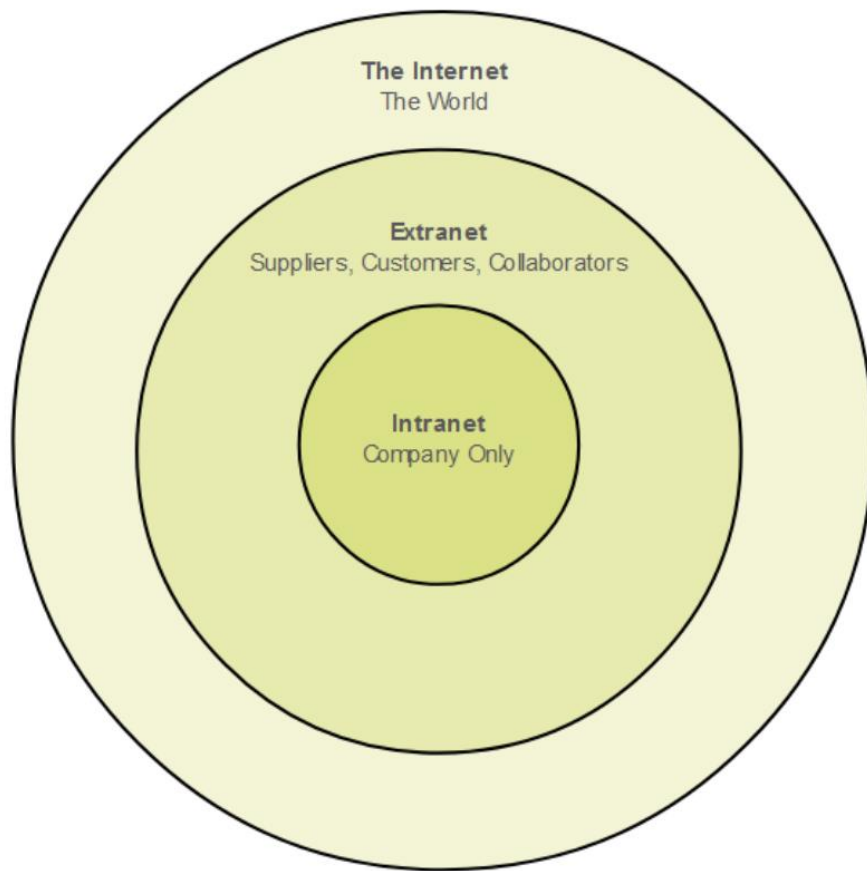
LAN	WAN
Interconnect end devices in a limited area.	Interconnect LANs over wide geographical areas.
Administered by a single organization or individual.	Typically administered by one or more service providers.
Provide high-speed bandwidth to internal devices.	Typically provide slower speed links between LANs.

- The Internet

- The internet is connected via a combination of LANs and WANs.
 - LANs are connected to each other using WANs
 - WANs may use copper wires, fiber optic cables, and wireless transmissions.
- The internet is not owned by any individual or group. The following groups were developed to help maintain structure on the internet:
 - IETF
 - ICANN
 - IAB

- Intranets and Extranets

- An intranet is a private collection of LANs and WANs internal to an organization that is meant to be accessible only to the organization's members or others with authorization.
- An organization might use an extranet to provide secure access to their network for individuals who work for a different organization that need access to their data on their network

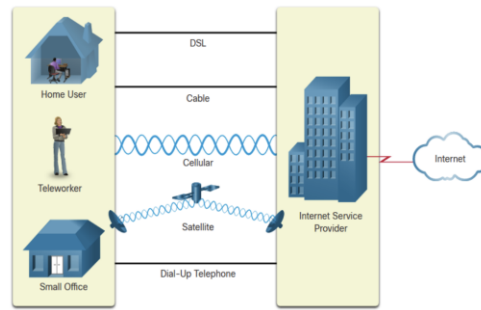


- - **Internet Connections**

- Internet Access Technologies

- Popular services for home users and small offices include broadband cable, broadband digital subscriber line (DSL), wireless WANs, and mobile services.
 - Organizations need faster connections to support IP phones, video conferencing and data center storage.
 - Business-class interconnections are usually provided by service providers (SP) and may include: business DSL, leased lines, and Metro Ethernet.

Home and Small Office Internet Connections

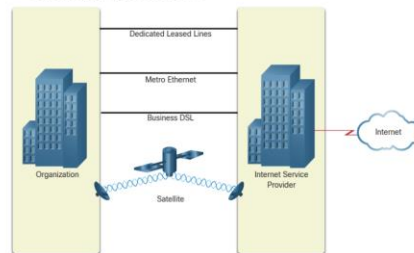


Connection	Description
Cable	high bandwidth, always on, internet offered by cable television service providers.
DSL	high bandwidth, always on, internet connection that runs over a telephone line.
Cellular	uses a cell phone network to connect to the internet.
Satellite	major benefit to rural areas without Internet Service Providers.
Dial-up telephone	an inexpensive, low bandwidth option using a modem.

Businesses Internet Connections

Corporate business connections may require:

- higher bandwidth
- dedicated connections
- managed services

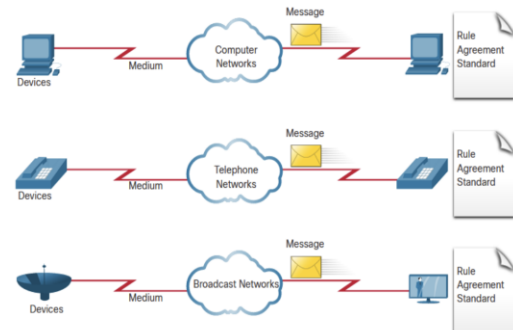


Type of Connection	Description
Dedicated Leased Line	These are reserved circuits within the service provider's network that connect distant offices with private voice and/or data networking.
Ethernet WAN	This extends LAN access technology into the WAN.
DSL	Business DSL is available in various formats including Symmetric Digital Subscriber Lines (SDSL).
Satellite	This can provide a connection when a wired solution is not available.

The Converging Network

Before converged networks, an organization would have been separately cabled for telephone, video, and data. Each of these networks would use different technologies to carry the signal.

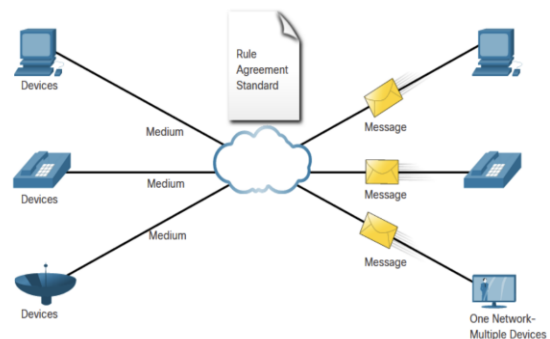
Each of these technologies would use a different set of rules and standards.



Converged data networks carry multiple services on one link including:

- data
- voice
- video

Converged networks can deliver data, voice, and video over the same network infrastructure. The network infrastructure uses the same set of rules and standards.



● Reliable Network

○ Network Architecture

- Network Architecture refers to the technologies that support the infrastructure that moves data across the network.
- There are four basic characteristics that the underlying architectures need to address to meet user expectations:
 - Fault Tolerance
 - Scalability
 - Quality of Service (QoS)
 - Security

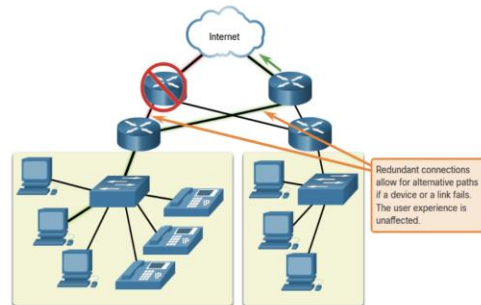
○ Fault Tolerance

A fault tolerant network limits the impact of a failure by limiting the number of affected devices. Multiple paths are required for fault tolerance.

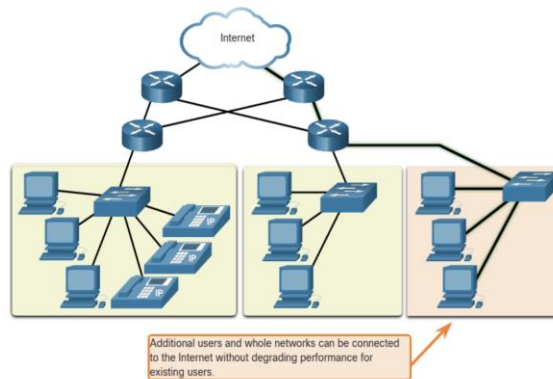
Reliable networks provide redundancy by implementing a packet switched network:

- Packet switching splits traffic into packets that are routed over a network.
- Each packet could theoretically take a different path to the destination.

This is not possible with circuit-switched networks which establish dedicated circuits.



○ Scalability

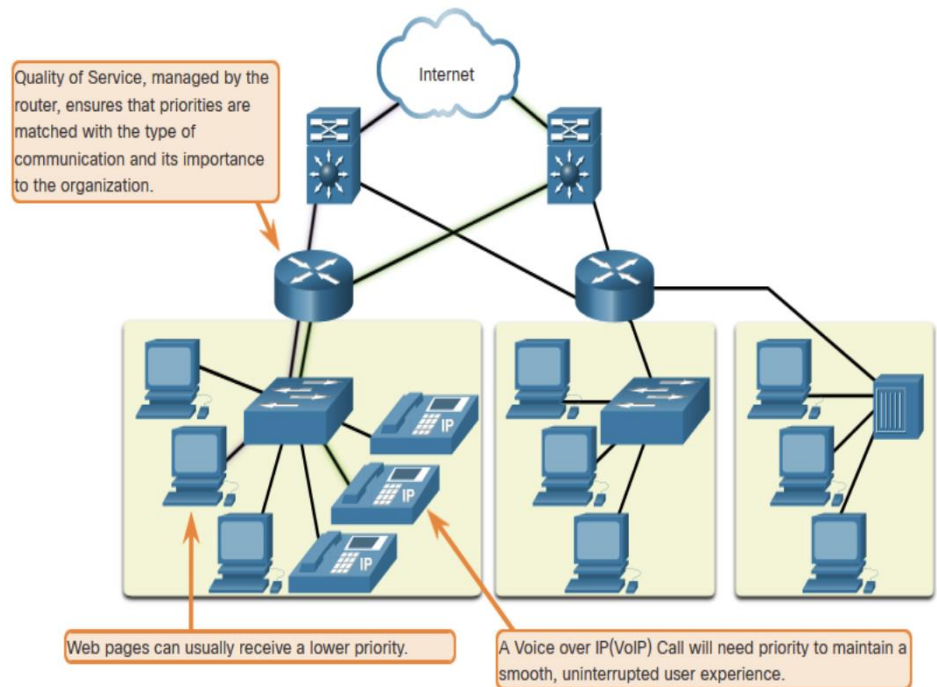


A scalable network can expand quickly and easily to support new users and applications without impacting the performance of services to existing users.

Network designers follow accepted standards and protocols in order to make the networks scalable.

○ Quality of Service

- Quality of Service (QoS) is the primary mechanism used to ensure reliable delivery of content for all users.
- With a QoS policy in place, the router can more easily manage the flow of data and voice traffic.



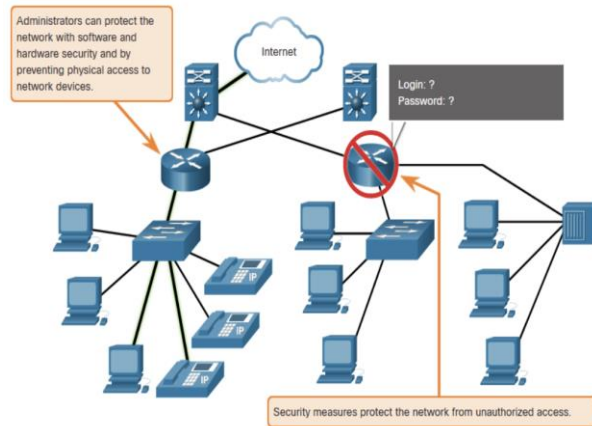
o Network Security

There are two main types of network security that must be addressed:

- Network infrastructure security
- Physical security of network devices
- Preventing unauthorized access to the devices
- Information Security
- Protection of the information or data transmitted over the network

Three goals of network security:

- Confidentiality – only intended recipients can read the data
- Integrity – assurance that the data has not been altered during transmission
- Availability – assurance of timely and reliable access to data for authorized users



• Network Trends

o Recent Trends

- The role of the network must adjust and continually transform in order to be able to keep up with new technologies and end user devices as they constantly come to the market.
- Several new networking trends that affect organizations and consumers:
 - Bring Your Own Device (BYOD)
 - Online collaboration
 - Video communications
 - Cloud computing

○ Bring Your Own Device (BYOD)

- Bring Your Own Device (BYOD) allows users to use their own devices giving them more opportunities and greater flexibility.
- BYOD allows end users to have the freedom to use personal tools to access information and communicate using their:
 - Laptops
 - Netbooks
 - Tablets
 - Smartphones
 - E-readers
- BYOD means any device, with any ownership, used anywhere.

○ Online Collaboration

- Collaborate and work with others over the network on joint projects.

○ Video Communication

- Video calls are made to anyone, regardless of where they are located.
- Video conferencing is a powerful tool for communicating with others.
- Video is becoming a critical requirement for effective collaboration.

○ Cloud Computing

- Cloud computing allows us to store personal files or backup our data on servers over the internet.
 - Applications can also be accessed using the Cloud.
 - Allows businesses to deliver to any device anywhere in the world
- Cloud computing is made possible by data centers.
 - Smaller companies that can't afford their own data centers, lease server and storage services from larger data center organizations in the Cloud.
- Four types of Clouds:
 - Public Clouds
 - Available to the general public through a pay-per-use model or for free.
 - Private Clouds
 - Intended for a specific organization or entity such as the government.
 - Hybrid Clouds
 - Made up of two or more Cloud types – for example, part custom and part public.
 - Each part remains a distinctive object but both are connected using the same architecture.
 - Community/Custom Clouds
 - Built to meet the needs of a specific industry, such as healthcare or media.
 - Can be private or public.

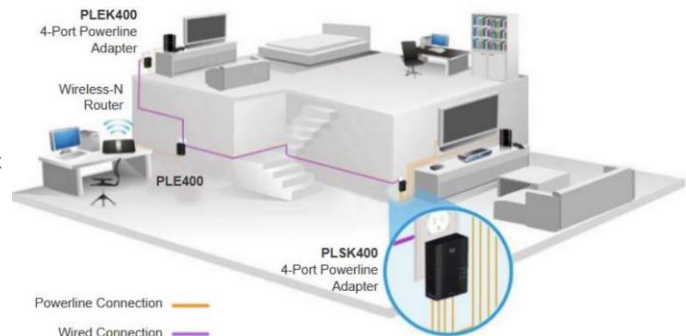
○ Technology Trends in the Home

- Smart home technology is a growing trend that allows

- technology to be integrated into every-day appliances which allows them to interconnect with other devices.
- Ovens might know what time to cook a meal for you by communicating with your calendar on what time you are scheduled to be home.
- Smart home technology is currently being developed for all rooms within a house.

○ Powerline Networking

- Powerline networking can allow devices to connect to a LAN where data network cables or wireless communications are not a viable option.
- Using a standard powerline adapter, devices can connect to the LAN wherever there is an electrical outlet by sending data on certain frequencies.
- Powerline networking is especially useful when wireless access points cannot reach all the devices in the home.



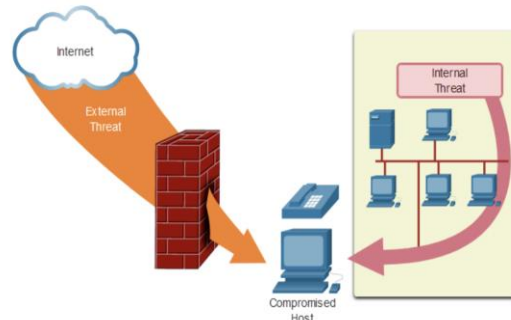
○ Wireless Broadband

- In addition to DSL and cable, wireless is another option used to connect homes and small businesses to the internet.
 - More commonly found in rural environments, a Wireless Internet Service Provider (WISP) is an ISP that connects subscribers to designated access points or hotspots.
 - Wireless broadband is another solution for the home and small businesses.
 - Uses the same cellular technology used by a smartphone.
 - An antenna is installed outside the house providing wireless or wired connectivity for devices in the home

• Network Security

◦ Security Threats

- Network security is an integral part of networking regardless of the size of the network.
- The network security that is implemented must take into account the environment while securing the data, but still allowing for quality of service that is expected of the network.
- Securing a network involves many protocols, technologies, devices, tools, and techniques in order to secure data and mitigate threats.
- Threat vectors might be external or internal.



■ External Threats:

- Viruses, worms, and Trojan horses
- Spyware and adware
- Zero-day attacks
- Threat Actor attacks
- Denial of service attacks
- Data interception and theft
- Identity theft

■ Internal Threats:

- Lost or Stolen devices
- Accidental misuse by employees
- Malicious employees

◦ Security Solutions

- Security must be implemented in multiple layers using more than one security solution. Network security components for home or small office network:
 - Antivirus and antispyware software should be installed on end devices.
 - Firewall filtering used to block unauthorized access to the network
- Larger networks have additional security requirements:
 - Dedicated firewall system
 - Access control lists (ACL)
 - Intrusion prevention systems (IPS)
 - Virtual private networks (VPN)

Basic Switch and End Device Configuration

Cisco IOS Access

- Operating Systems

- Shell

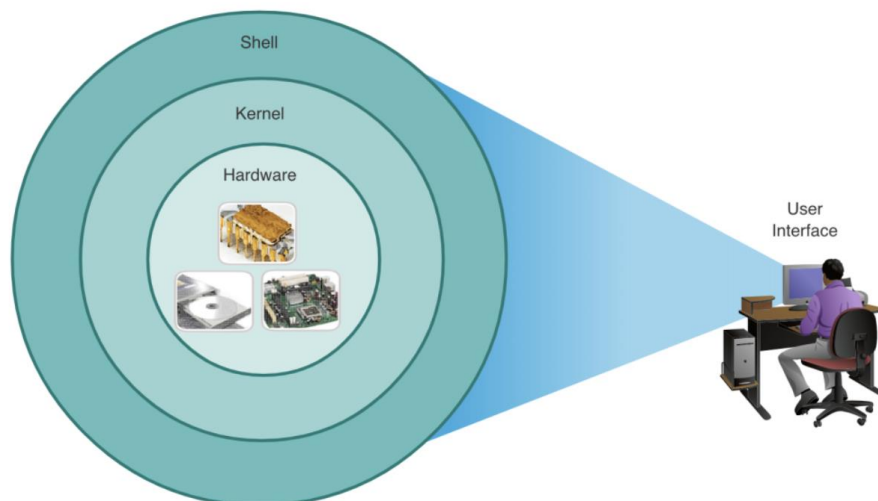
- The user interface that allows users to request specific tasks from the computer. These requests can be made either through the CLI or GUI interfaces.

- Kernel

- Communicates between the hardware and software of a computer and manages how hardware resources are used to meet software requirements.

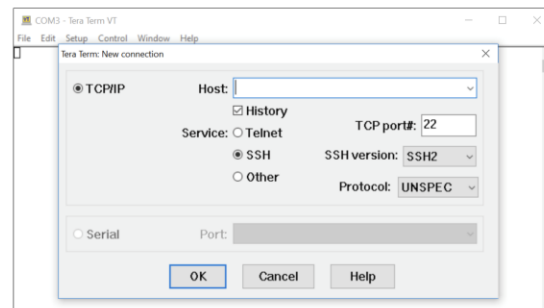
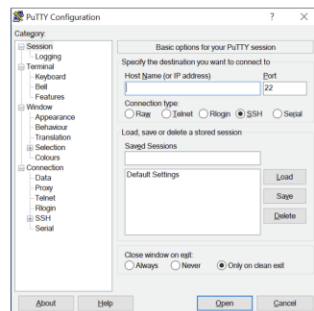
- Hardware

- The physical part of a computer including underlying electronics.



-
- GUI
 - A GUI allows the user to interact with the system using an environment of graphical icons, menus, and windows.
 - A GUI is more user-friendly and requires less knowledge of the underlying command structure that controls the system.
 - GUIs can fail, crash, or simply not operate as specified. For these reasons, network devices are typically accessed through a CLI.
- Purpose of OS
 - PC operating system enables a user to do the following:
 - Use a mouse to make selections and run programs
 - Enter text and text-based commands
 - CLI-based network operating system enables a network technician to do the following:
 - Use a keyboard to run CLI-based network programs
 - Use a keyboard to enter text and text-based commands

- View output on a monitor
- Access Methods
 - Console
 - A physical management port used to access a device in order to provide maintenance, such as performing the initial configurations.
 - Secure Shell (SSH)
 - Establishes a secure remote CLI connection to a device, through a virtual interface, over a network. (Note: This is the recommended method for remotely connecting to a device.)
 - Telnet
 - Establishes an insecure remote CLI connection to a device over the network. (Note: User authentication, passwords and commands are sent over the network in plaintext.)
- Terminal Emulation Programs
 - Terminal emulation programs are used to connect to a network device by either a console port or by an SSH/Telnet connection.
 - There are several terminal emulation programs to choose from such as PuTTY, Tera Term and SecureCRT.



● IOS Navigation

- Primary Command Modes
 - User EXEC Mode:
 - Allows access to only a limited number of basic monitoring commands
 - Identified by the CLI prompt that ends with the > symbol
 - Privileged EXEC Mode:
 - Allows access to all commands and features
 - Identified by the CLI prompt that ends with the # symbol

```
Router>

Switch>
```

- Privileged EXEC Mode:

- Allows access to all commands and features
- Identified by the CLI prompt that ends with the # symbol

```
Router#
```

```
Switch#
```

- Configuration Mode and Subconfiguration Modes

- Global Configuration Mode:

- Used to access configuration options on the device

```
Switch(config)#
```

- Line Configuration Mode:

- Used to configure console, SSH, Telnet or AUX access

```
Switch(config-line)#
```

- Interface Configuration Mode:

- Used to configure a switch port or router interface

```
Switch(config-if)#
```

- Navigation Between IOS Modes

- Privileged EXEC Mode:

- To move from user EXEC mode to privilege EXEC mode, use the **enable** command.

```
Switch> enable
Switch#
```

- Global Configuration Mode:

- To move in and out of global configuration mode, use the **configure terminal** command. To return to privilege EXEC mode, use the **exit** command.

```
Switch(config)#
Switch(config)#exit
Switch#
```

- Line Configuration Mode:

- To move in and out of line configuration mode, use the **line** command followed by the management line type. To return to global configuration mode, use the **exit** command.

```
Switch(config)#line console 0
Switch(config-line)#exit
Switch(config)#
```

- Subconfiguration Modes:

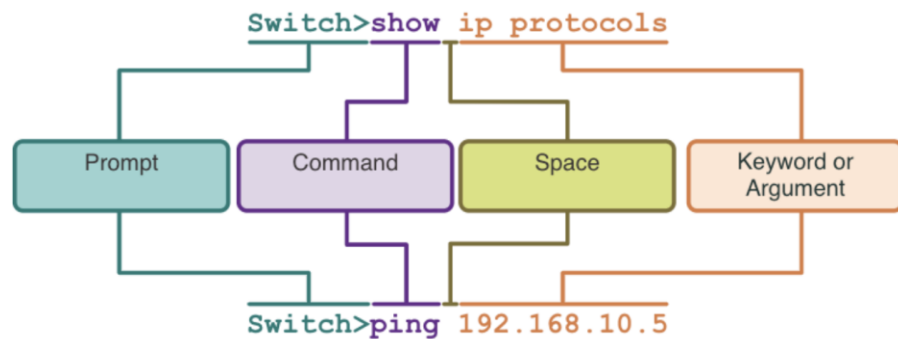
- To move out of any subconfiguration mode to get back to global configuration mode, use the **exit** command. To return to privilege EXEC mode, use the **end** command or key combination **Ctrl +Z**.
 - To move directly from one subconfiguration mode to another, type in the desired subconfiguration mode command. In the example, the command prompt changes from **(config-line)#** to **(config-if)#**.

```
Switch(config)#line console 0
Switch(config-line)#end
Switch#
```

```
Switch(config-line)#interface FastEthernet 0/1
Switch(config-if)#
```

• The Command Structure

◦ Basic IOS Command Structure



-
- Keyword
 - This is a specific parameter defined in the operating system (in the figure, ip protocols).
- Argument
 - This is not predefined; it is a value or variable defined by the user (in the figure, 192.168.10.5)

◦ IOS Command Syntax Check

A command might require one or more arguments. To determine the keywords and arguments required for a command, refer to the command syntax.

- Boldface text indicates commands and keywords that are entered as shown.
- Italic text indicates an argument for which the user provides the value.

Convention	Description
boldface	Boldface text indicates commands and keywords that you enter literally as shown.
<i>italics</i>	Italic text indicates arguments for which you supply values.
[x]	Square brackets indicate an optional element (keyword or argument).
{x}	Braces indicate a required element (keyword or argument).
[x {y z }]	Braces and vertical lines within square brackets indicate a required choice within an optional element. Spaces are used to clearly delineate parts of the command.

-
- The command syntax provides the pattern, or format, that must be used when entering a command.
- The command is **ping** and the user-defined argument is the *ip-address* of the destination device. For example, **ping 10.10.10.5**.
- The command is **traceroute** and the user-defined argument is the *ip-address* of the destination device. For example, **traceroute 192.168.254.254**.
- If a command is complex with multiple arguments, you may see it represented like this:

```
ping ip-address
```

```
traceroute ip-address
```

```
Switch(config-if)# switchport port-security aging { static | time time | type {absolute | inactivity}}
```

○ IOS Help Features

The IOS has two forms of help available: context-sensitive help and command syntax check.

- Context-sensitive help enables you to quickly find answers to these questions:
 - Which commands are available in each command mode?
 - Which commands start with specific characters or group of characters?
 - Which arguments and keywords are available to particular commands?
- Command syntax check verifies that a valid command was entered by the user.
 - If the interpreter cannot understand the command being entered, it will provide feedback describing what is wrong with the command.

```
Router#ping ?
WORD Ping destination address or hostname
ip      IP echo
ipv6    IPv6 echo
```

```
Switch#interface fastEthernet 0/1
^
% Invalid input detected at '^' marker.
```

○ Hot Keys and Shortcuts

- The IOS CLI provides hot keys and shortcuts that make configuring, monitoring, and troubleshooting easier.
- Commands and keywords can be shortened to the minimum number of characters that identify a unique selection. For example, the **configure** command can be shortened to **conf** because **configure** is the only command that begins with **conf**.

```
Router#con
% Ambiguous command: "con"
Router#con?
configure connect
```

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
```

The table below is a brief list of keystrokes to enhance command line editing.

Keystroke	Description
Tab	Completes a partial command name entry.
Backspace	Erases the character to the left of the cursor.
Left Arrow or Ctrl+B	Moves the cursor one character to the left.
Right Arrow or Ctrl+F	Moves the cursor one character to the right.
Up Arrow or Ctrl+P	Recalls the commands in the history buffer, beginning with the most recent commands.

○

The table below is a brief list of keystrokes to enhance command line editing.

Keystroke	Description
Tab	Completes a partial command name entry.
Backspace	Erases the character to the left of the cursor.
Left Arrow or Ctrl+B	Moves the cursor one character to the left.
Right Arrow or Ctrl+F	Moves the cursor one character to the right.
Up Arrow or Ctrl+P	Recalls the commands in the history buffer, beginning with the most recent commands.

Video – Hot Keys and Shortcuts

This video will cover the following:

- Tab key (tab completion)
- Command shortening
- Up and down arrow key
- CTRL + C
- CTRL + Z
- CTRL + Shift + 6
- CTRL + R

• Basic Device Configuration

○ Device Names

- The first configuration command on any device should be to give it a unique hostname.
- By default, all devices are assigned a factory default name. For example, a Cisco IOS switch is "Switch."
- Guideline for naming devices:
 - Start with a letter
 - Contain no spaces
 - End with a letter or digit
 - Use only letters, digits, and dashes
 - Be less than 64 characters in length

```
Switch# configure terminal
Switch(config)# hostname Sw-Floor-1
Sw-Floor-1(config)#
```

Note: To return the switch to the default prompt, use the **no hostname** global config command.

○ Password Guidelines

- The use of weak or easily guessed passwords are a security concern.
- All networking devices should limit administrative access by securing privileged EXEC, user EXEC, and remote Telnet access with passwords. In addition, all passwords should be encrypted and legal notifications provided.
- Password Guidelines:
 - Use passwords that are more than eight characters in length.
 - Use a combination of upper and lowercase letters, numbers, special characters, and/or numeric sequences.
 - Avoid using the same password for all devices.
 - Do not use common words because they are easily guessed.



Note: Most of the labs in this course use simple passwords such as **cisco** or **class**. These passwords are considered weak and easily guessable and should be avoided in production environments.

Configure Passwords

Securing user EXEC mode access:

- First enter line console configuration mode using the **line console 0** command in global configuration mode.
- Next, specify the user EXEC mode password using the **password password** command.
- Finally, enable user EXEC access using the **login** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line console 0
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

Securing privileged EXEC mode access:

- First enter global configuration mode.
- Next, use the **enable secret password** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# enable secret class
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

Securing VTY line access:

- First enter line VTY configuration mode using the **line vty 0 15** command in global configuration mode.
- Next, specify the VTY password using the **password password** command.
- Finally, enable VTY access using the **login** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line vty 0 15
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

- Note: VTY lines enable remote access using Telnet or SSH to the device. Many Cisco switches support up to 16 VTY lines that are numbered 0 to 15.

Encrypt Passwords

- The startup-config and running-config files display most passwords in plaintext.
- To encrypt all plaintext passwords, use the **service password-encryption** global config command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# service password-encryption
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

- Use the **show running-config** command to verify that the passwords on the device are now encrypted.

```
Sw-Floor-1# show running-config
!
!
!
line con 0
password 7 094F471A1A0A
login
!
!
Line vty 0 4
Password 7 03095A0F034F38435B49150A1819
Login
!
!
!
end
```

Banner Messages

- A banner message is important to warn unauthorized personnel from attempting to access the device.
- To create a banner message of the day on a network device, use the **banner motd # the message of the day #** global config command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# banner motd #Authorized Access Only!#
```

The banner will be displayed on attempts to access the device.

Press RETURN to get started.

Authorized Access Only!
User Access Verification
Password:

Note: The “#” in the command syntax is called the delimiting character. It is entered before and after the message.

● Save Configurations

○ Configuration Files

- There are two system files that store the device configuration:

- **startup-config** - This is the saved configuration file that is stored in NVRAM. It contains all the commands that will be used by the device upon startup or reboot. Flash does not lose its contents when the device is powered off.
- **running-config** - This is stored in Random Access Memory (RAM). It reflects the current configuration. Modifying a running configuration affects the operation of a Cisco device immediately. RAM is volatile memory. It loses all of its content when the device is powered off or restarted.
- To save changes made to the running configuration to the startup configuration file, use the **copy running-config startup-config** privileged EXEC mode command.

```
Router#show startup-config
Using 624 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

```
Router#show running-config
Building configuration...

Current configuration : 624 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

○ Alter the Running Configurations

If changes made to the running config do not have the desired effect and the running-config has not yet been saved, you can restore the device to its previous configuration. To do this you can:

- Remove the changed commands individually.
- Reload the device using the **reload** command in privilege EXEC mode. *Note: This will cause the device to briefly go offline, leading to network downtime.*

If the undesired changes were saved to the startup-config, it may be necessary to clear all the configurations using the **erase startup-config** command in privilege EXEC mode.

- After erasing the startup-config, reload the device to clear the running-config file from RAM.

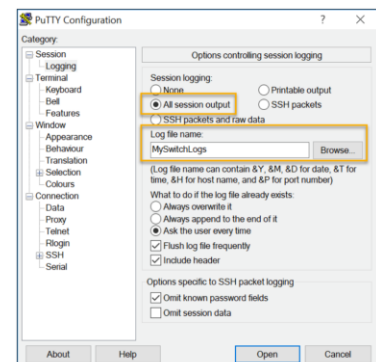
```
Router# reload
Proceed with reload? [confirm]
Initializing Hardware ...
```

```
Router# erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
Router#
```

○ Capture Configuration to a Text File

Configuration files can also be saved and archived to a text document.

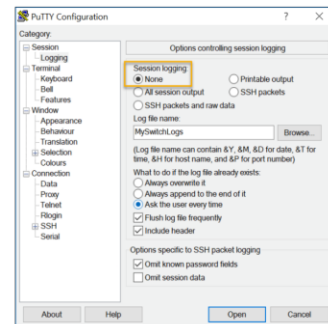
- **Step 1.** Open terminal emulation software, such as PuTTY or Tera Term, that is already connected to a switch.
- **Step 2.** Enable logging in to the terminal software and assign a name and file location to save the log file. The figure displays that **All session output** will be captured to the file specified (i.e., MySwitchLogs).



- **Step 3.** Execute the **show running-config** or **show startup-config** command at the privileged EXEC prompt. Text displayed in the terminal window will be placed into the chosen file.
- **Step 4.** Disable logging in the terminal software. The figure shows how to disable logging by choosing the **None** session logging option

Note: The text file created can be used as a record of how the device is currently implemented. The file could require editing before being used to restore a saved configuration to a device.

```
Switch# show running-config
Building configuration...
```



- **Ports and Addresses**

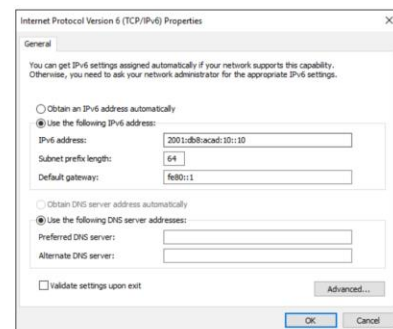
- IP Addresses

- The use of IP addresses is the primary means of enabling devices to locate one another and establish end-to-end communication on the internet.
- The structure of an IPv4 address is called dotted decimal notation and is represented by four decimal numbers between 0 and 255.
- An IPv4 subnet mask is a 32-bit value that differentiates the network portion of the address from the host portion. Coupled with the IPv4 address, the subnet mask determines to which subnet the device is a member.
- The default gateway address is the IP address of the router that the host will use to access remote networks, including the internet.



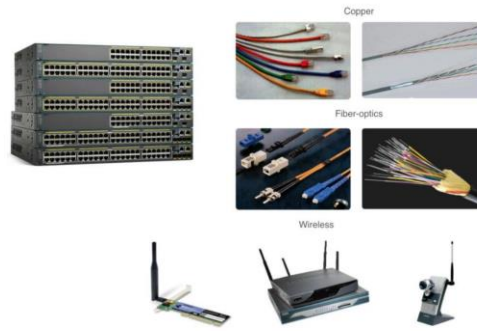
- IPv6 addresses are 128 bits in length and written as a string of hexadecimal values. Every four bits is represented by a single hexadecimal digit; for a total of 32 hexadecimal values. Groups of four hexadecimal digits are separated by a colon
 " :: "
- IPv6 addresses are not case-sensitive and can be written in either lowercase or uppercase.

Note: IP in this course refers to both the IPv4 and IPv6 protocols. IPv6 is the most recent version of IP and is replacing the more common IPv4.



○ Interface and Ports

- Network communications depend on end user device interfaces, networking device interfaces, and the cables that connect them.
- Types of network media include twisted-pair copper cables, fiber-optic cables, coaxial cables, or wireless.
- Different types of network media have different features and benefits. Some of the differences between various types of media include:
 - Distance the media can successfully carry a signal
 - Environment in which the media is to be installed
 - Amount of data and the speed at which it must be transmitted
 - Cost of the media and installation



● Configure IP Addressing

○ Manual IP Address Configuration for End Devices

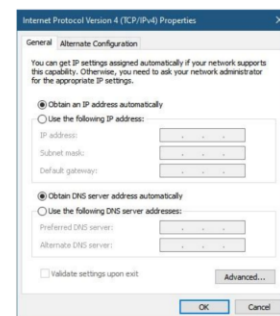
- End devices on the network need an IP address in order to communicate with other devices on the network.
- IPv4 address information can be entered into end devices manually, or automatically using Dynamic Host Configuration Protocol (DHCP).
- To manually configure an IPv4 address on a Windows PC, open the **Control Panel > Network Sharing Center > Change adapter settings** and choose the adapter. Next right-click and select **Properties** to display the **Local Area Connection Properties**.
- Next, click **Properties** to open the **Internet Protocol Version 4 (TCP/IPv4) Properties** window. Then configure the IPv4 address and subnet mask information, and default gateway.



Note: IPv6 addressing and configuration options are similar to IPv4.

○ Automatic IP Address Configuration for End Devices

- DHCP enables automatic IPv4 address configuration for every end device that is DHCP-enabled.
- End devices are typically by default using DHCP for automatic IPv4 address configuration.
- To configure DHCP on a Windows PC, open the **Control Panel > Network Sharing Center > Change adapter settings** and choose the adapter. Next right-click and select **Properties** to display the **Local Area Connection Properties**.
- Next, click **Properties** to open the **Internet Protocol Version 4 (TCP/IPv4) Properties** window, then select **Obtain an IP address automatically** and **Obtain DNS server address automatically**.



Note: IPv6 uses DHCPv6 and SLAAC (Stateless Address Autoconfiguration) for dynamic address allocation.

○ Switch Virtual Interface Configuration

To access the switch remotely, an IP address and a subnet mask must be configured on the SVI.

To configure an SVI on a switch:

- Enter the **interface vlan 1** command in global configuration mode.
- Next assign an IPv4 address using the **ip address** *ip-address subnet-mask* command.
- Finally, enable the virtual interface using the **no shutdown** command.

```
Switch# configure terminal
Switch(config)# interface vlan 1
Switch(config-if)# ip address 192.168.1.20 255.255.255.0
Switch(config-if)# no shutdown
```

● Verify Connectivity

○ Summary

- All end devices and network devices require an operating system (OS).
- Cisco IOS software separates management access into the following two command modes: User EXEC Mode and Privileged EXEC Mode.
- Global configuration mode is accessed before other specific configuration modes. From global config mode, the user can enter different subconfiguration modes.
- Each IOS command has a specific format or syntax and can only be executed in the appropriate mode.
- Basic device configurations- hostname, password, encrypt passwords and banner.
- There are two system files that store the device configuration: startup-config and running-config.
- IP addresses enable devices to locate one another and establish end-to-end communication on the internet. Each end device on a network must be configured with an IP address.

○ New Terms

• operating system (OS) • CLI • GUI • shell • kernel • hardware • console • Secure Shell (SSH) • Telnet • terminal emulation programs • user EXEC mode • privileged EXEC mode	• line configuration mode • interface configuration mode • Enable • configure terminal • exit • end • argument • keyword • command syntax • ping • traceroute • help command "?" • hot keys • hostname	• console • enable secret • VTY line • show running-config • banner motd • startup-config • running-config • reload • erase startup-config • DHCP • switch virtual interface (SVI) • ipconfig • show ip int brief
---	--	--

Protocols and Models & Physical Layer

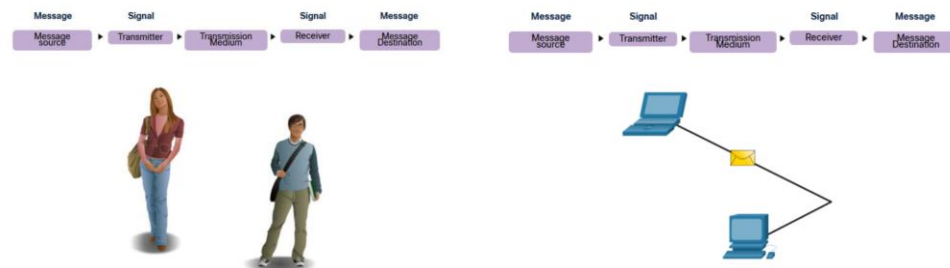
The Rules

- **Communications Fundamentals**

- Networks can vary in size and complexity. It is not enough to have a connection, devices must agree on “how” to communicate.
- There are three elements to any communication:
 - There will be a source (sender).
 - There will be a destination (receiver).
 - There will be a channel (media) that provides for the path of communications to occur.

- **Communications Protocols**

- All communications are governed by protocols.
- Protocols are the rules that communications will follow.
- These rules will vary depending on the protocol.



-
- **Rule Establishment**

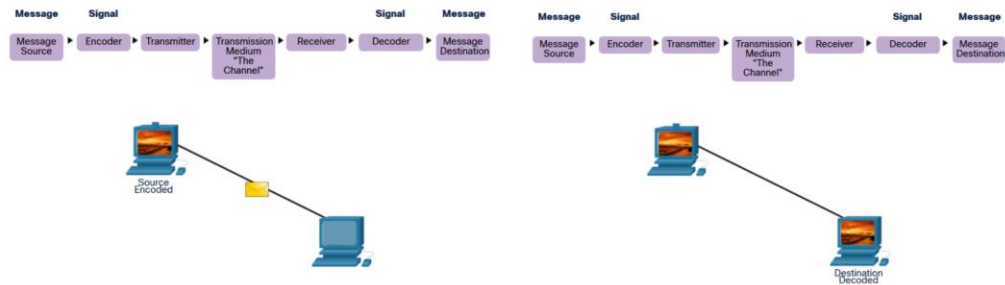
- Individuals must use established rules or agreements to govern the conversation.
- The first message is difficult to read because it is not formatted properly. The second shows the message properly formatted
- Protocols must account for the following requirements:
 - An identified sender and receiver
 - Common language and grammar
 - Speed and timing of delivery
 - Confirmation or acknowledgment requirements

- **Network Protocol Requirements**

- Common computer protocols must be in agreement and include the following requirements:
 - Message encoding
 - Message formatting and encapsulation
 - Message size
 - Message timing
 - Message delivery options

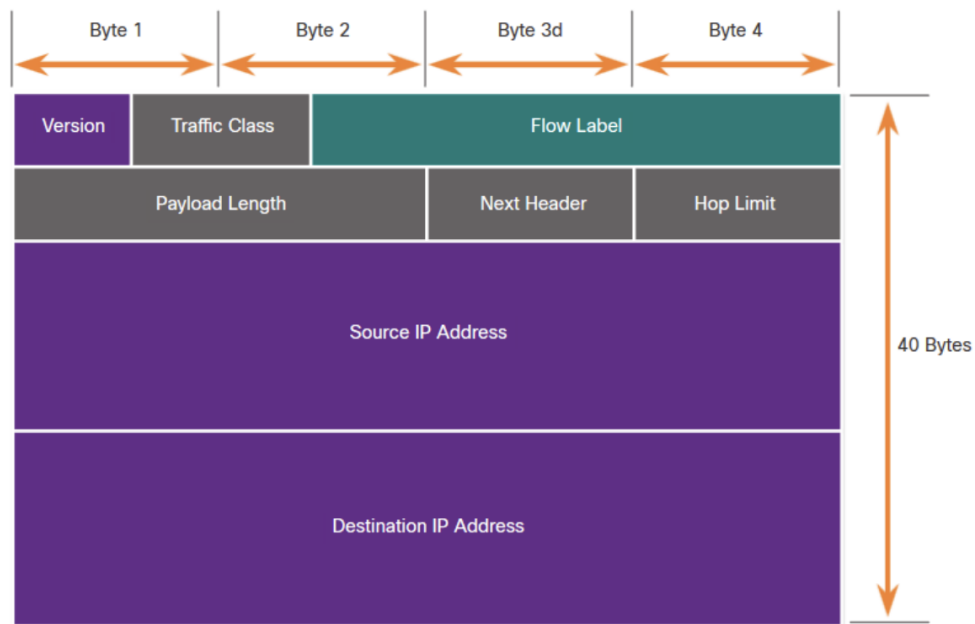
- **Message Encoding**

- Encoding is the process of converting information into another acceptable form for transmission.
- Decoding reverses this process to interpret the information.



● Message Formatting and Encapsulation

- When a message is sent, it must use a specific format or structure.
- Message formats depend on the type of message and the channel that is used to deliver the message.



● Message Size

- Encoding between hosts must be in an appropriate format for the medium.
 - Messages sent across the network are converted to bits
 - The bits are encoded into a pattern of light, sound, or electrical impulses.
 - The destination host must decode the signals to interpret the message.

● Message Timing

- Message timing includes the following:
 - Flow Control
 - Manages the rate of data transmission and defines how much information can be sent and the speed at which it can be delivered.
 - Response Timeout
 - Manages how long a device waits when it does not hear a reply from the destination.

- Access method
 - Determines when someone can send a message.
 - There may be various rules governing issues like “collisions”. This is when more than one device sends traffic at the same time and the messages become corrupt.
 - Some protocols are proactive and attempt to prevent collisions; other protocols are reactive and establish a recovery method after the collision occurs.
- Message Delivery Options
 - Message delivery may use one of the following methods:
 - Unicast
 - one to one communication
 - Multicast
 - one to many, typically not all
 - Broadcast
 - one to all
 - Note: Broadcasts are used in IPv4 networks, but are not an option for IPv6. Later we will also see “Anycast” as an additional delivery option for IPv6.



Protocols

● Network Protocol Overview

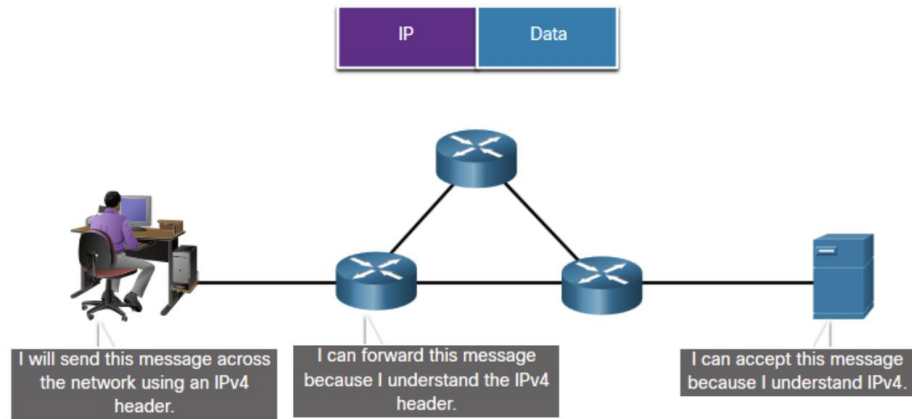
Network protocols define a common set of rules.

- Can be implemented on devices in:
 - Software
 - Hardware
 - Both
- Protocols have their own:
 - Function
 - Format
 - Rules

Protocol Type	Description
Network Communications	enable two or more devices to communicate over one or more networks
Network Security	secure data to provide authentication, data integrity, and data encryption
Routing	enable routers to exchange route information, compare path information, and select best path
Service Discovery	used for the automatic detection of devices or services

● Network Protocol Functions

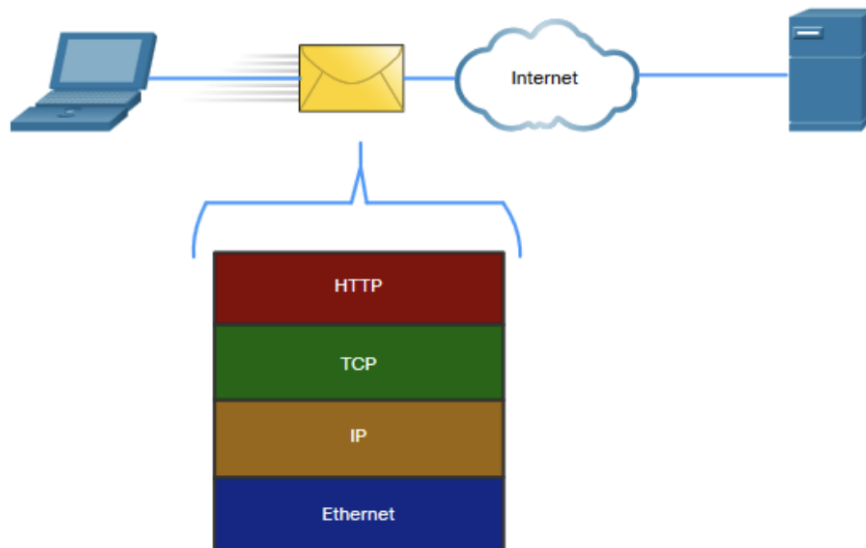
- Devices use agreed-upon protocols to communicate.
- Protocols may have one or more functions.



Function	Description
Addressing	Identifies sender and receiver
Reliability	Provides guaranteed delivery
Flow Control	Ensures data flows at an efficient rate
Sequencing	Uniquely labels each transmitted segment of data
Error Detection	Determines if data became corrupted during transmission
Application Interface	Process-to-process communications between network applications

● Protocol Interaction

- Networks require the use of several protocols.
- Each protocol has its own function and format.



Protocol	Function
Hypertext Transfer Protocol (HTTP)	▪ Governs the way a web server and a web client interact ▪ Defines content and format
Transmission Control Protocol (TCP)	▪ Manages the individual conversations ▪ Provides guaranteed delivery ▪ Manages flow control
Internet Protocol (IP)	Delivers messages globally from the sender to the receiver
Ethernet	Delivers messages from one NIC to another NIC on the same Ethernet Local Area Network (LAN)

Protocol Suites

- Network Protocol Suites

Protocols must be able to work with other protocols.

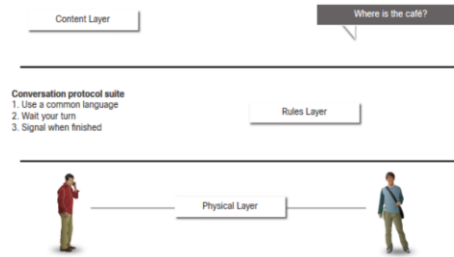
Protocol suite:

- A group of inter-related protocols necessary to perform a communication function
- Sets of rules that work together to help solve a problem

The protocols are viewed in terms of layers:

- Higher Layers
- Lower Layers- concerned with moving data and provide services to upper layers

© 1997-2000 CISCO



Protocol suites are sets of rules that work together to help solve a problem.

© Cisco and/or its affiliates. All rights reserved. Cisco Confidential

22

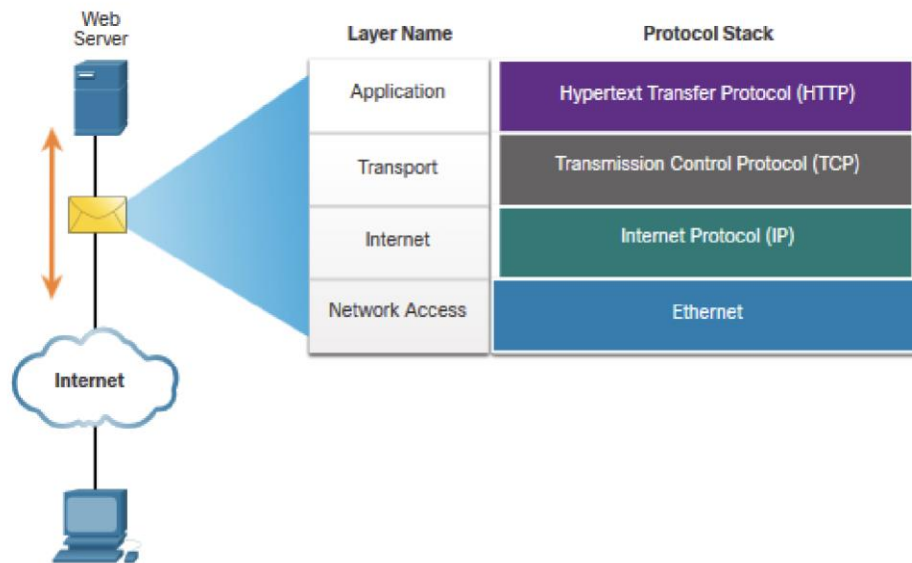
- Evolution of Protocol Suites

- There are several protocol suites.
 - Internet Protocol Suite or TCP/IP- The most common protocol suite and maintained by the Internet Engineering Task Force (IETF)
 - Open Systems Interconnection (OSI) protocols- Developed by the International Organization for Standardization (ISO) and the International Telecommunications Union (ITU)
 - AppleTalk- Proprietary suite release by Apple Inc.
 - Novell NetWare- Proprietary suite developed by Novell Inc.

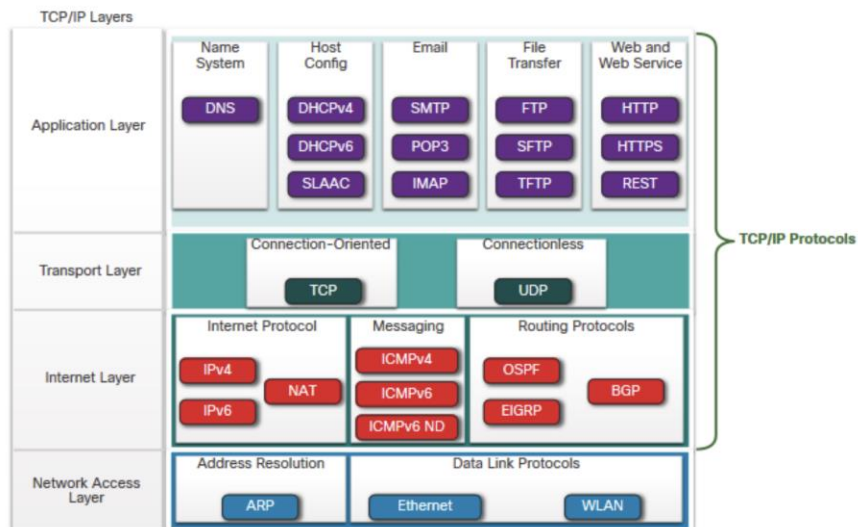
TCP/IP Layer Name	TCP/IP	ISO	AppleTalk	Novell Netware
Application	HTTP DNS DHCP FTP	ACSE ROSE TRSE SESE	AFP	NDS
Transport	TCP UDP	TP0 TP1 TP2 TP3 TP4	ATP AEP NBP RTMP	SPX
Internet	IPv4 IPv6 ICMPv4 ICMPv6	CONP/CMNS CLNP/CLNS	AARP	IPX
Network Access	Ethernet ARP WLAN			

- TCP/IP Protocol Example

- TCP/IP protocols operate at the application, transport, and internet layers.
- The most common network access layer LAN protocols are Ethernet and WLAN (wireless LAN).



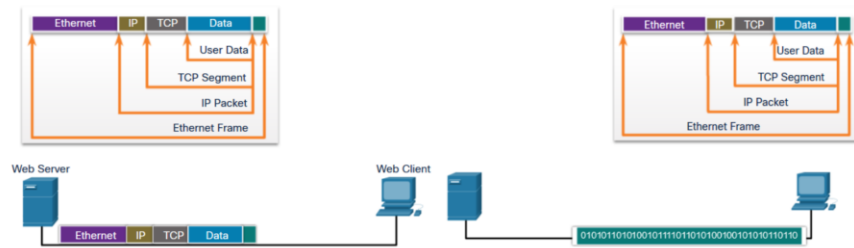
-
- **TCP/IP Protocol Suite**
 - TCP/IP is the protocol suite used by the internet and includes many protocols.
 - TCP/IP is:
 - An open standard protocol suite that is freely available to the public and can be used by any vendor
 - A standards-based protocol suite that is endorsed by the networking industry and approved by a standards organization to ensure interoperability



- TCP/IP Communication Process

- A web server encapsulating and sending a web page to a client.

- A client de-encapsulating the web page for the web browser



Standards Organizations

- Open Standard



Open standards encourage:

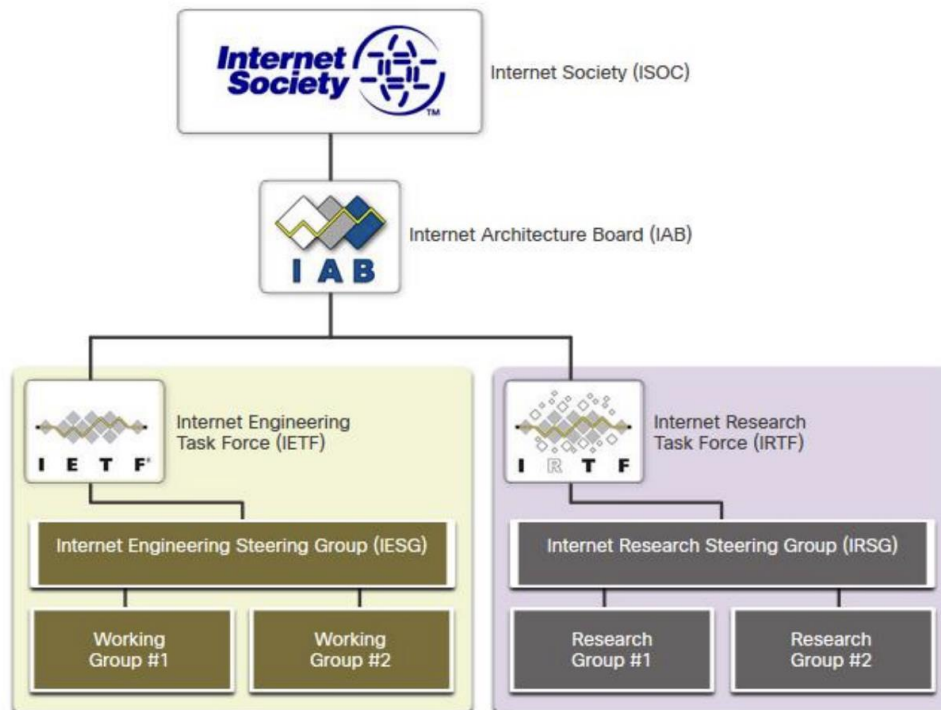
- interoperability
- competition
- innovation

Standards organizations are:

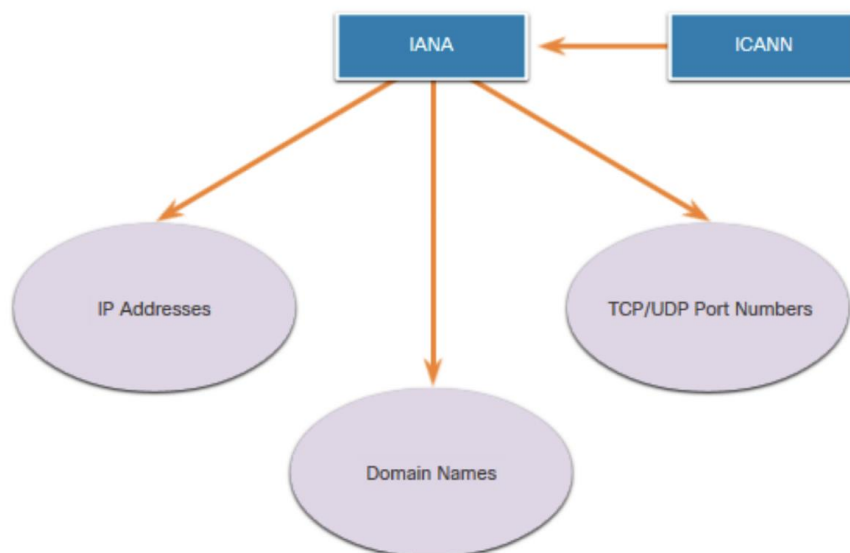
- vendor-neutral
- non-profit organizations
- established to develop and promote the concept of open standards.

- Internet Standards

- Internet Society (ISOC)
 - Promotes the open development and evolution of internet
- Internet Architecture Board (IAB)
 - Responsible for management and development of internet standards
- Internet Engineering Task Force (IETF)
 - Develops, updates, and maintains internet and TCP/IP technologies
- Internet Research Task Force (IRTF)
 - Focused on long-term research related to internet and TCP/IP protocols



-
- Standards organizations involved with the development and support of TCP/IP
 - Internet Corporation for Assigned Names and Numbers (ICANN)
 - Coordinates IP address allocation, the management of domain names, and assignment of other information
 - Internet Assigned Numbers Authority (IANA)
 - Oversees and manages IP address allocation, domain name management, and protocol identifiers for ICANN



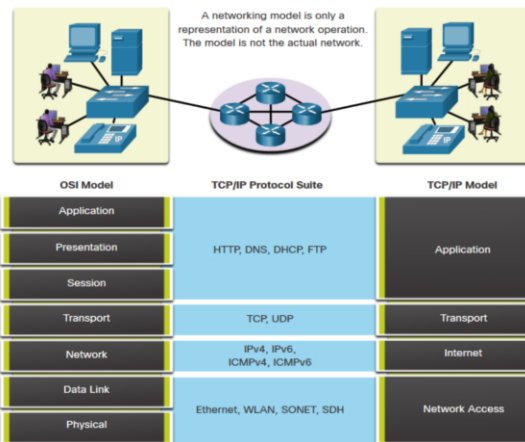
○

- **Electronic and Communications Standards**

- **Institute of Electrical and Electronics Engineers (IEEE)**, pronounced "I-triple-E")
 - dedicated to creating standards in power and energy, healthcare, telecommunications, and networking
- **Electronic Industries Alliance (EIA)** - develops standards relating to electrical wiring, connectors, and the 19-inch racks used to mount networking equipment
- **Telecommunications Industry Association (TIA)** - develops communication standards in radio equipment, cellular towers, Voice over IP (VoIP) devices, satellite communications, and more
- **International Telecommunications Union-Telecommunication Standardization Sector (ITU-T)** - defines standards for video compression, Internet Protocol Television (IPTV), and broadband communications, such as a digital subscriber line (DSL)

Reference Mode

- **The Benefits of Using a Layered Model**



Complex concepts such as how a network operates can be difficult to explain and understand. For this reason, a layered model is used.

Two layered models describe network operations:

- Open System Interconnection (OSI) Reference Model
- TCP/IP Reference Model

- These are the benefits of using a layered model:
 - Assist in protocol design because protocols that operate at a specific layer have defined information that they act upon and a defined interface to the layers above and below
 - Foster competition because products from different vendors can work together
 - Prevent technology or capability changes in one layer from affecting other layers above and below
 - Provide a common language to describe networking functions and capabilities

- The OSI Reference Model

OSI Model Layer	Description
7 - Application	Contains protocols used for process-to-process communications.
6 - Presentation	Provides for common representation of the data transferred between application layer services.
5 - Session	Provides services to the presentation layer and to manage data exchange.
4 - Transport	Defines services to segment, transfer, and reassemble the data for individual communications.
3 - Network	Provides services to exchange the individual pieces of data over the network.
2 - Data Link	Describes methods for exchanging data frames over a common media.
1 - Physical	Describes the means to activate, maintain, and de-activate physical connections.

○

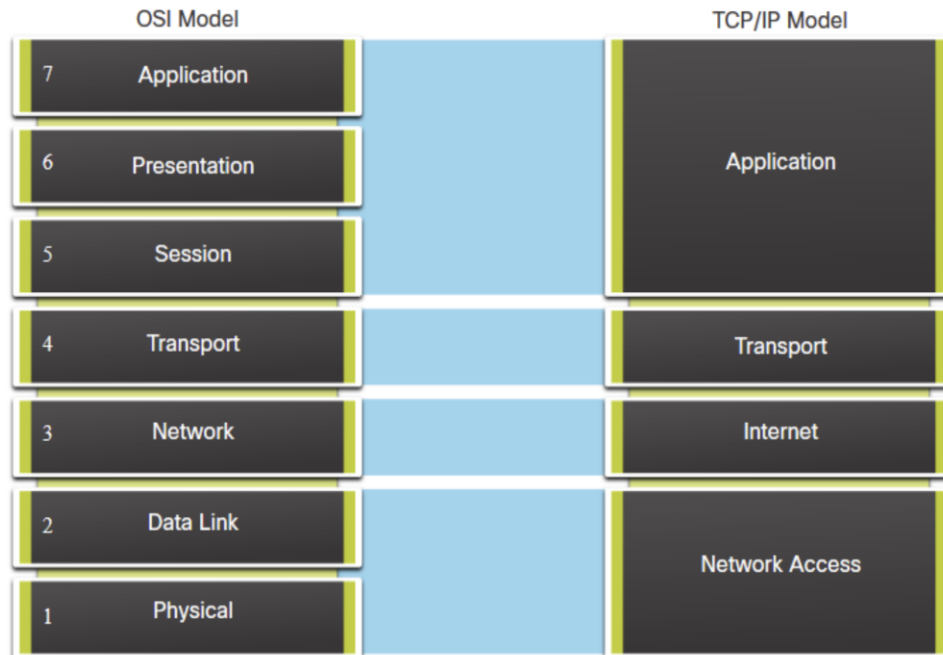
- The TCP/IP Reference Model

TCP/IP Model Layer	Description
Application	Represents data to the user, plus encoding and dialog control.
Transport	Supports communication between various devices across diverse networks.
Internet	Determines the best path through the network.
Network Access	Controls the hardware devices and media that make up the network.

○

- OSI and TCP/IP Model Comparison

- The OSI model divides the network access layer and the application layer of the TCP/IP model into multiple layers.
- The TCP/IP protocol suite does not specify which protocols to use when transmitting over a physical medium.
- OSI Layers 1 and 2 discuss the necessary procedures to access the media and the physical means to send data over a network.



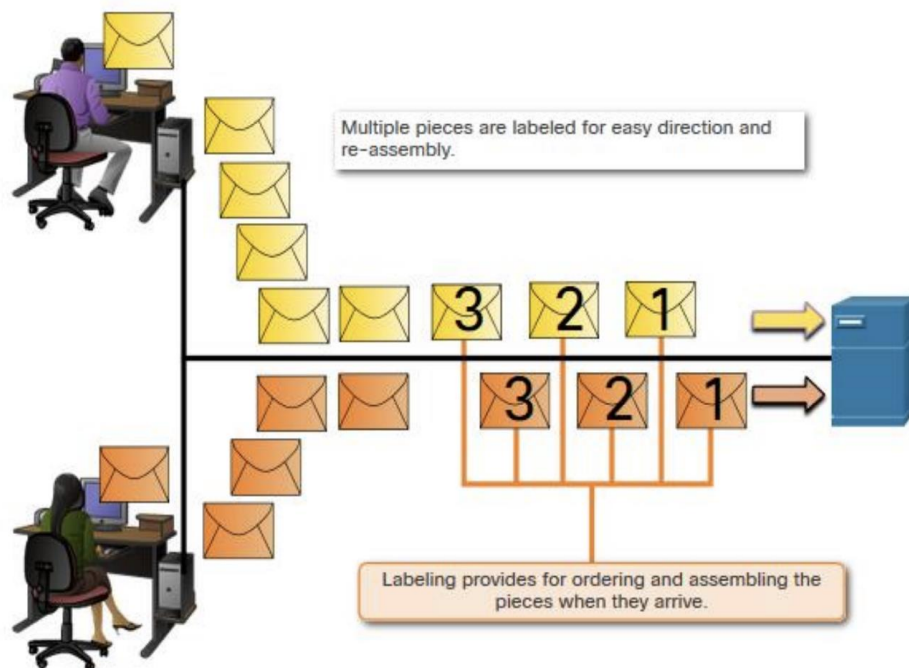
Data Encapsulation

- Segmenting Messages

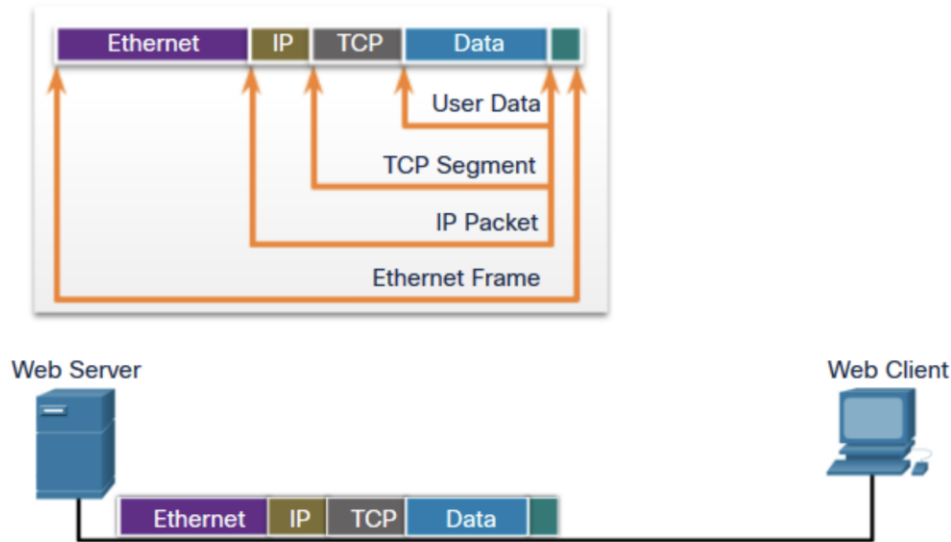
- Segmenting is the process of breaking up messages into smaller units. Multiplexing is the process of taking multiple streams of segmented data and interleaving them together.
- Segmenting messages has two primary benefits:
 - Increases speed
 - Large amounts of data can be sent over the network without tying up a communications link.
 - Increases efficiency
 - Only segments which fail to reach the destination need to be retransmitted, not the entire data stream.

- Sequencing

- Sequencing messages is the process of numbering the segments so that the message may be reassembled at the destination.
- TCP is responsible for sequencing the individual segments.



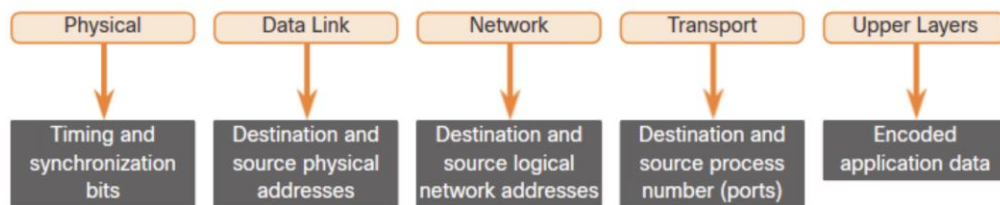
-
- **Protocol Data Units**
 - Encapsulation is the process where protocols add their information to the data.
 - At each stage of the process, a PDU has a different name to reflect its new functions.
 - There is no universal naming convention for PDUs, in this course, the PDUs are named according to the protocols of the TCP/IP suite.
 - PDUs passing down the stack are as follows:
 - 1. Data (Data Stream)
 - 2. Segment
 - 3. Packet
 - 4. Frame
 - 5. Bits (Bit Stream)
- **Encapsulation Example**
 - Encapsulation is a top down process.
 - The level above does its process and then passes it down to the next level of the model. This process is repeated by each layer until it is sent out as a bit stream.



-
- **De-encapsulation Example**
 - Data is de-encapsulated as it moves up the stack.
 - When a layer completes its process, that layer strips off its header and passes it up to the next level to be processed. This is repeated at each layer until it is a data stream that the application can process.
 - 1. Received as Bits (Bit Stream)
 - 2. Frame
 - 3. Packet
 - 4. Segment
 - 5. Data (Data Stream)

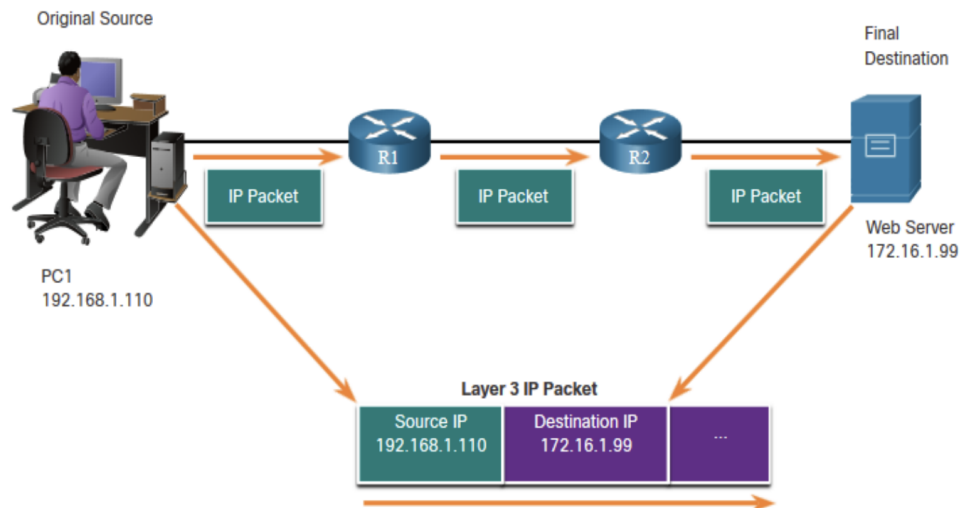
Data Access

- **Addresses**
 - Both the data link and network layers use addressing to deliver data from source to destination.
 - Network layer source and destination addresses
 - Responsible for delivering the IP packet from original source to the final destination.
 - Data link layer source and destination addresses
 - Responsible for delivering the data link frame from one network interface card (NIC) to another NIC on the same network.

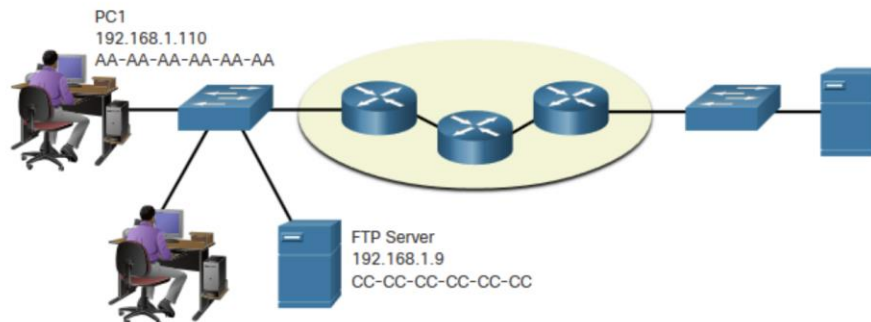
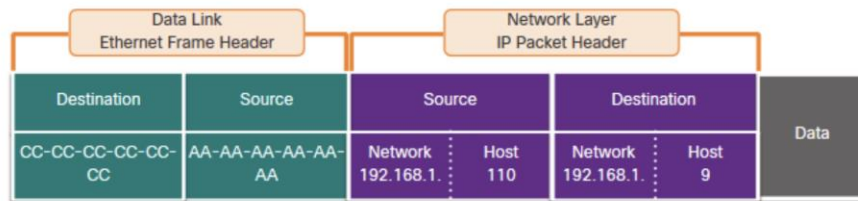


-
- **Layer 3 Logical Address**
 - The IP packet contains two IP addresses:
 - Source IP address

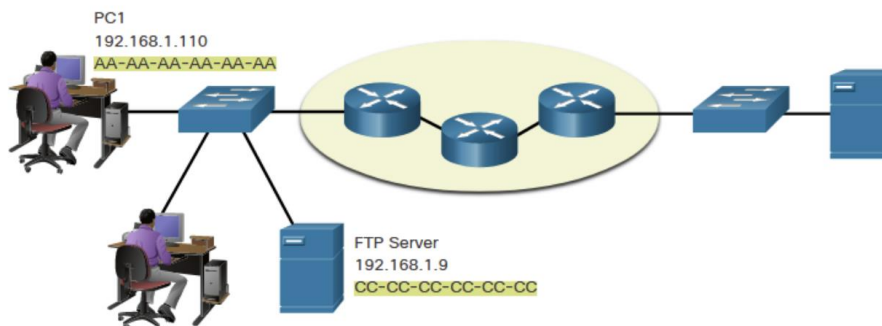
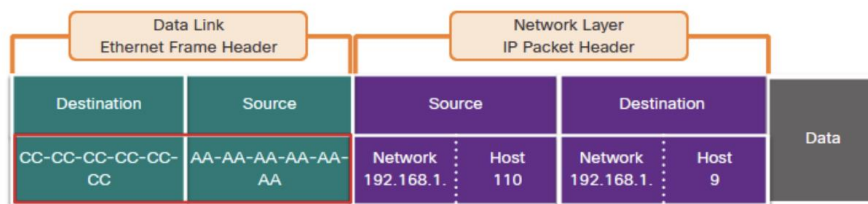
- The IP address of the sending device, original source of the packet.
- Destination IP address
 - The IP address of the receiving device, final destination of the packet.
- These addresses may be on the same link or remote.



-
- An IP address contains two parts:
 - Network portion (IPv4) or Prefix (IPv6)
 - The left-most part of the address indicates the network group which the IP address is a member.
 - Each LAN or WAN will have the same network portion.
 - Host portion (IPv4) or Interface ID (IPv6)
 - The remaining part of the address identifies a specific device within the group.
 - This portion is unique for each device on the network.
- **Devices on the Same Network**
 - When devices are on the same network the source and destination will have the same number in network portion of the address.
 - PC1 – 192.168.1.110
 - FTP Server – 192.168.1.9



-
- **Role of the Data Link Layer Addresses: Same IP Network**
 - When devices are on the same Ethernet network the data link frame will use the actual MAC address of the destination NIC.
 - MAC addresses are physically embedded into the Ethernet NIC and are local addressing.
 - The Source MAC address will be that of the originator on the link.
 - The Destination MAC address will always be on the same link as the source, even if the ultimate destination is remote.



-
- **Devices on a Remote Network**
 - What happens when the actual (ultimate) destination is not on the same LAN and is remote?
 - What happens when PC1 tries to reach the Web Server?
 - Does this impact the network and data link layers?

- **Role of the Network Layer Addresses**

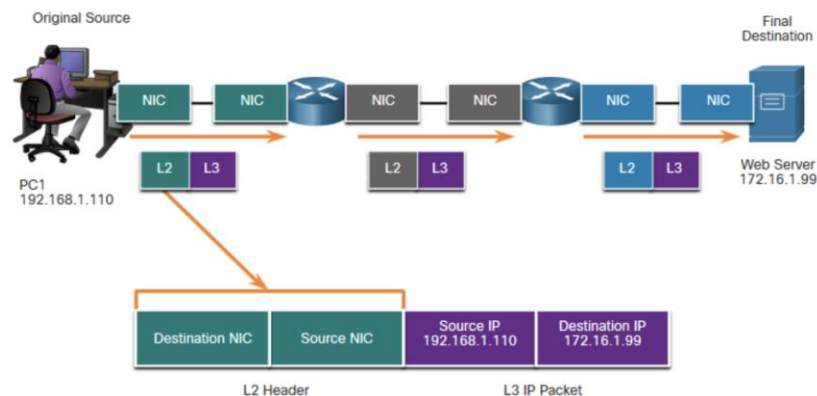
- When the source and destination have a different network portion, this means they are on different networks.
- PC1 – 192.168.1
- Web Server – 172.16.1

- **Role of the Data Link Layer Addresses: Different IP Networks**

- When the final destination is remote, Layer 3 will provide Layer 2 with the local default gateway IP address, also known as the router address.
 - The default gateway (DGW) is the router interface IP address that is part of this LAN and will be the “door” or “gateway” to all other remote locations.
 - All devices on the LAN must be told about this address or their traffic will be confined to the LAN only.
 - Once Layer 2 on PC1 forwards to the default gateway (Router), the router then can start the routing process of getting the information to actual destination.
 - The data link addressing is local addressing so it will have a source and destination for each link.
 - The MAC addressing for the first segment is :
 - Source – AA-AA-AA-AA-AA-AA (PC1) Sends the frame.
 - Destination – 11-11-11-11-11 (R1- Default Gateway MAC) Receives the frame.
- Note: While the L2 local addressing will change from link to link or hop to hop, the L3 addressing remains the same.

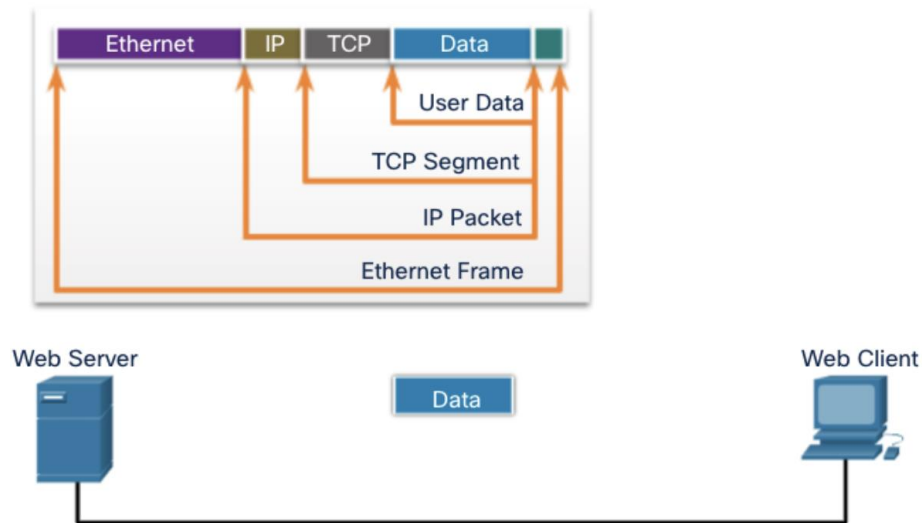
- **Data Link Addresses**

- Since data link addressing is local addressing, it will have a source and destination for each segment or hop of the journey to the destination.
 - The MAC addressing for the first segment is:
 - Source – (PC1 NIC) sends frame
 - Destination – (First Router- DGW interface) receives frame



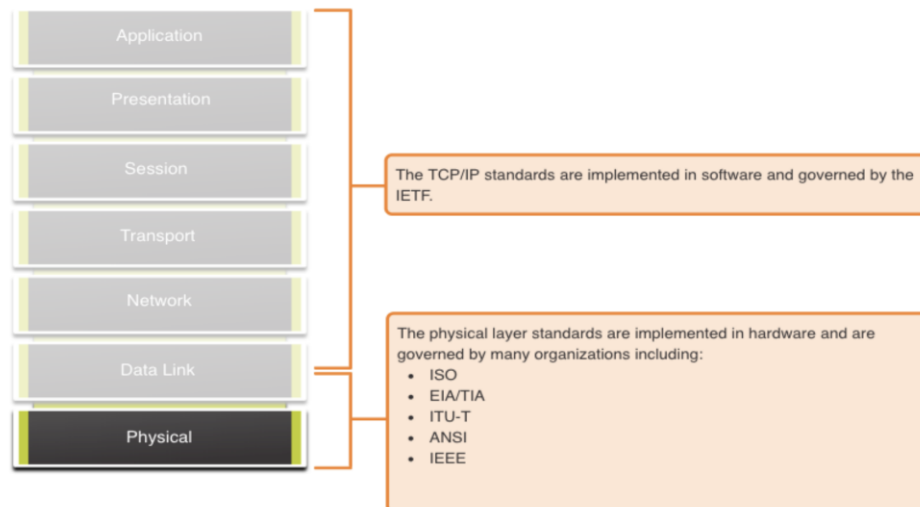
- The MAC addressing for the second hop is:
 - Source – (First Router- exit interface) sends frame
 - Destination – (Second Router) receives frame

- Accepts a complete frame from the Data Link Layer and encodes it as a series of signals that are transmitted to the local media
- This is the last step in the encapsulation process.
- The next device in the path to the destination receives the bits and re-encapsulates the frame, then decides what to do with it.



Physical Layer Characteristics

- Physical Layer Standards



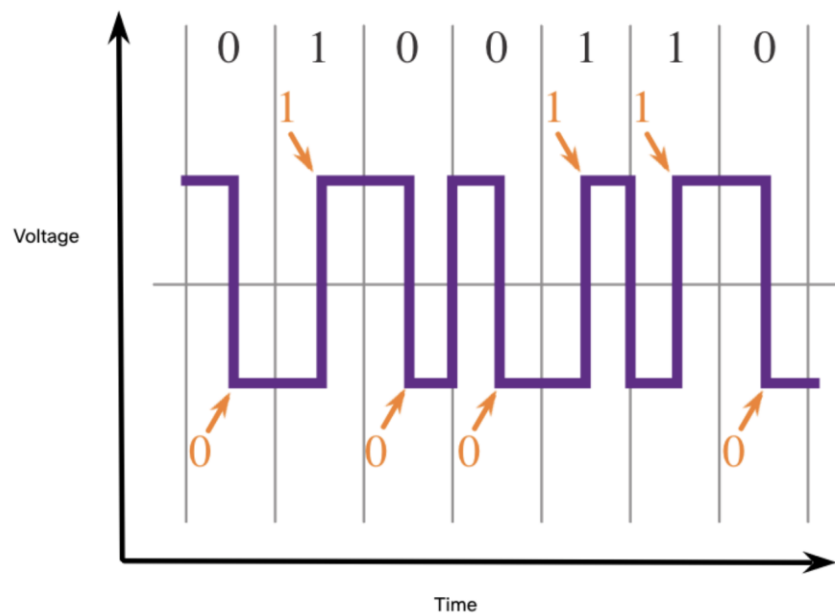
- Physical Components

- Physical Layer Standards address three functional areas:
 - Physical Components
 - Encoding
 - Signaling
- The Physical Components are the hardware devices, media, and other connectors that transmit the signals that represent the bits.

- Hardware components like NICs, interfaces and connectors, cable materials, and cable designs are all specified in standards associated with the physical layer.

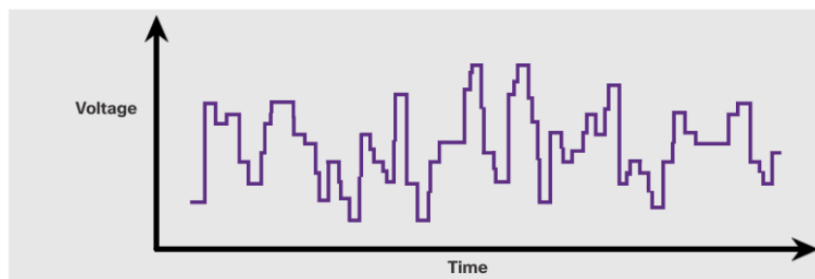
- **Encoding**

- Encoding converts the stream of bits into a format recognizable by the next device in the network path.
- This 'coding' provides predictable patterns that can be recognized by the next device.
- Examples of encoding methods include Manchester (shown in the figure), 4B/5B, and 8B/10B.



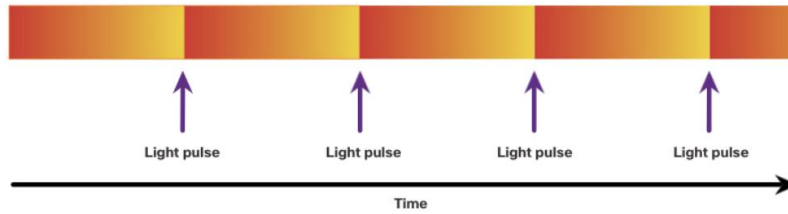
-
- **Signaling**

- The signaling method is how the bit values, "1" and "0" are represented on the physical medium.
- The method of signaling will vary based on the type of medium being used.

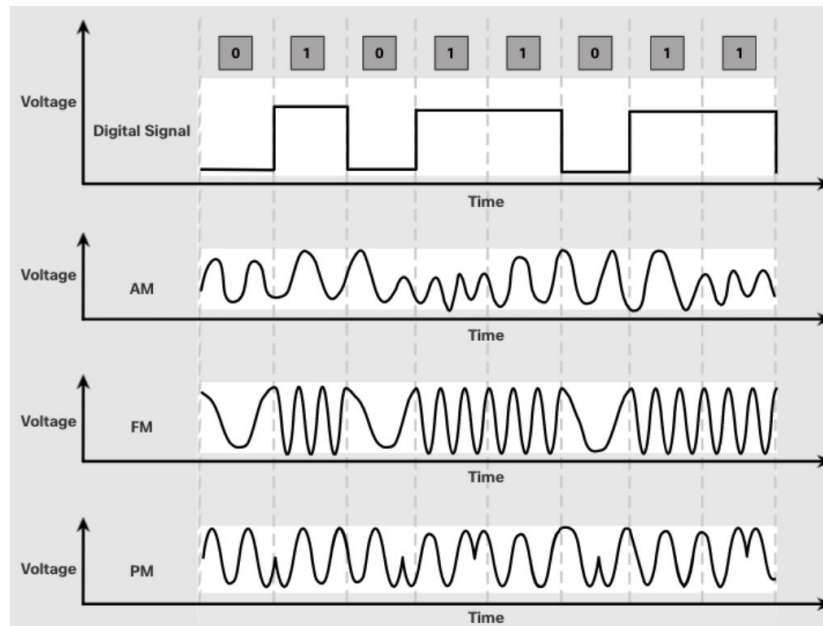


Electrical Signals Over Copper Cable

-



Light Pulses Over Fiber-Optic Cable



Microwave Signals Over Wireless

● Bandwidth

- Bandwidth is the capacity at which a medium can carry data.
- Digital bandwidth measures the amount of data that can flow from one place to another in a given amount of time; how many bits can be transmitted in a second.
- Physical media properties, current technologies, and the laws of physics play a role in determining available bandwidth.

Unit of Bandwidth	Abbreviation	Equivalence
Bits per second	bps	1 bps = fundamental unit of bandwidth
Kilobits per second	Kbps	1 Kbps = 1,000 bps = 10^3 bps
Megabits per second	Mbps	1 Mbps = 1,000,000 bps = 10^6 bps
Gigabits per second	Gbps	1 Gbps = 1,000,000,000 bps = 10^9 bps
Terabits per second	Tbps	1 Tbps = 1,000,000,000,000 bps = 10^{12} bps

● Bandwidth Terminology

- Latency

- Amount of time, including delays, for data to travel from one given point to another
- Throughput
 - The measure of the transfer of bits across the media over a given period of time
- Goodput
 - The measure of usable data transferred over a given period of time
 - $\text{Goodput} = \text{Throughput} - \text{traffic overhead}$

Copper Cabling

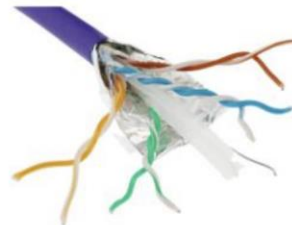
● Characteristics of Copper Cabling

- Copper cabling is the most common type of cabling used in networks today. It is inexpensive, easy to install, and has low resistance to electrical current flow.
- Limitations:
 - Attenuation – the longer the electrical signals have to travel, the weaker they get.
 - The electrical signal is susceptible to interference from two sources, which can distort and corrupt the data signals (Electromagnetic Interference (EMI) and Radio Frequency Interference (RFI) and Crosstalk).
- Mitigation:
 - Strict adherence to cable length limits will mitigate attenuation.
 - Some kinds of copper cable mitigate EMI and RFI by using metallic shielding and grounding.
 - Some kinds of copper cable mitigate crosstalk by twisting opposing circuit pair wires together.

● Types of Copper Cabling



Unshielded Twisted-Pair (UTP) Cable



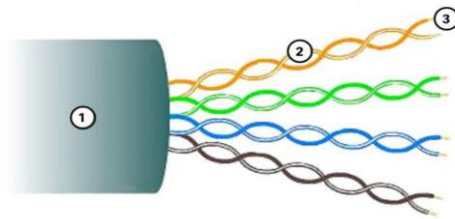
Shielded Twisted-Pair (STP) Cable



Coaxial Cable

○

- Unshielded Twisted Pair (UTP)



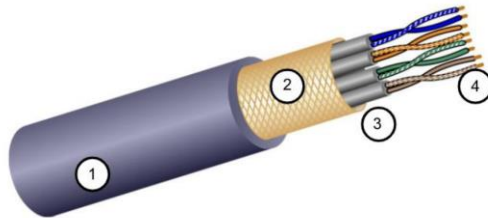
- UTP is the most common networking media.
- Terminated with RJ-45 connectors
- Interconnects hosts with intermediary network devices.

Key Characteristics of UTP

1. The outer jacket protects the copper wires from physical damage.
2. Twisted pairs protect the signal from interference.
3. Color-coded plastic insulation electrically isolates the wires from each other and identifies each pair.

- Shielded Twisted Pair (STP)

- Better noise protection than UTP
- More expensive than UTP
- Harder to install than UTP
- Terminated with RJ-45 connectors
- Interconnects hosts with intermediary network devices
- Key Characteristics of STP 1.
 - The outer jacket protects the copper wires from physical damage
 - 2. Braided or foil shield provides EMI/RFI protection
 - 3. Foil shield for each pair of wires provides EMI/RFI protection
 - 4. Color-coded plastic insulation electrically isolates the wires from each other and identifies each pair



- Coaxial Cable

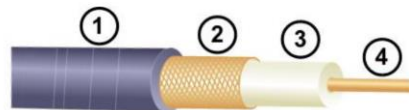
Consists of the following:

1. Outer cable jacket to prevent minor physical damage
2. A woven copper braid, or metallic foil, acts as the second wire in the circuit and as a shield for the inner conductor.
3. A layer of flexible plastic insulation
4. A copper conductor is used to transmit the electronic signals.

There are different types of connectors used with coax cable.

Commonly used in the following situations:

- Wireless installations - attach antennas to wireless devices
- Cable internet installations - customer premises wiring



UTP Cabling

- Properties of UTP Cabling

- UTP has four pairs of color-coded copper wires twisted together and encased in a flexible plastic sheath. No shielding is used. UTP relies on the following properties to limit crosstalk:
 - Cancellation - Each wire in a pair of wires uses opposite polarity. One wire is negative, the other wire is positive. They are twisted together and the magnetic fields effectively cancel each other and outside EMI/RFI.
 - Variation in twists per foot in each wire - Each wire is twisted a different amount, which helps prevent crosstalk amongst the wires in the cable.

- UTP Cabling Standards and Connectors

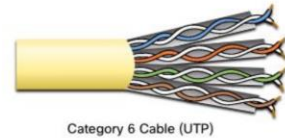
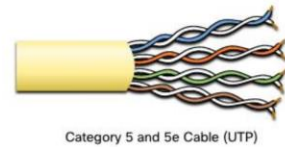
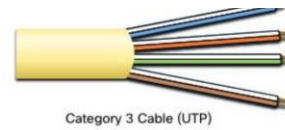
Standards for UTP are established by the TIA/EIA. TIA/EIA-568 standardizes elements like:

- Cable Types
- Cable Lengths
- Connectors
- Cable Termination
- Testing Methods

Electrical standards for copper cabling are established by the IEEE, which rates cable according to its performance.

Examples include:

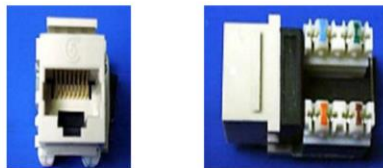
- Category 3
- Category 5 and 5e
- Category 6



RJ-45 Connector



Poorly terminated UTP cable

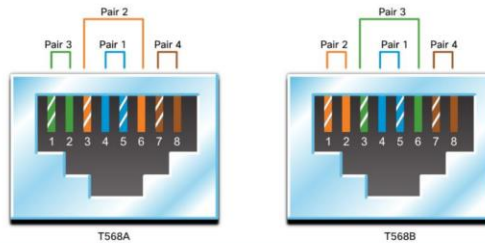


RJ-45 Socket



Properly terminated UTP cable

- Straight-through and Crossover UTP Cables



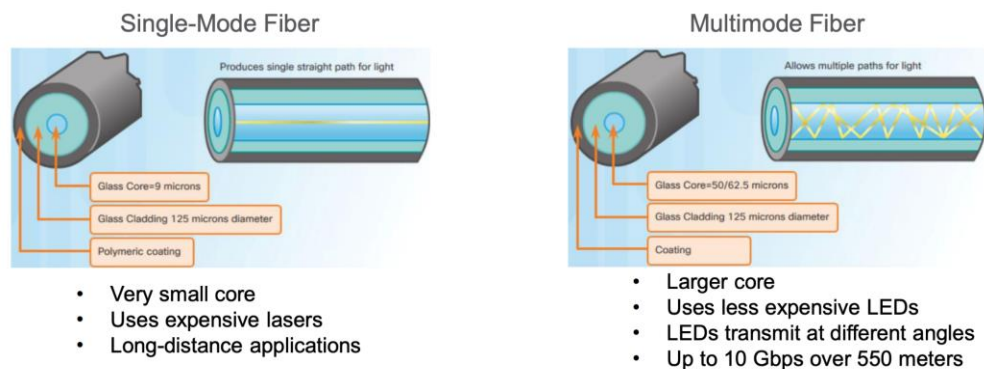
Cable Type	Standard	Application
Ethernet Straight-through	Both ends T568A or T568B	Host to Network Device
Ethernet Crossover *	One end T568A, other end T568B	Host-to-Host, Switch-to-Switch, Router-to-Router
* Considered Legacy due to most NICs using Auto-MDIX to sense cable type and complete connection		
Rollover	Cisco Proprietary	Host serial port to Router or Switch Console Port, using an adapter

Fiber-Optic Cabling

- Properties of Fiber-Optic Cabling

- Not as common as UTP because of the expense involved
- Ideal for some networking scenarios
- Transmits data over longer distances at higher bandwidth than any other networking media
- Less susceptible to attenuation, and completely immune to EMI/RFI
- Made of flexible, extremely thin strands of very pure glass
- Uses a laser or LED to encode bits as pulses of light
- The fiber-optic cable acts as a wave guide to transmit light between the two ends with minimal signal loss

- Types of Fiber Media



Dispersion refers to the spreading out of a light pulse over time. Increased dispersion means increased loss of signal strength. MMF has greater dispersion than SMF, with a the maximum cable distance for MMF is 550 meters.

- Fiber-Optic Cabling Usage

- Fiber-optic cabling is now being used in four types of industry:
 - 1. Enterprise Networks
 - Used for backbone cabling applications and interconnecting infrastructure devices

- 2. Fiber-to-the-Home (FTTH)
 - Used to provide always-on broadband services to homes and small businesses
- 3. Long-Haul Networks
 - Used by service providers to connect countries and cities
- 4. Submarine Cable Networks
 - Used to provide reliable high-speed, high-capacity solutions capable of surviving in harsh undersea environments at up to transoceanic distances.
- Our focus in this course is the use of fiber within the enterprise.

● Fiber-Optic Connectors



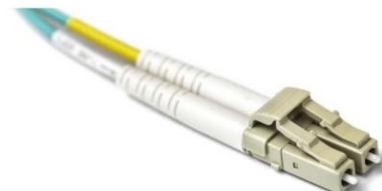
Straight-Tip (ST) Connectors



Lucent Connector (LC) Simplex Connectors



Subscriber Connector (SC) Connectors



Duplex Multimode LC Connectors

● Fiber Patch Cords



SC-SC MM Patch Cord



LC-LC SM Patch Cord



ST-LC MM Patch Cord



ST-SC SM Patch Cord

○ A yellow jacket is for single-mode fiber cables and orange (or aqua) for multimode fiber cables.

● Fiber versus Copper

- Optical fiber is primarily used as backbone cabling for high-traffic, point-to-point connections between data distribution facilities and for the interconnection of buildings in multi-building campuses.

Implementation Issues	UTP Cabling	Fiber-Optic Cabling
Bandwidth supported	10 Mb/s - 10 Gb/s	10 Mb/s - 100 Gb/s
Distance	Relatively short (1 - 100 meters)	Relatively long (1 - 100,000 meters)
Immunity to EMI and RFI	Low	High (Completely immune)
Immunity to electrical hazards	Low	High (Completely immune)
Media and connector costs	Lowest	Highest
Installation skills required	Lowest	Highest
Safety precautions	Lowest	Highest

Wireless Media

• Properties of Wireless Media

- It carries electromagnetic signals representing binary digits using radio or microwave frequencies. This provides the greatest mobility option. Wireless connection numbers continue to increase.
- Some of the limitations of wireless:
 - Coverage area
 - Effective coverage can be significantly impacted by the physical characteristics of the deployment location.
 - Interference
 - Wireless is susceptible to interference and can be disrupted by many common devices.
 - Security
 - Wireless communication coverage requires no access to a physical strand of media, so anyone can gain access to the transmission.
 - Shared medium
 - WLANs operate in half-duplex, which means only one device can send or receive at a time. Many users accessing the WLAN simultaneously results in reduced bandwidth for each user.

• Types of Wireless Media

The IEEE and telecommunications industry standards for wireless data communications cover both the data link and physical layers. In each of these standards, physical layer specifications dictate:

- Data to radio signal encoding methods
- Frequency and power of transmission
- Signal reception and decoding requirements
- Antenna design and construction

Wireless Standards:

- **Wi-Fi (IEEE 802.11)** - Wireless LAN (WLAN) technology
- **Bluetooth (IEEE 802.15)** - Wireless Personal Area network (WPAN) standard
- **WiMAX (IEEE 802.16)** - Uses a point-to-multipoint topology to provide broadband wireless access
- **Zigbee (IEEE 802.15.4)** - Low data-rate, low power-consumption communications, primarily for Internet of Things (IoT) applications

- **Wireless LAN**

- In general, a Wireless LAN (WLAN) requires the following devices:
 - Wireless Access Point (AP)
 - Concentrate wireless signals from users and connect to the existing copper-based network infrastructure
 - Wireless NIC Adapters
 - Provide wireless communications capability to network hosts
- There are a number of WLAN standards. When purchasing WLAN equipment, ensure compatibility, and interoperability.
- Network Administrators must develop and apply stringent security policies and processes to protect WLANs from unauthorized access and damage.

Number Systems & Data Link Layer

Binary Number System

- **Binary and IPv4 Addresses**

- Binary numbering system consists of 1s and 0s, called bits
- Decimal numbering system consists of digits 0 through 9
- Hosts, servers, and network equipment using binary addressing to identify each other.
- Each address is made up of a string of 32 bits, divided into four sections called octets.
- Each octet contains 8 bits (or 1 byte) separated by a dot.
- For ease of use by people, this dotted notation is converted to dotted decimal.

- **Binary Positional Notation**

- Positional notation means that a digit represents different values depending on the “position” the digit occupies in the sequence of numbers.
- The decimal positional notation system operates as shown in the tables below.

Radix	10	10	10	10
Position in Number	3	2	1	0
Calculate	(10^3)	(10^2)	(10^1)	(10^0)
Position Value	1000	100	10	1



	Thousands	Hundreds	Tens	Ones
Positional Value	1000	100	10	1
Decimal Number (1234)	1	2	3	4
Calculate	1×1000	2×100	3×10	4×1
Add them up...	1000	+ 200	+ 30	+ 4
Result	1,234			

-
- The binary positional notation system operates as shown in the tables below.

Radix	2	2	2	2	2	2	2	2
Position in Number	7	6	5	4	3	2	1	0
Calculate	(2 ⁷)	(2 ⁶)	(2 ⁵)	(2 ⁴)	(2 ³)	(2 ²)	(2 ¹)	(2 ⁰)
Position Value	128	64	32	16	8	4	2	1



Positional Value	128	64	32	16	8	4	2	1
Binary Number (11000000)	1	1	0	0	0	0	0	0
Calculate	1x128	1x64	0x32	0x16	0x8	0x4	0x2	0x1
Add Them Up...	128	+ 64	+ 0	+ 0	+ 0	+ 0	+ 0	+ 0
Result	192							

● Convert Binary to Decimal

Convert 11000000.10101000.00001011.00001010 to decimal.

Positional Value	128	64	32	16	8	4	2	1
Binary Number (11000000)	1	1	0	0	0	0	0	0
Calculate	1x128	1x64	0x32	0x16	0x8	0x4	0x2	0x1
Add Them Up...	128	+ 64	+ 0	+ 0	+ 0	+ 0	+ 0	+ 0
Binary Number (10101000)	1	0	1	0	1	0	0	0
Calculate	1x128	0x64	1x32	0x16	1x8	0x4	0x2	0x1
Add Them Up...	128	+ 0	+ 32	+ 0	+ 8	+ 0	+ 0	+ 0
Binary Number (00001011)	0	0	0	0	1	0	1	1
Calculate	0x128	0x64	0x32	0x16	1x8	0x4	1x2	1x1
Add Them Up...	0	+ 0	+ 0	+ 0	+ 8	+ 0	+ 2	+ 1
Binary Number (00001010)	0	0	0	0	1	0	1	0
Calculate	0x128	0x64	0x32	0x16	1x8	0x4	1x2	0x1
Add Them Up...	0	+ 0	+ 0	+ 0	+ 8	+ 0	+ 2	+ 0

➡ 192

➡ 168

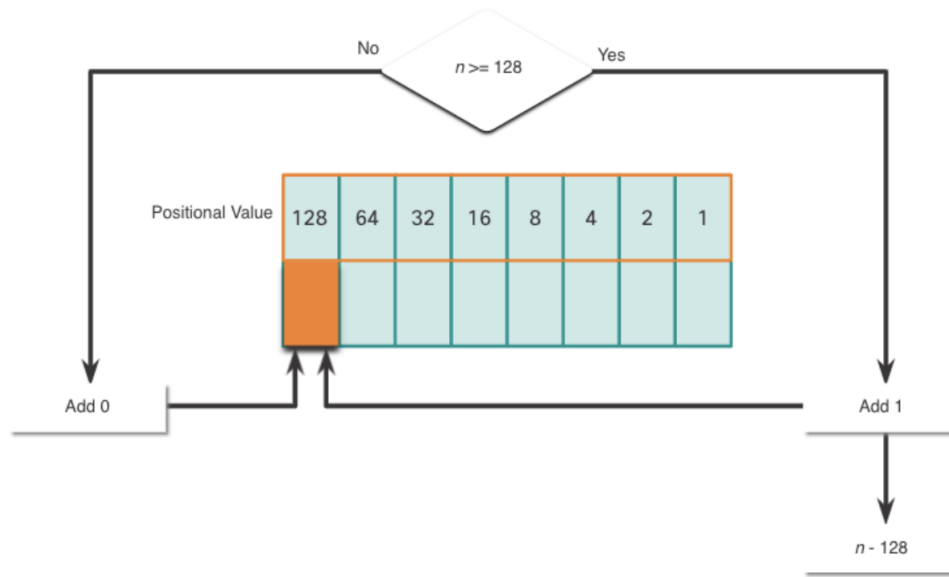
➡ 11

➡ 10

192.168.11.10

● Decimal to Binary Conversion

- The binary positional value table is useful in converting a dotted decimal IPv4 address to binary.
- Start in the 128 position (the most significant bit). Is the decimal number of the octet (n) equal to or greater than 128?
- If no, record a binary 0 in the 128 positional value and move to the 64 positional value.
- If yes, record a binary 1 in the 128 positional value, subtract 128 from the decimal number, and move to the 64 positional value.
- Repeat these steps through the 1 positional value.



- Decimal to Binary Conversion Example

Convert decimal 168 to binary

- Is 168 > 128?
 - Yes, enter 1 in 128 position and subtract 128 (168-128=40)
- Is 40 > 64?
 - No, enter 0 in 64 position and move on
- Is 40 > 32?
 - Yes, enter 1 in 32 position and subtract 32 (40-32=8)
- Is 8 > 16?
 - No, enter 0 in 16 position and move on
- Is 8 > 8?
 - Equal. Enter 1 in 8 position and subtract 8 (8-8=0)
- No values left. Enter 0 in remaining binary positions

128	64	32	16	8	4	2	1
1	0	1	0	1	0	0	0

Decimal 168 is written as 10101000 in binary

- IPv4 Addresses

- Routers and computers only understand binary, while humans work in decimal. It is important for you to gain a thorough understanding of these two numbering systems and how they are used in networking.



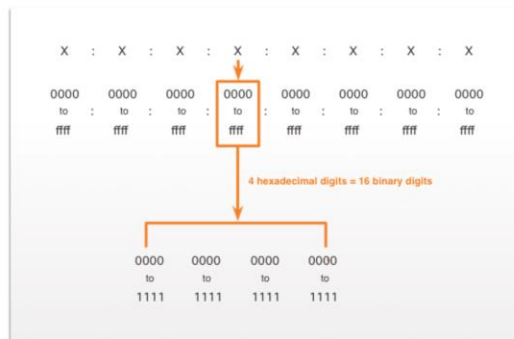
Hexadecimal Number System

Hexadecimal and IPv6 Addresses

- To understand IPv6 addresses, you must be able to convert hexadecimal to decimal and vice versa.
- Hexadecimal is a base sixteen numbering system, using the digits 0 through 9 and letters A to F.
- It is easier to express a value as a single hexadecimal digit than as four binary bit.
- Hexadecimal is used to represent IPv6 addresses and MAC addresses.

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

- IPv6 addresses are 128 bits in length. Every 4 bits is represented by a single hexadecimal digit. That makes the IPv6 address a total of 32 hexadecimal values.
- The figure shows the preferred method of writing out an IPv6 address, with each X representing four hexadecimal values.
- Each four hexadecimal character group is referred to as a hextet.



Decimal to Hexadecimal Conversions

- Follow the steps listed to convert decimal numbers to hexadecimal values:
 - Convert the decimal number to 8-bit binary strings.
 - Divide the binary strings in groups of four starting from the rightmost position.
 - Convert each four binary numbers into their equivalent hexadecimal digit.
- For example, 168 converted into hex using the three-step process.
 - 168 in binary is 10101000.
 - 10101000 in two groups of four binary digits is 1010 and 1000.
 - 1010 is hex A and 1000 is hex 8, so 168 is A8 in hexadecimal.

Hexadecimal to Decimal Conversions

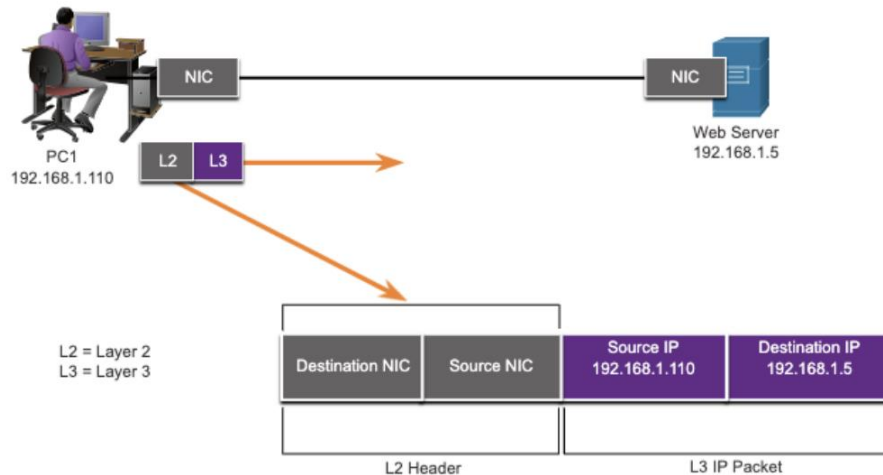
- Follow the steps listed to convert hexadecimal numbers to decimal values:
 - Convert the hexadecimal number to 4-bit binary strings.
 - Create 8-bit binary grouping starting from the rightmost position.
 - Convert each 8-bit binary grouping into their equivalent decimal digit.
- For example, D2 converted into decimal using the three-step process:
 - D2 in 4-bit binary strings is 1101 and 0010.
 - 1101 and 0010 is 11010010 in an 8-bit grouping.
 - 11010010 in binary is equivalent to 210 in decimal, so D2 is 210 in decimal.

Data Link Layer

Purpose of the Data Link Layer

- The Data Link Layer

- The Data Link layer is responsible for communications between end-device network interface cards.
- It allows upper layer protocols to access the physical layer media and encapsulates Layer 3 packets (IPv4 and IPv6) into Layer 2 Frames.
- It also performs error detection and rejects corrupts frames.

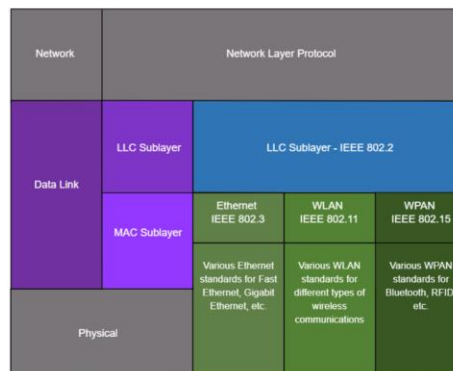


-
- IEEE 802 LAN/MAN Data Link Sublayers

IEEE 802 LAN/MAN standards are specific to the type of network (Ethernet, WLAN, WPAN, etc).

The Data Link Layer consists of two sublayers. **Logical Link Control (LLC)** and **Media Access Control (MAC)**.

- The LLC sublayer communicates between the networking software at the upper layers and the device hardware at the lower layers.
- The MAC sublayer is responsible for data encapsulation and media access control.



-
- Providing Access to Media

- Packets exchanged between nodes may experience numerous data link layers and media transitions.
- At each hop along the path, a router performs four basic Layer 2 functions:
 - Accepts a frame from the network medium.
 - De-encapsulates the frame to expose the encapsulated packet.
 - Re-encapsulates the packet into a new frame.
 - Forwards the new frame on the medium of the next network segment.

- Data Link Layer Standards

Data link layer protocols are defined by engineering organizations:

- Institute for Electrical and Electronic Engineers (IEEE).
- International Telecommunications Union (ITU).
- International Organizations for Standardization (ISO).
- American National Standards Institute (ANSI).



Topologies

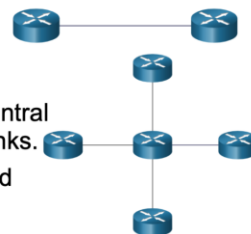
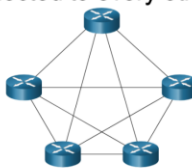
- Physical and Logical Topologies

- The topology of a network is the arrangement and relationship of the network devices and the interconnections between them. There are two types of topologies used when describing networks:
 - Physical topology – shows physical connections and how devices are interconnected.
 - Logical topology – identifies the virtual connections between devices using device interfaces and IP addressing schemes.

- WAN Topologies

There are three common physical WAN topologies:

- **Point-to-point** – the simplest and most common WAN topology. Consists of a permanent link between two endpoints.
- **Hub and spoke** – similar to a star topology where a central site interconnects branch sites through point-to-point links.
- **Mesh** – provides high availability but requires every end system to be connected to every other end system.



© 2019 Cisco and/or its affiliates. All rights reserved. Cisco Confidential

- Point-to-Point WAN Topology

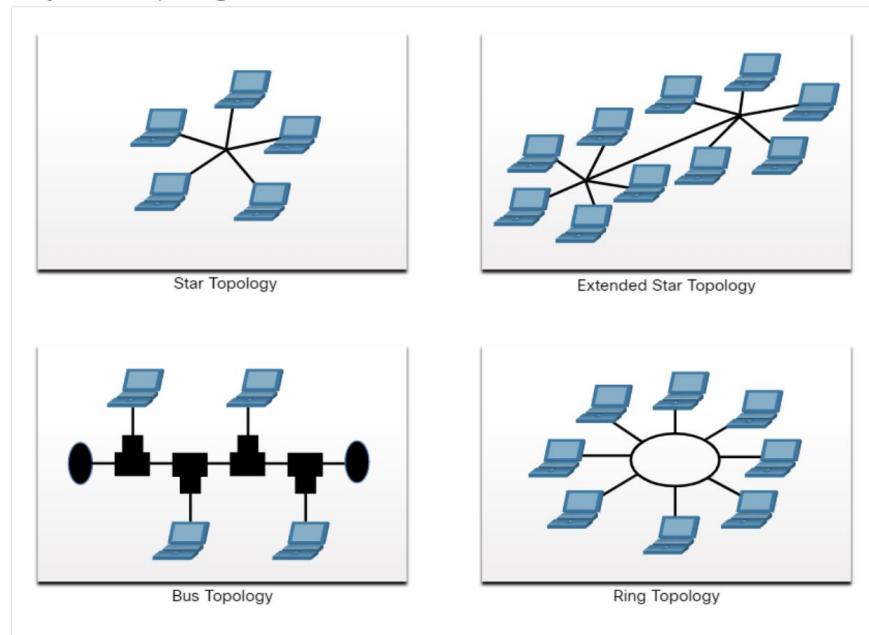
- Physical point-to-point topologies directly connect two nodes.
- The nodes may not share the media with other hosts.
- Because all frames on the media can only travel to or from the two nodes, Point-to-Point WAN protocols can be very simple.



- LAN Topologies

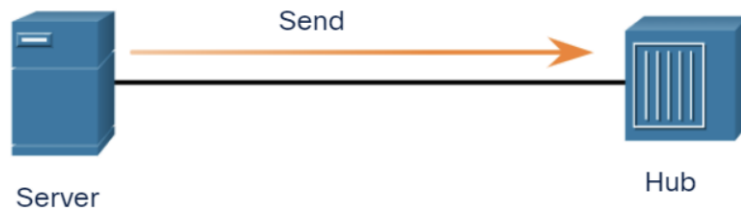
- End devices on LANs are typically interconnected using a star or extended star topology. Star and extended star topologies are easy to install, very scalable and easy to troubleshoot.
- Early Ethernet and Legacy Token Ring technologies provide two additional topologies:
- Bus
 - All end systems chained together and terminated on each end.
- Ring
 - Each end system is connected to its respective neighbors to form a ring.

Physical Topologies

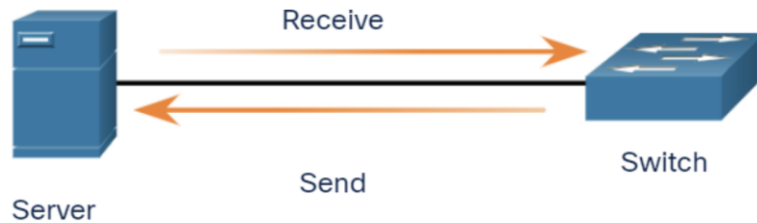


-
- Half and Full Duplex Communication

- Half-duplex communication
 - Only allows one device to send or receive at a time on a shared medium.
 - Used on WLANs and legacy bus topologies with Ethernet hubs.



- Full-duplex communication
 - Allows both devices to simultaneously transmit and receive on a shared medium.
 - Ethernet switches operate in full-duplex mode.



-
- Access Control Methods

- Contention-based access
 - All nodes operating in half-duplex, competing for use of the medium.
- Examples are:

- Carrier sense multiple access with collision detection (CSMA/CD) as used on legacy bus-topology Ethernet.
- Carrier sense multiple access with collision avoidance (CSMA/CA) as used on Wireless LANs.

- Controlled access
 - Deterministic access where each node has its own time on the medium.
 - Used on legacy networks such as Token Ring and ARCNET.

- Contention-Based Access – CSMA/CD

- CSMA/CD
 - Used by legacy Ethernet LANs.
 - Operates in half-duplex mode where only one device sends or receives at a time.
 - Uses a collision detection process to govern when a device can send and what happens if multiple devices send at the same time.
- CSMA/CD collision detection process:
 - Devices transmitting simultaneously will result in a signal collision on the shared media.
 - Devices detect the collision.
 - Devices wait a random period of time and retransmit data.

- Contention-Based Access – CSMA/CA

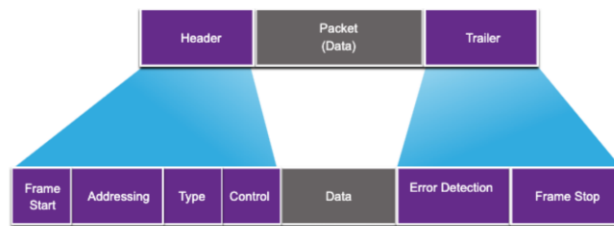
- CSMA/CA
 - Used by IEEE 802.11 WLANs.
 - Operates in half-duplex mode where only one device sends or receives at a time.
 - Uses a collision avoidance process to govern when a device can send and what happens if multiple devices send at the same time.
- CSMA/CA collision avoidance process:
 - When transmitting, devices also include the time duration needed for the transmission.
 - Other devices on the shared medium receive the time duration information and know how long the medium will be unavailable.

Data Link Frame

- The Frame

- Data is encapsulated by the data link layer with a header and a trailer to form a frame.
- A data link frame has three parts:
 - Header
 - Data
 - Trailer
- The fields of the header and trailer vary according to data link layer protocol.
- The amount of control information carried within the frame varies according to access control information and logical topology

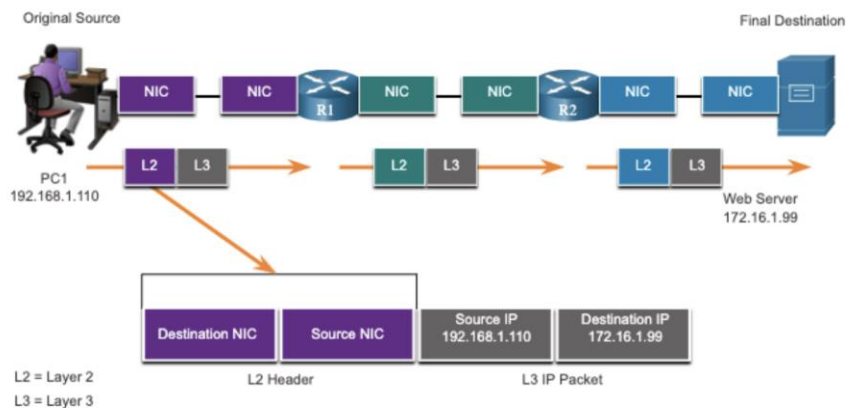
- Frame Fields



Field	Description
Frame Start and Stop	Identifies beginning and end of frame
Addressing	Indicates source and destination nodes
Type	Identifies encapsulated Layer 3 protocol
Control	Identifies flow control services
Data	Contains the frame payload
Error Detection	Used for determine transmission errors

- Layer 2 Addresses

- Also referred to as a physical address.
- Contained in the frame header.
- Used only for local delivery of a frame on the link.
- Updated by each device that forwards the frame.



- LAN and WAN Frames

- The logical topology and physical media determine the data link protocol used:

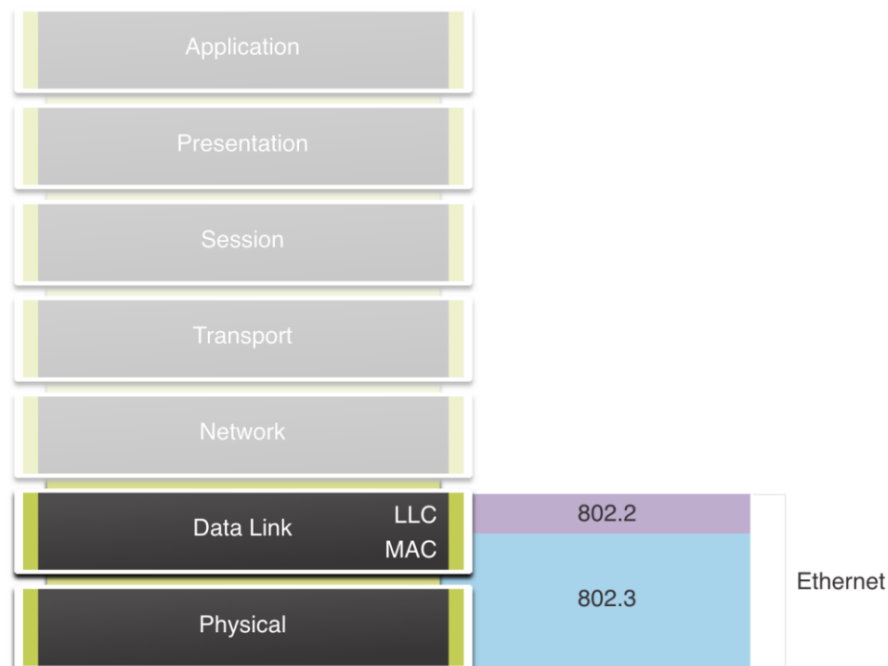
- Ethernet
- 802.11 Wireless
- Point-to-Point (PPP)
- High-Level Data Link Control (HDLC)
- Frame-Relay
- Each protocol performs media access control for specified logical topologies.

Ethernet Switching & Network Layer

Ethernet Frames

- Ethernet Encapsulation

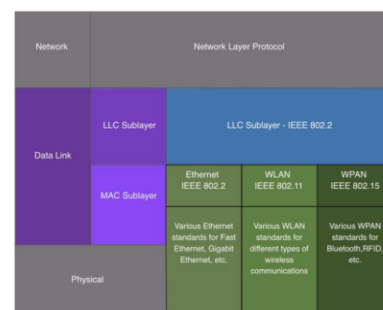
- Ethernet operates in the data link layer and the physical layer.
- It is a family of networking technologies defined in the IEEE 802.2 and 802.3 standards.



-
- Data Link Sublayers

The 802 LAN/MAN standards, including Ethernet, use two separate sublayers of the data link layer to operate:

- **LLC Sublayer:** (IEEE 802.2) Places information in the frame to identify which network layer protocol is used for the frame.
- **MAC Sublayer:** (IEEE 802.3, 802.11, or 802.15) Responsible for data encapsulation and media access control, and provides data link layer addressing.

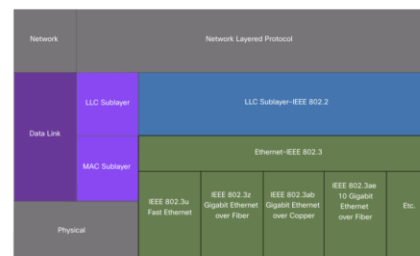


- **MAC Sublayer**

- The MAC sublayer is responsible for data encapsulation and accessing the media.
- Data Encapsulation
 - IEEE 802.3 data encapsulation includes the following:
 - 1. Ethernet frame - This is the internal structure of the Ethernet frame.
 - 2. Ethernet Addressing - The Ethernet frame includes both a source and destination MAC address to deliver the Ethernet frame from Ethernet NIC to Ethernet NIC on the same LAN.
 - 3. Ethernet Error detection - The Ethernet frame includes a frame check sequence (FCS) trailer used for error detection.

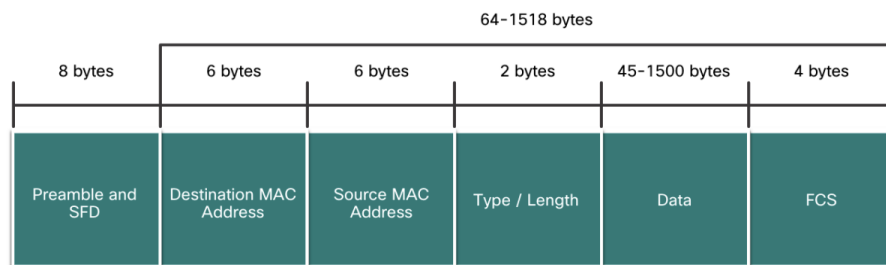
Media Access

- The IEEE 802.3 MAC sublayer includes the specifications for different Ethernet communications standards over various types of media including copper and fiber.
- Legacy Ethernet using a bus topology or hubs, is a shared, half-duplex medium. Ethernet over a half-duplex medium uses a contention-based access method, carrier sense multiple access/collision detection (CSMA/CD).
- Ethernet LANs of today use switches that operate in full-duplex. Full-duplex communications with Ethernet switches do not require access control through CSMA/CD.



- **Ethernet Frame Fields**

- The minimum Ethernet frame size is 64 bytes and the maximum is 1518 bytes. The preamble field is not included when describing the size of the frame.
- Any frame less than 64 bytes in length is considered a “collision fragment” or “runt frame” and is automatically discarded. Frames with more than 1500 bytes of data are considered “jumbo” or “baby giant frames”.
- If the size of a transmitted frame is less than the minimum, or greater than the maximum, the receiving device drops the frame. Dropped frames are likely to be the result of collisions or other unwanted signals. They are considered invalid. Jumbo frames are usually supported by most Fast Ethernet and Gigabit Ethernet switches and NICs.



so Confidential

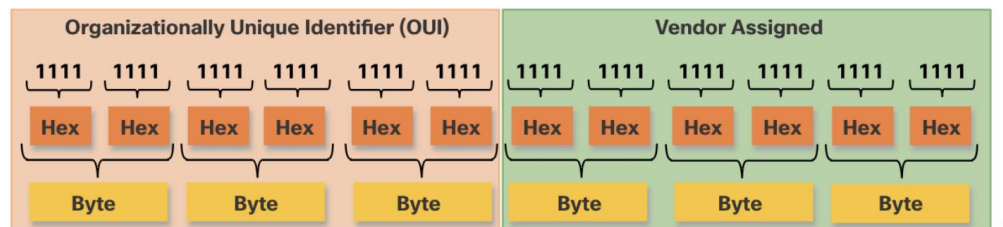
Ethernet MAC Address

- **MAC Address and Hexadecimal**

- An Ethernet MAC address consists of a 48-bit binary value, expressed using 12 hexadecimal values.
- Given that 8 bits (one byte) is a common binary grouping, binary 00000000 to 11111111 can be represented in hexadecimal as the range 00 to FF,
- When using hexadecimal, leading zeroes are always displayed to complete the 8-bit representation. For example the binary value 0000 1010 is represented in hexadecimal as 0A.
- Hexadecimal numbers are often represented by the value preceded by 0x (e.g., 0x73) to distinguish between decimal and hexadecimal values in documentation.
- Hexadecimal may also be represented by a subscript 16, or the hex number followed by an H (e.g., 73H).

- **Ethernet MAC Address**

- In an Ethernet LAN, every network device is connected to the same, shared media. MAC addressing provides a method for device identification at the data link layer of the OSI model.
- An Ethernet MAC address is a 48-bit address expressed using 12 hexadecimal digits. Because a byte equals 8 bits, we can also say that a MAC address is 6 bytes in length.
- All MAC addresses must be unique to the Ethernet device or Ethernet interface. To ensure this, all vendors that sell Ethernet devices must register with the IEEE to obtain a unique 6 hexadecimal (i.e., 24-bit or 3-byte) code called the organizationally unique identifier (OUI).
- An Ethernet MAC address consists of a 6 hexadecimal vendor OUI code followed by a 6 hexadecimal vendor-assigned value.

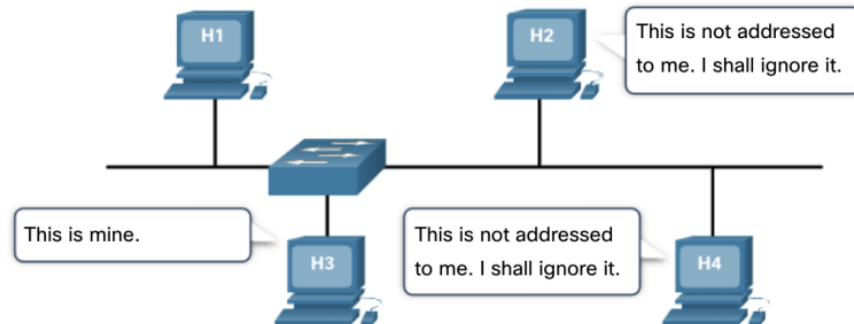


-
- **Frame Processing**

- When a device is forwarding a message to an Ethernet network, the Ethernet header include a Source MAC address and a Destination MAC address.
- When a NIC receives an Ethernet frame, it examines the destination MAC address to see if it matches the physical MAC address that is stored in RAM. If there is no match, the device discards the frame. If there is a match, it passes the frame up the OSI layers, where the de-encapsulation process takes place.
- Note:
 - Ethernet NICs will also accept frames if the destination MAC address is a broadcast or a multicast group of which the host is a member.

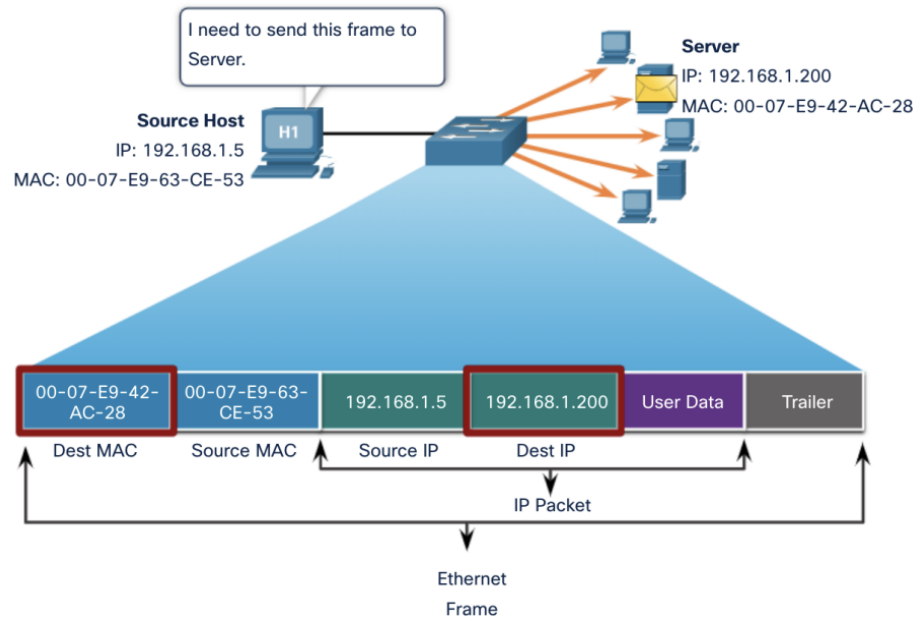
- Any device that is the source or destination of an Ethernet frame, will have an Ethernet NIC and therefore, a MAC address. This includes workstations, servers, printers, mobile devices, and routers.

Destination Address	Source Address	Data
CC:CC:CC:CC:CC:CC	AA:AA:AA:AA:AA:AA	Encapsulated data
Frame Addressing		

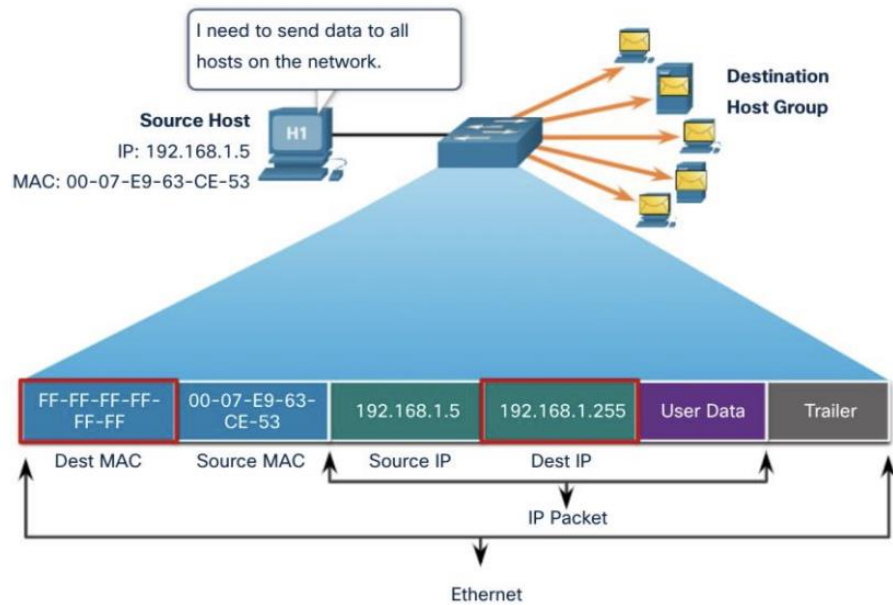


- Unicast MAC Address

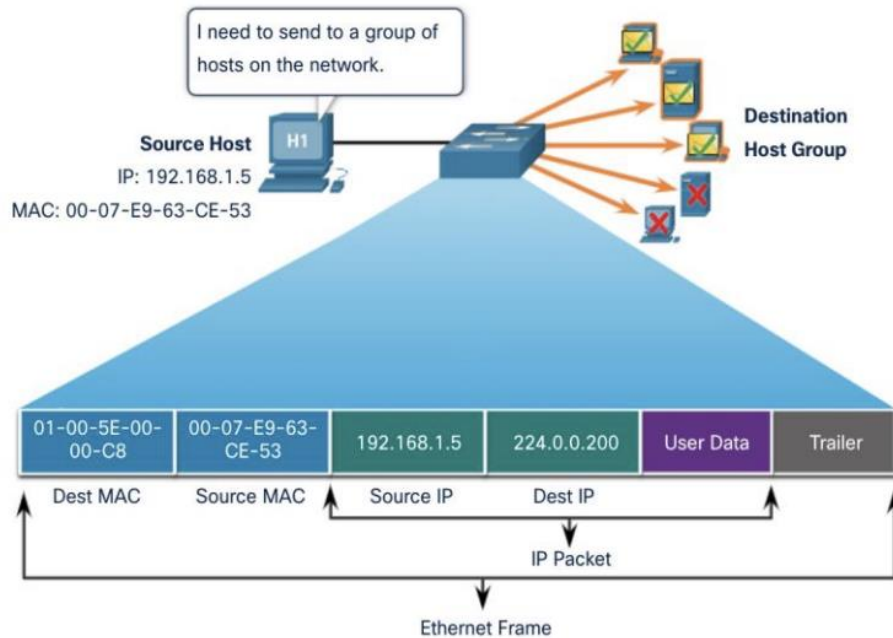
- In Ethernet, different MAC addresses are used for Layer 2 unicast, broadcast, and multicast communications.
 - A unicast MAC address is the unique address that is used when a frame is sent from a single transmitting device to a single destination device.
 - The process that a source host uses to determine the destination MAC address associated with an IPv4 address is known as Address Resolution Protocol (ARP). The process that a source host uses to determine the destination MAC address associated with an IPv6 address is known as Neighbor Discovery (ND).
- Note: The source MAC address must always be a unicast.



-
- **Broadcast MAC Address**
 - An Ethernet broadcast frame is received and processed by every device on the Ethernet LAN. The features of an Ethernet broadcast are as follows:
 - It has a destination MAC address of FF-FF-FFFF-FF-FF in hexadecimal (48 ones in binary).
 - It is flooded out all Ethernet switch ports except the incoming port. It is not forwarded by a router.
 - If the encapsulated data is an IPv4 broadcast packet, this means the packet contains a destination IPv4 address that has all ones (1s) in the host portion. This numbering in the address means that all hosts on that local network (broadcast domain) will receive and process the packet.



-
- **Multicast MAC Address**
 - An Ethernet multicast frame is received and processed by a group of devices that belong to the same multicast group.
 - There is a destination MAC address of 01-00-5E when the encapsulated data is an IPv4 multicast packet and a destination MAC address of 33-33 when the encapsulated data is an IPv6 multicast packet.
 - There are other reserved multicast destination MAC addresses for when the encapsulated data is not IP, such as Spanning Tree Protocol (STP).
 - It is flooded out all Ethernet switch ports except the incoming port, unless the switch is configured for multicast snooping. It is not forwarded by a router, unless the router is configured to route multicast packets.
 - Because multicast addresses represent a group of addresses (sometimes called a host group), they can only be used as the destination of a packet. The source will always be a unicast address.
 - As with the unicast and broadcast addresses, the multicast IP address requires a corresponding multicast MAC address.



The MAC Address Table

• Switch Fundamentals

- A Layer 2 Ethernet switch uses Layer 2 MAC addresses to make forwarding decisions. It is completely unaware of the data (protocol) being carried in the data portion of the frame, such as an IPv4 packet, an ARP message, or an IPv6 ND packet. The switch makes its forwarding decisions based solely on the Layer 2 Ethernet MAC addresses.
- An Ethernet switch examines its MAC address table to make a forwarding decision for each frame, unlike legacy Ethernet hubs that repeat bits out all ports except the incoming port.
- When a switch is turned on, the MAC address table is empty
- Note: The MAC address table is sometimes referred to as a content addressable memory (CAM) table.

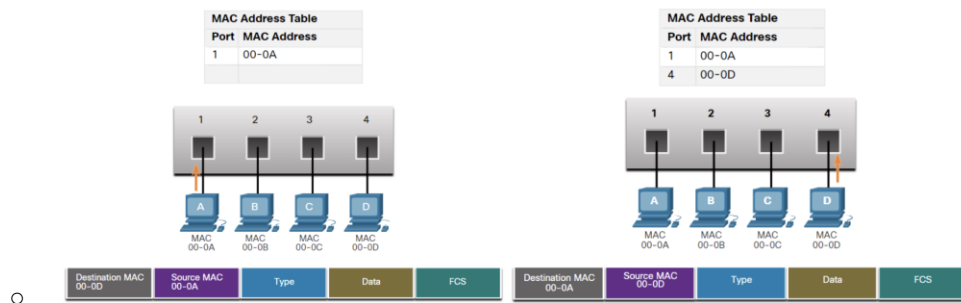
• Switch Learning and Forwarding

- Examine the Source MAC Address (Learn)
- Every frame that enters a switch is checked for new information to learn. It does this by examining the source MAC address of the frame and the port number where the frame entered the switch. If the source MAC address does not exist, it is added to the table along with the incoming port number. If the source MAC address does exist, the switch updates the refresh timer for that entry. By default, most Ethernet switches keep an entry in the table for 5 minutes.
- Note:
 - If the source MAC address does exist in the table but on a different port, the switch treats this as a new entry. The entry is replaced using the same MAC address but with the more current port number

- Find the Destination MAC Address (Forward) If the destination MAC address is a unicast address, the switch will look for a match between the destination MAC address of the frame and an entry in its MAC address table. If the destination MAC address is in the table, it will forward the frame out the specified port. If the destination MAC address is not in the table, the switch will forward the frame out all ports except the incoming port. This is called an unknown unicast.
- Note:
 - If the destination MAC address is a broadcast or a multicast, the frame is also flooded out all ports except the incoming port.

● Filtering Frames

- As a switch receives frames from different devices, it is able to populate its MAC address table by examining the source MAC address of every frame. When the MAC address table of the switch contains the destination MAC address, it is able to filter the frame and forward out a single port.



Switch Speeds and Forwarding Methods

● Frame Forwarding Methods on Cisco Switches

- Switches use one of the following forwarding methods for switching data between network ports:
- Store-and-forward switching
 - This frame forwarding method receives the entire frame and computes the CRC. If the CRC is valid, the switch looks up the destination address, which determines the outgoing interface. Then the frame is forwarded out of the correct port.
 - Cut-through switching
 - This frame forwarding method forwards the frame before it is entirely received. At a minimum, the destination address of the frame must be read before the frame can be forwarded.
- A big advantage of store-and-forward switching is that it determines if a frame has errors before propagating the frame. When an error is detected in a frame, the switch discards the frame. Discarding frames with errors reduces the amount of bandwidth consumed by corrupt data.
- Store-and-forward switching is required for quality of service (QoS) analysis on converged networks where frame classification for traffic prioritization is necessary. For example, voice over IP (VoIP) data streams need to have priority over web-browsing traffic.

- **Cut-Through Switching**

- In cut-through switching, the switch acts upon the data as soon as it is received, even if the transmission is not complete. The switch buffers just enough of the frame to read the destination MAC address so that it can determine to which port it should forward out the data. The switch does not perform any error checking on the frame. There are two variants of cut-through switching:
 - Fast-forward switching
 - Offers the lowest level of latency by immediately forwarding a packet after reading the destination address. Because fast-forward switching starts forwarding before the entire packet has been received, there may be times when packets are relayed with errors. The destination NIC discards the faulty packet upon receipt. Fast Forward switching is the typical cut-through method of switching.
 - Fragment-free switching
 - A compromise between the high latency and high integrity of store-and-forward switching and the low latency and reduced integrity of fast-forward switching, the switch stores and performs an error check on the first 64 bytes of the frame before forwarding. Because most network errors and collisions occur during the first 64 bytes, this ensures that a collision has not occurred before forwarding the frame.

- **Memory Buffering on Switches**

An Ethernet switch may use a buffering technique to store frames before forwarding them or when the destination port is busy because of congestion.

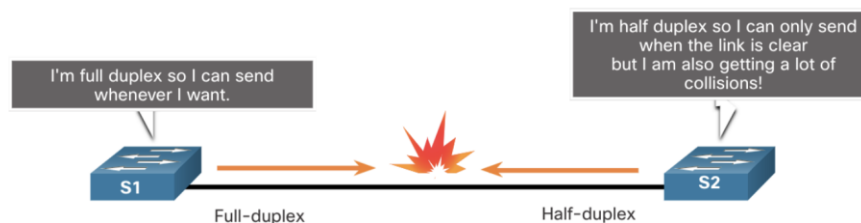
Method	Description
Port-based memory	• Frames are stored in queues that are linked to specific incoming and outgoing ports. • A frame is transmitted to the outgoing port only when all the frames ahead in the queue have been successfully transmitted. • It is possible for a single frame to delay the transmission of all the frames in memory because of a busy destination port. • This delay occurs even if the other frames could be transmitted to open destination ports.
Shared memory	• Deposits all frames into a common memory buffer shared by all switch ports and the amount of buffer memory required by a port is dynamically allocated. • The frames in the buffer are dynamically linked to the destination port enabling a packet to be received on one port and then transmitted on another port, without moving it to a different queue.

- Shared memory buffering also results in larger frames that can be transmitted with fewer dropped frames. This is important with asymmetric switching which allows for different data rates on different ports. Therefore, more bandwidth can be dedicated to certain ports (e.g., server port).

- **Duplex and Speed Settings**

- Two of the most basic settings on a switch are the bandwidth (“speed”) and duplex settings for each individual switch port. It is critical that the duplex and bandwidth settings match between the switch port and the connected devices.
- There are two types of duplex settings used for communications on an Ethernet network:
 - Full-duplex - Both ends of the connection can send and receive simultaneously.
 - Half-duplex - Only one end of the connection can send at a time.

- Autonegotiation is an optional function found on most Ethernet switches and NICs. It enables two devices to automatically negotiate the best speed and duplex capabilities.
- Note:
 - Gigabit Ethernet ports only operate in full-duplex.
- Duplex mismatch is one of the most common causes of performance issues on 10/100 Mbps Ethernet links. It occurs when one port on the link operates at half duplex while the other port operates at full-duplex.
- This can occur when one or both ports on a link are reset, and the auto negotiation process does not result in both link partners having the same configuration.
- It also can occur when users reconfigure one side of a link and forget to reconfigure the other. Both sides of a link should have auto negotiation on, or both sides should have it off. Best practice is to configure both Ethernet switch ports as full-duplex.



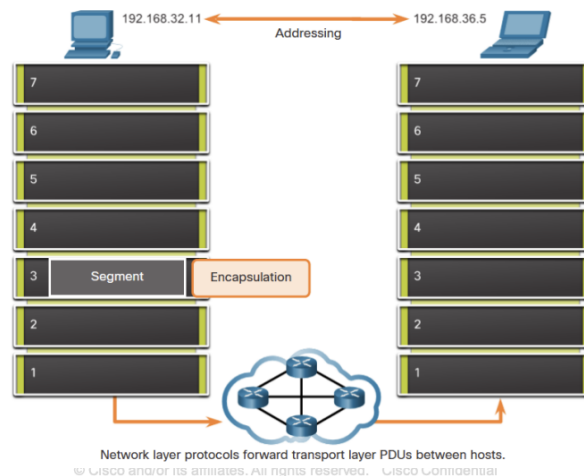
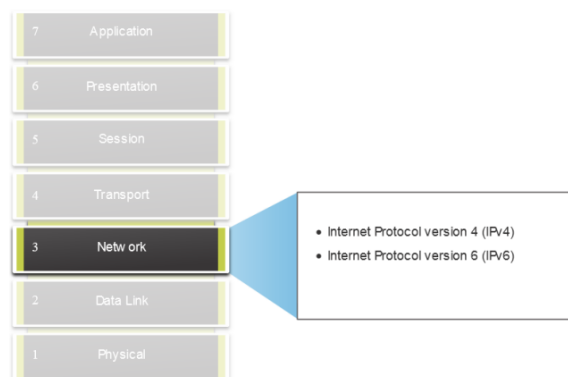
● Auto-MDIX

- Connections between devices once required the use of either a crossover or straight through cable. The type of cable required depended on the type of interconnecting devices.
- Note:
 - A direct connection between a router and a host requires a cross-over connection.
- Most switch devices now support the automatic medium-dependent interface crossover (auto-MDIX) feature. When enabled, the switch automatically detects the type of cable attached to the port and configures the interfaces accordingly.
- The auto-MDIX feature is enabled by default on switches running Cisco IOS Release 12.2(18)SE or later. However, the feature could be disabled. For this reason, you should always use the correct cable type and not rely on the auto-MDIX feature.
- Auto-MDIX can be re-enabled using the `mdix auto` interface configuration command.

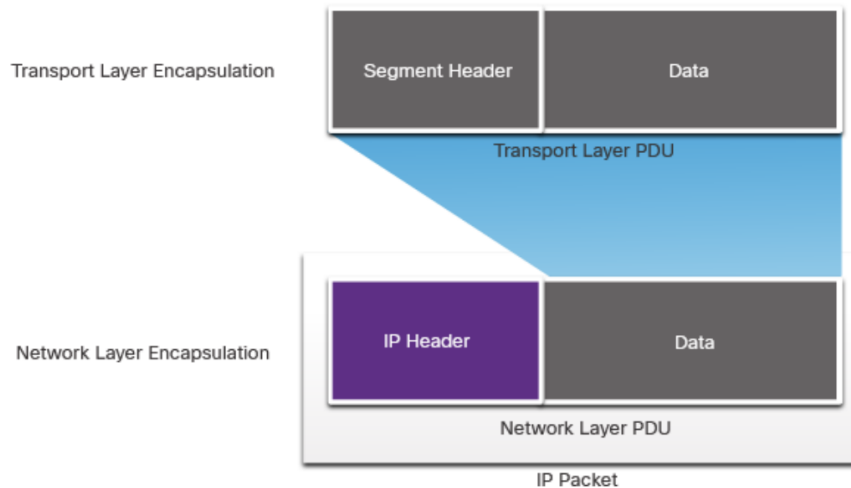
Network Layer

Network Layer Characteristics

- The Network Layer
 - Provides services to allow end devices to exchange data
 - IP version 4 (IPv4) and IP version 6 (IPv6) are the principle network layer communication protocols.
 - The network layer performs four basic operations:
 - Addressing end devices
 - Encapsulation
 - Routing
 - De-encapsulation



-
- IP Encapsulation
 - IP encapsulates the transport layer segment.
 - IP can use either an IPv4 or IPv6 packet and not impact the layer 4 segment.
 - IP packet will be examined by all layer 3 devices as it traverses the network.
 - The IP addressing does not change from source to destination.
 - Note:
 - NAT will change addressing, but will be discussed in a later module.



-
- **Characteristics of IP**

- IP is meant to have low overhead and may be described as:
 - Connectionless
 - Best Effort
 - Media Independent

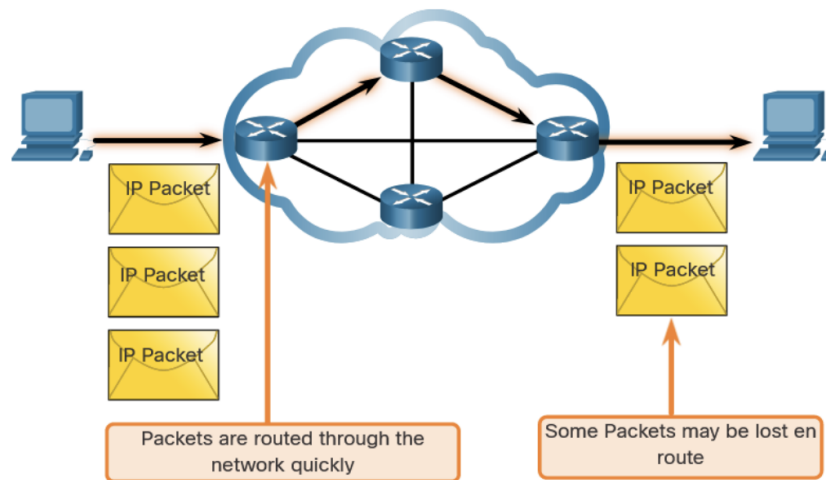
- **Connectionless**

- IP is Connectionless
 - IP does not establish a connection with the destination before sending the packet.
 - There is no control information needed (synchronizations, acknowledgments, etc.).
 - The destination will receive the packet when it arrives, but no pre-notifications are sent by IP.
 - If there is a need for connection-oriented traffic, then another protocol will handle this (typically TCP at the transport layer).

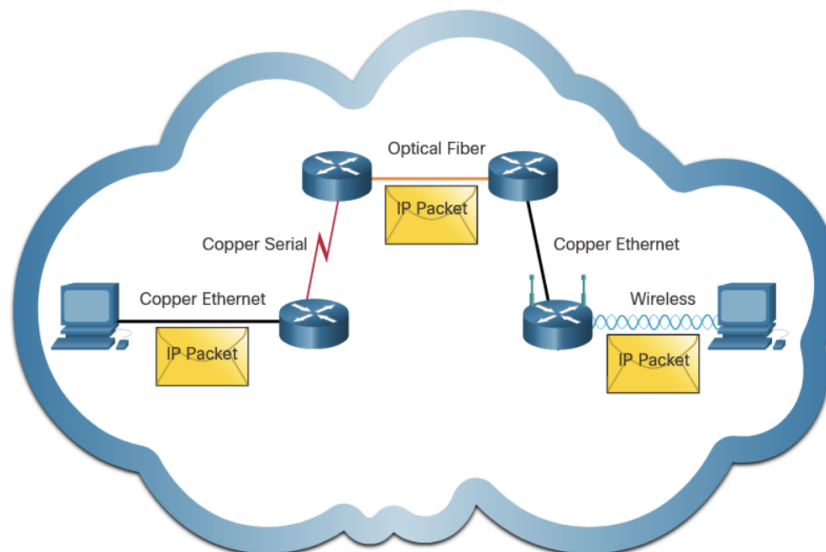


-
- **Best Effort**

- IP is Best Effort
 - IP will not guarantee delivery of the packet.
 - IP has reduced overhead since there is no mechanism to resend data that is not received.
 - IP does not expect acknowledgements.
 - IP does not know if the other device is operational or if it received the packet.



-
- **Media Independent**
 - IP is unreliable:
 - It cannot manage or fix undelivered or corrupt packets.
 - IP cannot retransmit after an error.
 - IP cannot realign out of sequence packets.
 - IP must rely on other protocols for these functions.
 - IP is media Independent:
 - IP does not concern itself with the type of frame required at the data link layer or the media type at the physical layer.
 - IP can be sent over any media type: copper, fiber, or wireless.



-
- The network layer will establish the Maximum Transmission Unit (MTU).
 - Network layer receives this from control information sent by the data link layer.
 - The network then establishes the MTU size.

- Fragmentation is when Layer 3 splits the IPv4 packet into smaller units.
 - Fragmenting causes latency.
 - IPv6 does not fragment packets.
 - Example: Router goes from Ethernet to a slow WAN with a smaller MTU

IPv4 Packet

● IPv4 Packet Header

- IPv4 is the primary communication protocol for the network layer.
- The network header has many purposes:
 - It ensures the packet is sent in the correct direction (to the destination).
 - It contains information for network layer processing in various fields.
 - The information in the header is used by all layer 3 devices that handle the packet

● IPv4 Packet Header Fields

- The IPv4 network header characteristics:
 - It is in binary.
 - Contains several fields of information
 - Diagram is read from left to right, 4 bytes per line
 - The two most important fields are the source and destination.
- Protocols may have one or more functions.



Significant fields in the IPv4 header:

Function	Description
Version	This will be for v4, as opposed to v6, a 4 bit field= 0100
Differentiated Services	Used for QoS: DiffServ – DS field or the older IntServ – ToS or Type of Service
Header Checksum	Detect corruption in the IPv4 header
Time to Live (TTL)	Layer 3 hop count. When it becomes zero the router will discard the packet.
Protocol	I.D.s next level protocol: ICMP, TCP, UDP, etc.
Source IPv4 Address	32 bit source address
Destination IPv4 Address	32 bit destination address

○

IPv6 Packets

- Limitations of IPv4

- IPv4 has three major limitations:
 - IPv4 address depletion – We have basically run out of IPv4 addressing.
 - Lack of end-to-end connectivity – To make IPv4 survive this long, private addressing and NAT were created. This ended direct communications with public addressing.
 - Increased network complexity – NAT was meant as a temporary solution and creates issues on the network as a side effect of manipulating the network headers addressing. NAT causes latency and troubleshooting issues.

- IPv6 Overview

- IPv6 was developed by Internet Engineering Task Force (IETF).
- IPv6 overcomes the limitations of IPv4.
- Improvements that IPv6 provides:
- Increased address space
 - based on 128 bit address, not 32 bits
- Improved packet handling
 - simplified header with fewer fields
- Eliminates the need for NAT
 - since there is a huge amount of addressing, there is no need to use private addressing internally and be mapped to a shared public address

IPv4 and IPv6 Address Space Comparison

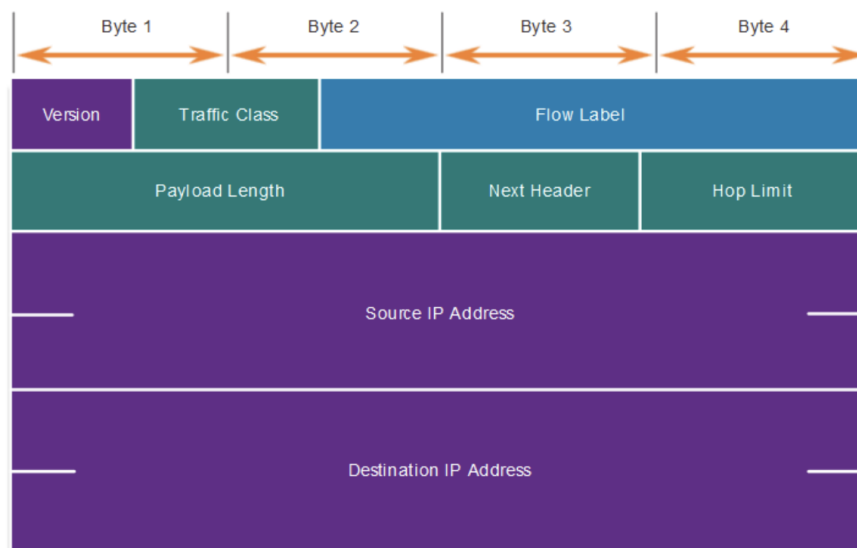
Number Name	Scientific Notation	Number of Zeros
1 Thousand	10^3	1,000
1 Million	10^6	1,000,000
1 Billion	10^9	1,000,000,000
1 Trillion	10^{12}	1,000,000,000,000
1 Quadrillion	10^{15}	1,000,000,000,000,000
1 Quintillion	10^{18}	1,000,000,000,000,000,000
1 Sextillion	10^{21}	1,000,000,000,000,000,000,000
1 Septillion	10^{24}	1,000,000,000,000,000,000,000,000
1 Octillion	10^{27}	1,000,000,000,000,000,000,000,000,000
1 Nonillion	10^{30}	1,000,000,000,000,000,000,000,000,000,000
1 Decillion	10^{33}	1,000,000,000,000,000,000,000,000,000,000,000
1 Undecillion	10^{36}	1,000,000,000,000,000,000,000,000,000,000,000,000

Legend

- There are 4 billion IPv4 addresses
- There are 340 undecillion IPv6 addresses

IPv4 Packet Header Fields in the IPv6 Packet Header

- The IPv6 header is simplified, but not smaller.
- The header is fixed at 40 Bytes or octets long.
- Several IPv4 fields were removed to improve performance.
- Some IPv4 fields were removed to improve performance:
 - Flag
 - Fragment Offset
 - Header Checksum



- IPv6 Packet Header

Significant fields in the IPv6 header:

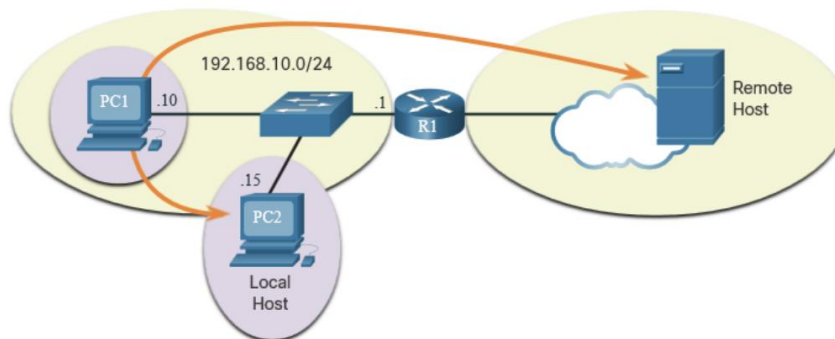
Function	Description
Version	This will be for v6, as opposed to v4, a 4 bit field= 0110
Traffic Class	Used for QoS: Equivalent to DiffServ – DS field
Flow Label	Informs device to handle identical flow labels the same way, 20 bit field
Payload Length	This 16-bit field indicates the length of the data portion or payload of the IPv6 packet
Next Header	I.D.s next level protocol: ICMP, TCP, UDP, etc.
Hop Limit	Replaces TTL field Layer 3 hop count
Source IPv4 Address	128 bit source address
Destination IPv4 Address	128 bit destination address

-
- IPv6 packet may also contain extension headers (EH).
- EH headers characteristics:
 - provide optional network layer information
 - are optional
 - are placed between IPv6 header and the payload
 - may be used for fragmentation, security, mobility support, etc.
- Note: Unlike IPv4, routers do not fragment IPv6 packets.

How a Host Routes

- Host Forwarding Decision

- Packets are always created at the source.
- Each host devices creates their own routing table.
- A host can send packets to the following:
 - Itself – 127.0.0.1 (IPv4), ::1 (IPv6)
 - Local Hosts – destination is on the same LAN
 - Remote Hosts – devices are not on the same LAN



-
- The Source device determines whether the destination is local or remote
- Method of determination:
 - IPv4 – Source uses its own IP address and Subnet mask, along with the destination IP address
 - IPv6 – Source uses the network address and prefix advertised by the local router

- Local traffic is dumped out the host interface to be handled by an intermediary device.
- Remote traffic is forwarded directly to the default gateway on the LAN.
- **Default Gateway**
 - A router or layer 3 switch can be a default-gateway.
 - Features of a default gateway (DGW):
 - It must have an IP address in the same range as the rest of the LAN.
 - It can accept data from the LAN and is capable of forwarding traffic off of the LAN.
 - It can route to other networks.
 - If a device has no default gateway or a bad default gateway, its traffic will not be able to leave the LAN.
- **A Host Routes to the Default Gateway**
 - The host will know the default gateway (DGW) either statically or through DHCP in IPv4.
 - IPv6 sends the DGW through a router solicitation (RS) or can be configured manually.
 - A DGW is static route which will be a last resort route in the routing table. • All device on the LAN will need the DGW of the router if they intend to send traffic remotely.
- **Host Routing Tables**
 - On Windows, route print or netstat -r to display the PC routing table
 - Three sections displayed by these two commands:
 - Interface List – all potential interfaces and MAC addressing
 - IPv4 Routing Table
 - IPv6 Routing Table



IPv4 Routing Table for PC1

```

C:\Users\PC1> netstat -r

IPv4 Route Table
=====
Active Routes:

```

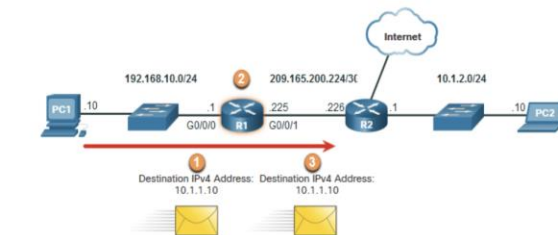
Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	192.168.10.1	192.168.10.10	25
127.0.0.0	255.0.0.0	On-link	127.0.0.1	306
127.0.0.1	255.255.255.255	On-link	127.0.0.1	306
127.255.255.255	255.255.255.255	On-link	127.0.0.1	306
192.168.10.0	255.255.255.0	On-link	192.168.10.10	281
192.168.10.10	255.255.255.255	On-link	192.168.10.10	281
192.168.10.255	255.255.255.255	On-link	192.168.10.10	281
224.0.0.0	240.0.0.0	On-link	127.0.0.1	306
224.0.0.0	240.0.0.0	On-link	192.168.10.10	281
255.255.255.255	255.255.255.255	On-link	127.0.0.1	306
255.255.255.255	255.255.255.255	On-link	192.168.10.10	281

○

Introduction to Routing

Router Packet Forwarding Decision

What happens when the router receives the frame from the host device?



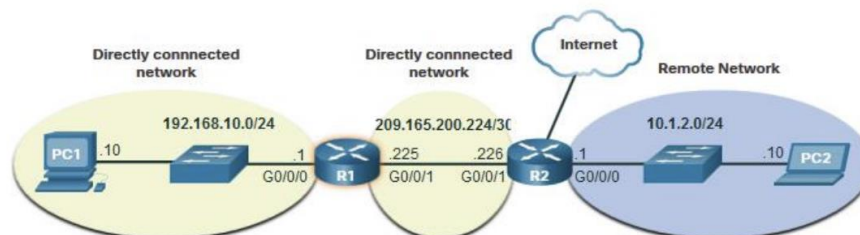
1. Packet arrives on the Gigabit Ethernet 0/0/0 interface of router R1. R1 de-encapsulates the Layer 2 Ethernet header and trailer.
2. Router R1 examines the destination IPv4 address of the packet and searches for the best match in its IPv4 routing table. The route entry indicates that this packet is to be forwarded to router R2.
3. Router R1 encapsulates the packet into a new Ethernet header and trailer, and forwards the packet to the next hop router R2.

R1 Routing Table

Route	Next Hop or Exit Interface
192.168.10.0 /24	G0/0/0
209.165.200.224/30	G0/0/1
10.1.1.0/24	via R2
Default Route 0.0.0.0/0	via R2

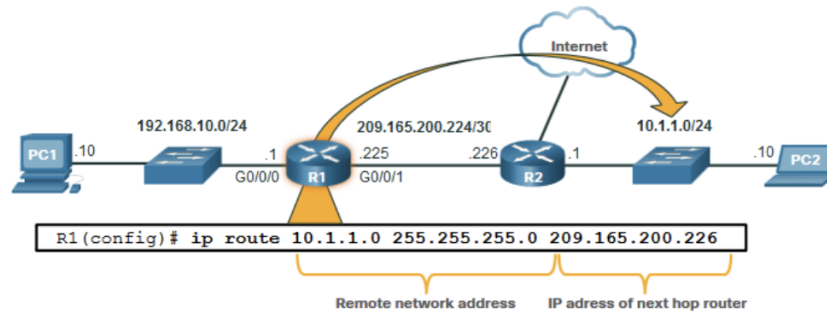
IP Router Routing Table

- There three types of routes in a router's routing table:
 - Directly Connected
 - These routes are automatically added by the router, provided the interface is active and has addressing.
 - Remote
 - These are the routes the router does not have a direct connection and may be learned:
 - Manually – with a static route
 - Dynamically – by using a routing protocol to have the routers share their information with each other
 - Default Route
 - This forwards all traffic to a specific direction when there is not a match in the routing table

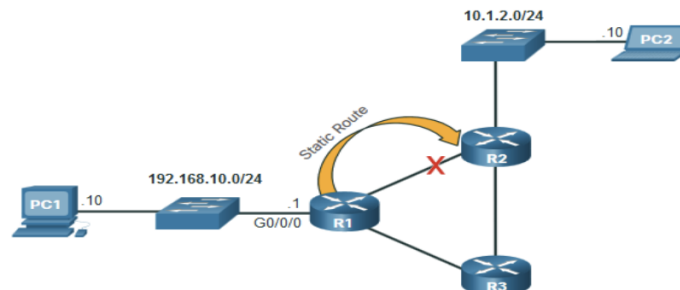


Static Routing

- Static Route Characteristics:
 - Must be configured manually
 - Must be adjusted manually by the administrator when there is a change in the topology
 - Good for small non-redundant networks
 - Often used in conjunction with a dynamic routing protocol for configuring a default route



R1 is manually configured with a static route to reach the 10.1.1.0/24 network. If this path changes, R1 will require a new static route.



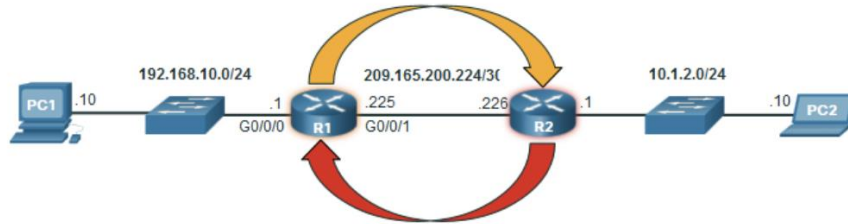
If the route from R1 via R2 is no longer available, a new static route via R3 would need to be configured. A static route does not automatically adjust for topology changes.

© Cisco and/or its affiliates. All rights reserved. Cisco Confidential

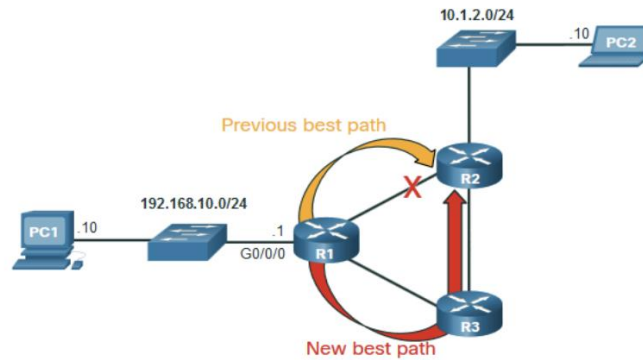
65

• Dynamic Routing

- Dynamic Routes Automatically:
 - Discover remote networks
 - Maintain up-to-date information
 - Choose the best path to the destination
 - Find new best paths when there is a topology change
- Dynamic routing can also share static default routes with the other routers.



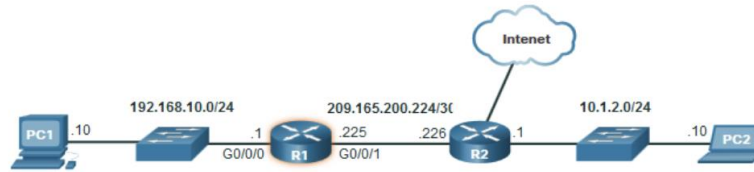
- R1 is using the routing protocol OSPF to let R2 know about the 192.168.10.0/24 network.
- R2 is using the routing protocol OSPF to let R1 know about the 10.1.1.0/24 network.



R1, R2, and R3 are using the dynamic routing protocol OSPF. If there is a network topology change, they can automatically adjust to find a new best path.

● Introduction to an IPv4 Routing Table

- The show ip route command shows the following route sources:
 - L - Directly connected local interface IP address
 - C – Directly connected network
 - S – Static route was manually configured by an administrator
 - O – OSPF
 - D – EIGRP
- This command shows types of routes:
 - Directly Connected – C and L
 - Remote Routes – O, D, etc.
 - Default Routes – S*



```

R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 209.165.200.226 to network 0.0.0.0
S*   0.0.0.0/0 [1/0] via 209.165.200.226, GigabitEthernet0/0/1
     10.0.0.0/24 is subnetted, 1 subnets
O     10.1.1.0 [110/2] via 209.165.200.226, 00:02:45, GigabitEthernet0/0/1
C     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L     192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
L     209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C     209.165.200.224/30 is directly connected, GigabitEthernet0/0/1
L     209.165.200.225/32 is directly connected, GigabitEthernet0/0/1
R1#

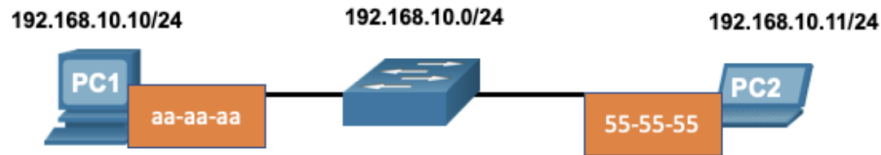
```

○

Address Resolution & Basic Router Configuration

MAC and IP

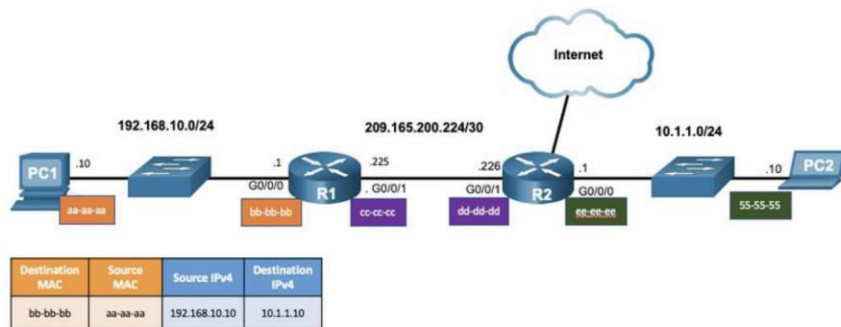
- Destination on Same Network
 - There are two primary addresses assigned to a device on an Ethernet LAN:
 - Layer 2 physical address (the MAC address) – Used for NIC to NIC communications on the same Ethernet network.
 - Layer 3 logical address (the IP address) – Used to send the packet from the source device to the destination device.
 - Layer 2 addresses are used to deliver frames from one NIC to another NIC on the same network. If a destination IP address is on the same network, the destination MAC address will be that of the destination device.



Destination MAC	Source MAC	Source IPv4	Destination IPv4
55-55-55	aa-aa-aa	192.168.10.10	192.168.10.11

- Destination on Remote Network

- When the destination IP address is on a remote network, the destination MAC address is that of the default gateway.
 - ARP is used by IPv4 to associate the IPv4 address of a device with the MAC address of the device NIC.
 - ICMPv6 is used by IPv6 to associate the IPv6 address of a device with the MAC address of the device NIC.

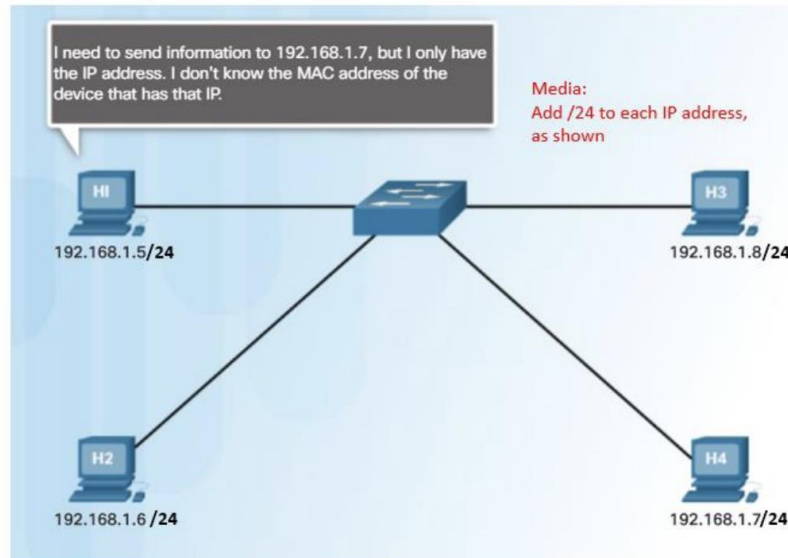


Destination MAC	Source MAC	Source IPv4	Destination IPv4
bb-bb-bb	aa-aa-aa	192.168.10.10	10.1.1.10

ARP

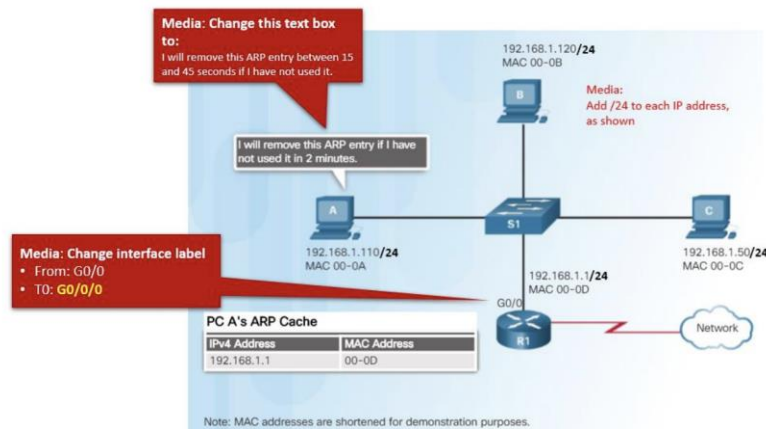
- ARP Overview

- A device uses ARP to determine the destination MAC address of a local device when it knows its IPv4 address. ARP provides two basic functions:
 - Resolving IPv4 addresses to MAC addresses
 - Maintaining an ARP table of IPv4 to MAC address mappings



-
- **ARP Functions**
 - To send a frame, a device will search its ARP table for a destination IPv4 address and a corresponding MAC address.
 - If the packet's destination IPv4 address is on the same network, the device will search the ARP table for the destination IPv4 address.
 - If the destination IPv4 address is on a different network, the device will search the ARP table for the IPv4 address of the default gateway.
 - If the device locates the IPv4 address, its corresponding MAC address is used as the destination MAC address in the frame.
 - If no ARP table entry is found, then the device sends an ARP request.
- **ARP Requests**
 - The ARP request is encapsulated in an Ethernet frame using the following header information:
 - Destination MAC address – This is a broadcast address FF-FF-FF-FF-FF-FF.
 - Source MAC address – This is MAC address of the sender of the ARP request.
 - Type - ARP messages have a type field of 0x806. This informs the receiving NIC that the data portion of the frame needs to be passed to the ARP process.
 - Because ARP requests are broadcasts, they are flooded out all ports by the switch, except the receiving port. All Ethernet NICs on the LAN process broadcasts and must deliver the ARP request to its operating system for processing. Every device must process the ARP request to see if the target IPv4 address matches its own. A router will not forward broadcasts out other interfaces.
- **ARP Operations**
 - Only the device with the target IPv4 address associated with the ARP request will respond with an ARP reply. The ARP reply is encapsulated in an Ethernet frame using the following header information:

- Destination MAC address – This is the MAC address of the sender of the ARP request.
 - Source MAC address – This is the MAC address of the sender of the ARP reply.
 - Type - ARP messages have a type field of 0x806.
- Only the device that originally sent the ARP request will receive the unicast ARP reply. After the ARP reply is received, the device will add the IPv4 address and the corresponding MAC address to its ARP table. Packets destined for that IPv4 address can now be encapsulated in frames using its corresponding MAC address. If no device responds to the ARP request, the packet is dropped because a frame cannot be created.
- **ARP Role**
 - The IPv4 address of the default gateway is stored in the IPv4 configuration of the hosts. When a host creates a packet for a destination, it compares the destination IPv4 address and its own IPv4 address to determine if the two IPv4 addresses are located on the same Layer 3 network. If the destination host is not on its same network, the source checks its ARP table for an entry with the IPv4 address of the default gateway. If there is not an entry, it uses the ARP process to determine a MAC address of the default gateway.
- **Removing Entries from an ARP Table**
 - Entries in the ARP table are not permanent and are removed when an ARP cache timer expires after a specified period of time.
 - The duration of the ARP cache timer differs depending on the operating system.
 - ARP table entries can also be removed manually by the administrator.



- ARP Tables on Networking Devices

- The **show ip arp** command displays the ARP table on a Cisco router.
- The **arp -a** command displays the ARP table on a Windows 10 PC.

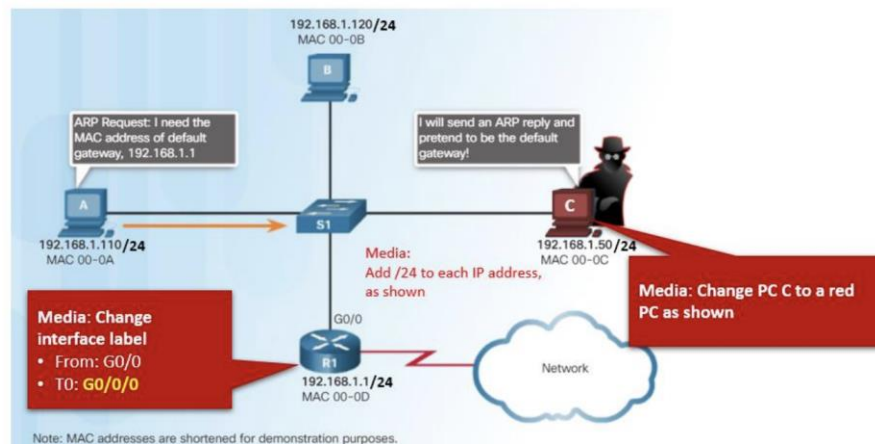
```
R1# show ip arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 192.168.10.1      -          a0e0.af0d.e140 ARPA   GigabitEthernet0/0/0
```

```
C:\Users\PC> arp -a

Interface: 192.168.1.124 --- 0x10
Internet Address      Physical Address    Type
192.168.1.1           c8-d7-19-cc-a0-86   dynamic
192.168.1.101         08-3e-0c-f5-f7-77   dynamic
```

- ARP Issues – ARP Broadcasting and ARP Spoofing

- ARP requests are received and processed by every device on the local network.
- Excessive ARP broadcasts can cause some reduction in performance.
- ARP replies can be spoofed by a threat actor to perform an ARP poisoning attack.
- Enterprise level switches include mitigation techniques to protect against ARP attacks.

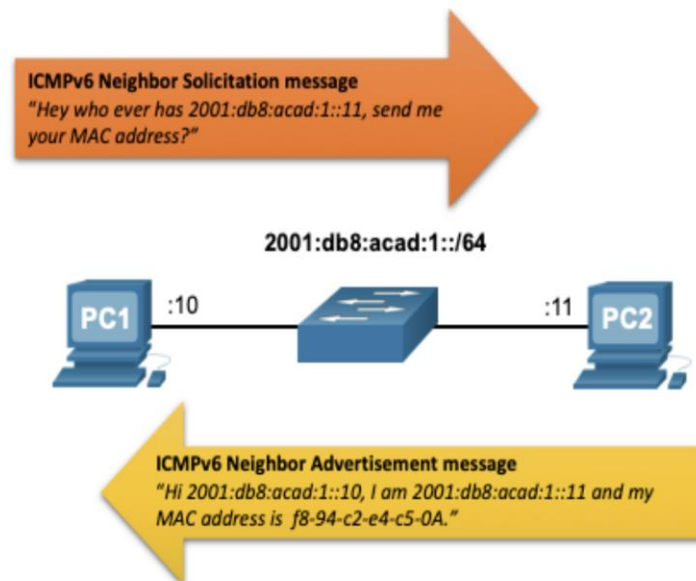


IPv6 Neighbor Discovery

- IPv6 Neighbor Discovery Messages

- IPv6 Neighbor Discovery (ND) protocol provides:
 - Address resolution
 - Router discovery
 - Redirection services
 - ICMPv6 Neighbor Solicitation (NS) and Neighbor Advertisement (NA) messages are used for device-to-device messaging such as address resolution.
 - ICMPv6 Router Solicitation (RS) and Router Advertisement (RA) messages are used for messaging between devices and routers for router discovery.

- ICMPv6 redirect messages are used by routers for better next-hop selection.
- IPv6 Neighbor Discovery – Address Resolution
 - IPv6 devices use ND to resolve the MAC address of a known IPv6 address.
 - ICMPv6 Neighbor Solicitation messages are sent using special Ethernet and IPv6 multicast addresses.



○

Basic Router Configuration

Configure Initial Router Settings

- Basic Router Configuration Steps

- Configure the device name.
- Secure privileged EXEC mode.
- Secure user EXEC mode.
- Secure remote Telnet / SSH access.
- Encrypt all plaintext passwords.
- Provide legal notification and save the configuration.

```
Router(config)# hostname hostname
```

```
Router(config)# enable secret password
```

```
Router(config)# line console 0
Router(config-line)# password password
Router(config-line)# login
```

```
Router(config)# line vty 0 4
Router(config-line)# password password
Router(config-line)# login
Router(config-line)# transport input {ssh | telnet}
```

```
Router(config)# service password encryption
```

```
Router(config)# banner motd # message #
Router(config)# end
Router# copy running-config startup-config
```

○

- Basic Router Configuration Example

- Commands for basic router configuration on R1.
- Configuration is saved to NVRAM.

```
R1(config)# hostname R1
R1(config)# enable secret class
R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# transport input ssh telnet
R1(config-line)# exit
R1(config)# service password encryption
R1(config)# banner motd #
Enter TEXT message. End with a new line and the #
*****
WARNING: Unauthorized access is prohibited!
*****
R1(config)# exit
R1# copy running-config startup-config
```

Configure Interfaces

- Configure Router Interfaces

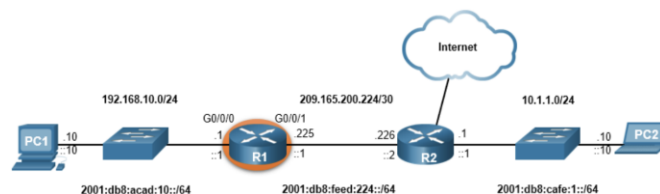
- Configuring a router interface includes issuing the following commands:

```
Router(config)# interface type-and-number
Router(config-if)# description description-text
Router(config-if)# ip address ipv4-address subnet-mask
Router(config-if)# ipv6 address ipv6-address/prefix-length
Router(config-if)# no shutdown
```

- It is a good practice to use the description command to add information about the network connected to the interface.
 - The no shutdown command activates the interface.
 - To enable IPv6 routing, enter the command ipv6 unicast-routing:
Router(config)# ipv6 unicast-routing

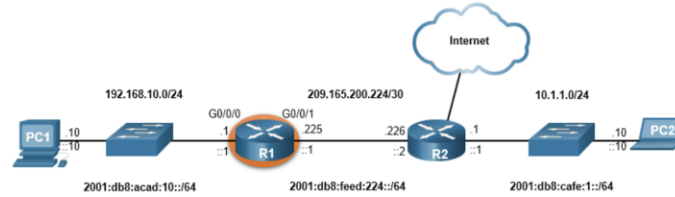
- Configure Router Interfaces Example

- The commands to configure interface G0/0/0 on R1 are shown here:



```
R1(config)# interface gigabitEthernet 0/0/0
R1(config-if)# description Link to LAN
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:10::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
*Aug 1 01:43:53.435: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to down
*Aug 1 01:43:56.447: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to up
*Aug 1 01:43:57.447: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0,
changed state to up
```

- The commands to configure interface G0/0/1 on R1 are shown here:



```
R1(config)# interface gigabitEthernet 0/0/1
R1(config-if)# description Link to R2
R1(config-if)# ip address 209.165.200.225 255.255.255.252
R1(config-if)# ipv6 address 2001:db8:feed:224::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
*Aug 1 01:46:29.170: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to down
*Aug 1 01:46:32.171: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to up
*Aug 1 01:46:33.171: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1,
changed state to up
```

• Verify Interface Configuration

- To verify interface configuration use the show ip interface brief and show ipv6 interface brief commands shown here:

```
R1# show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.10.1 YES manual up up
GigabitEthernet0/0/1 209.165.200.225 YES manual up up
Vlan1 unassigned YES unset administratively down down
```

```
R1# show ipv6 interface brief
GigabitEthernet0/0/0 [up/up]
FE80::201:C9FF:FE89:4501
2001:DB8:ACAD:10::1
GigabitEthernet0/0/1 [up/up]
FE80::201:C9FF:FE89:4502
2001:DB8:FEED:224::1
Vlan1 [administratively down/down]
unassigned
R1#
```

• Configure Verification Commands

- The table summarizes show commands used to verify interface configuration.

Commands	Description
show ip interface brief show ipv6 interface brief	Displays all interfaces, their IP addresses, and their current status.
show ip route show ipv6 route	Displays the contents of the IP routing tables stored in RAM.
show interfaces	Displays statistics for all interfaces on the device. Only displays the IPv4 addressing information.
show ip interfaces	Displays the IPv4 statistics for all interfaces on a router.
show ipv6 interfaces	Displays the IPv6 statistics for all interfaces on a router.

- View status of all interfaces with the show ip interface brief and show ipv6 interface brief commands, shown here:

```
R1# show ip interface brief
Interface          IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0 192.168.10.1    YES manual up            up
GigabitEthernet0/0/1 209.165.200.225 YES manual up            up
Vlan1              unassigned      YES unset  administratively down down
R1#
```

```
R1# show ipv6 interface brief
GigabitEthernet0/0/0 [up/up]
FE80::201:C9FF:FE89:4501
2001:DB8:ACAD:10::1
GigabitEthernet0/0/1 [up/up]
FE80::201:C9FF:FE89:4502
2001:DB8:FEED:224::1
Vlan1 [administratively down/down]
unassigned
R1#
```

-
- Display the contents of the IP routing tables with the show ip route and show ipv6 route commands as shown here:

```
R1# show ip route
< output omitted>
Gateway of last resort is not set
C    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
C    209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
L    209.165.200.225/32 is directly connected, GigabitEthernet0/0/1
R1#
```

```
R1# show ipv6 route
<output omitted>
C    2001:DB8:ACAD:10::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L    2001:DB8:ACAD:10::1/128 [0/0]
    via GigabitEthernet0/0/0, receive
C    2001:DB8:FEED:224::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L    2001:DB8:FEED:224::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
L    FF00::/8 [0/0]
    via Null0, receive
R1#
```

-
- Display statistics for all interfaces with the show interfaces command, as shown here:

```

R1# show interfaces gig0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Hardware is ISR4321-2x1GE, address is a0e0.af0d.e140 (bia a0e0.af0d.e140)
  Description: Link to LAN
  Internet address is 192.168.10.1/24
  MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive not supported
  Full Duplex, 100Mbps, link type is auto, media type is RJ45
  output flow-control is off, input flow-control is off
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:01, output 00:00:35, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    1180 packets input, 109486 bytes, 0 no buffer
      Received 84 broadcasts (0 IP multicasts)
      0 runts, 0 giants, 0 throttles

<output omitted>

R1#

```

-
- Display IPv4 statistics for router interfaces with the show ip interface command, as shown here

```

R1# show ip interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet address is 192.168.10.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing Common access list is not set
  Outgoing access list is not set
  Inbound Common access list is not set
  Inbound access list is not set
  Proxy ARP is enabled
  Local Proxy ARP is disabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachable are always sent
  ICMP mask replies are never sent
  IP fast switching is enabled
  IP Flow switching is disabled

<output omitted>

R1#

```

-
- Display IPv6 statistics for router interfaces with the show ipv6 interface command shown here:

```

R1# show ipv6 interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is
  FE80::868A:8DFF:FE44:49B0
  No Virtual link-local address(es):
  Description: Link to LAN
  Global unicast address(es):
    2001:DB8:ACAD:10::1, subnet is 2001:DB8:ACAD:10::/64
  Joined group address(es):
    FF02::1
    FF02::1:FF00:1
    FF02::1:FF44:49B0
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND NS retransmit interval is 1000 milliseconds

R1#

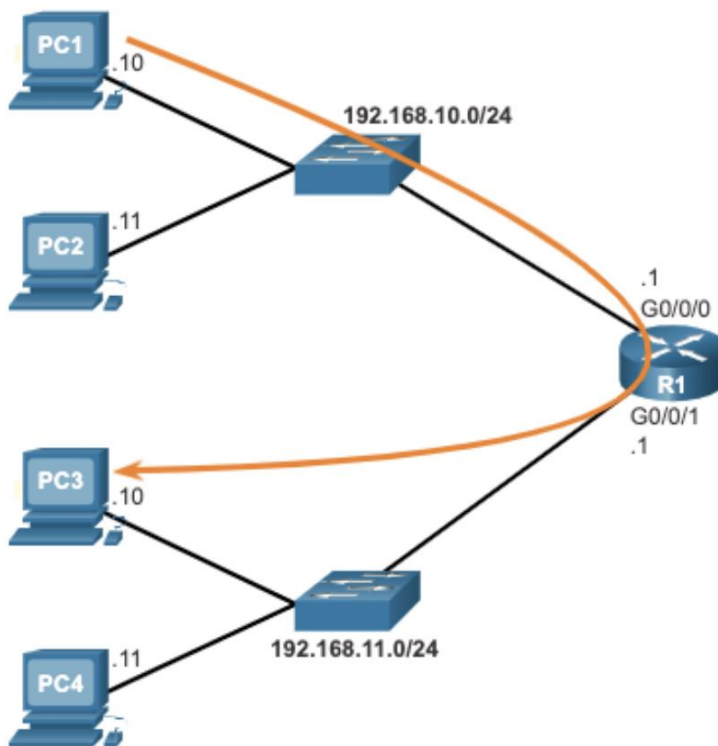
```

○

Configure the Default Gateway

- Default Gateway on a Host

- The default gateway is used when a host sends a packet to a device on another network.
- The default gateway address is generally the router interface address attached to the local network of the host.
- To reach PC3, PC1 addresses a packet with the IPv4 address of PC3, but forwards the packet to its default gateway, the G0/0/0 interface of R1.

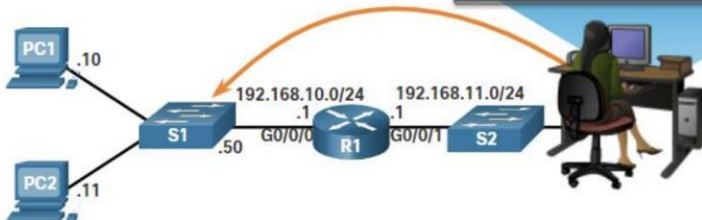


-
- Note: The IP address of the host and the router interface must be in the same network.

- **Default Gateway on a Switch**

- A switch must have a default gateway address configured to remotely manage the switch from another network.
- To configure an IPv4 default gateway on a switch, use the ip defaultgateway ip-address global configuration command.

```
S1# show running-config
Building configuration...
!
<Output Omitted>
service password-encryption
!
hostname S1
!
interface Vlan1
  ip address 192.168.10.50.255.255.0
!
<Output Omitted>
!
ip default-gateway 192.168.10.1
<Output Omitted>
```



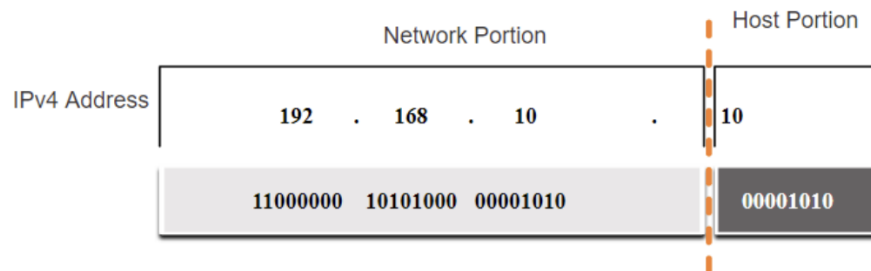
-

IPv4 Addressing

IPv4 Address Structure

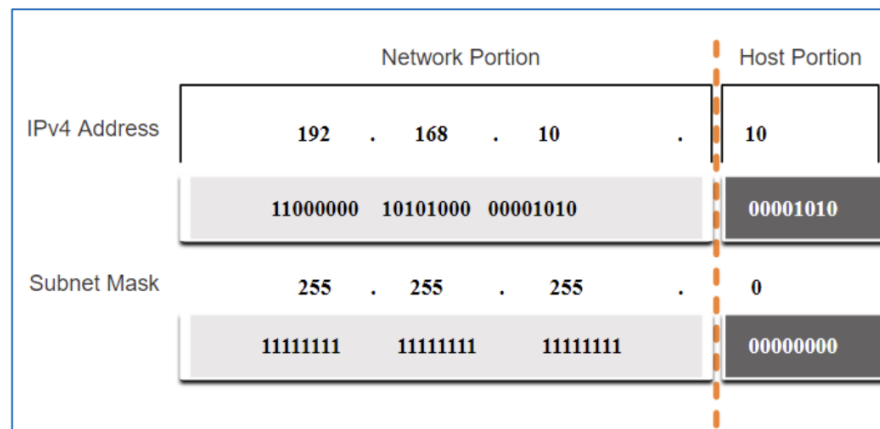
- **Network and Host Portions**

- An IPv4 address is a 32-bit hierarchical address that is made up of a network portion and a host portion.
- When determining the network portion versus the host portion, you must look at the 32-bit stream.
- A subnet mask is used to determine the network and host portions.



-
- **The Subnet Mask**

- To identify the network and host portions of an IPv4 address, the subnet mask is compared to the IPv4 address bit for bit, from left to right.
- The actual process used to identify the network and host portions is called **ANDing**.



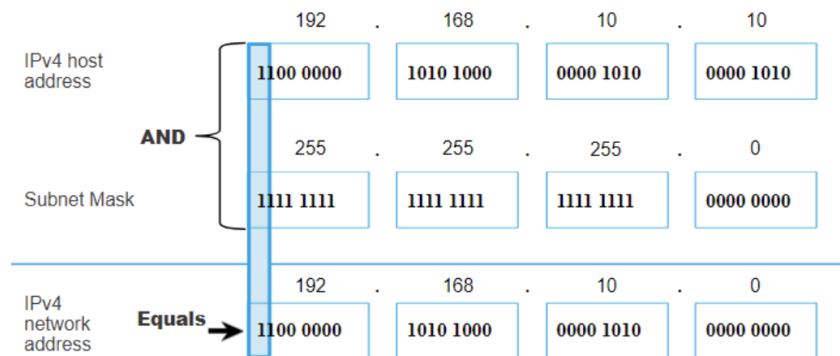
-
- **The Prefix Length**

- A prefix length is a less cumbersome method used to identify a subnet mask address.
- The prefix length is the number of bits set to 1 in the subnet mask.
It is written in “slash notation” therefore, count the number of bits in the subnet mask and prepend it with a slash.

Subnet Mask	32-bit Address	Prefix Length
255.0.0.0	11111111.00000000.00000000.00000000	/8
255.255.0.0	11111111.11111111.00000000.00000000	/16
255.255.255.0	11111111.11111111.11111111.00000000	/24
255.255.255.128	11111111.11111111.11111111.10000000	/25
255.255.255.192	11111111.11111111.11111111.11000000	/26
255.255.255.224	11111111.11111111.11111111.11100000	/27
255.255.255.240	11111111.11111111.11111111.11110000	/28
255.255.255.248	11111111.11111111.11111111.11111000	/29
255.255.255.252	11111111.11111111.11111111.11111100	/30

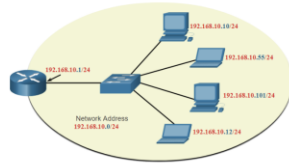
- **Determining the Network: Logical AND**

- A logical AND Boolean operation is used in determining the network address.
- Logical AND is the comparison of two bits where only a 1 AND 1 produces a 1 and any other combination results in a 0.
- $1 \text{ AND } 1 = 1$, $0 \text{ AND } 1 = 0$, $1 \text{ AND } 0 = 0$, $0 \text{ AND } 0 = 0$
- 1 = True and 0 = False
- To identify the network address, the host IPv4 address is logically ANDed, bit by bit, with the subnet mask to identify the network address.



- **Network, Host, and Broadcast Addresses**

- Within each network are three types of IP addresses:
- Network address
- Host addresses
- Broadcast address



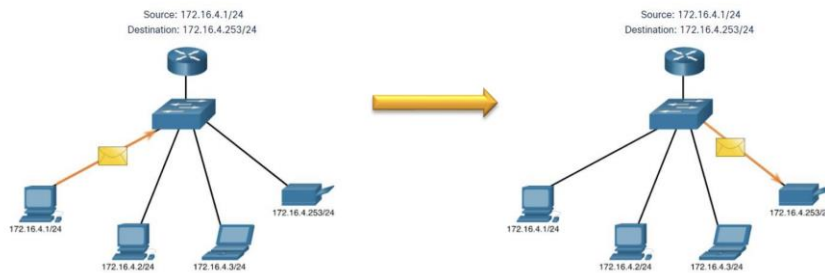
	Network Portion			Host Portion	Host Bits
Subnet mask 255.255.255.0 or /24	255	255	255	0	00000000
Network address 192.168.10.0 or /24	192	168	10	0	All 0s
First address 192.168.10.1 or /24	192	168	10	1	All 0s and a 1
Last address 192.168.10.254 or /24	192	168	10	254	All 1s and a 0
Broadcast address 192.168.10.255 or /24	192	168	10	255	All 1s

○

IPv4 Unicast, Broadcast, and Multicast

• Unicast

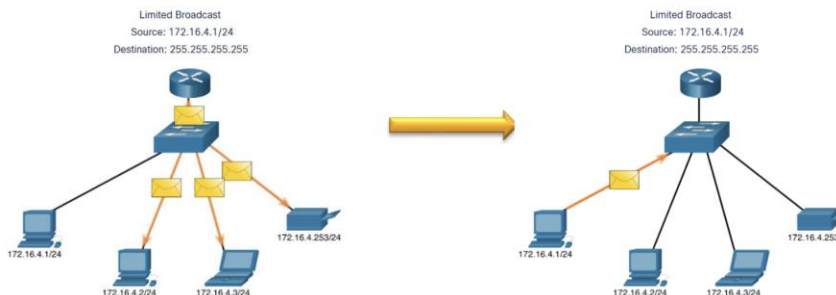
- Unicast transmission is sending a packet to one destination IP address.
- For example, the PC at 172.16.4.1 sends a unicast packet to the printer at 172.16.4.253



○

• Broadcast

- Broadcast transmission is sending a packet to all other destination IP addresses.
- For example, the PC at 172.16.4.1 sends a broadcast packet to all IPv4 hosts.



○

• Multicast

- Multicast transmission is sending a packet to a multicast address group.
For example, the PC at 172.16.4.1 sends a multicast packet to the multicast group address 224.10.10.5.



○

Types of IPv4 Addresses

- **Public and Private IPv4 Addresses**

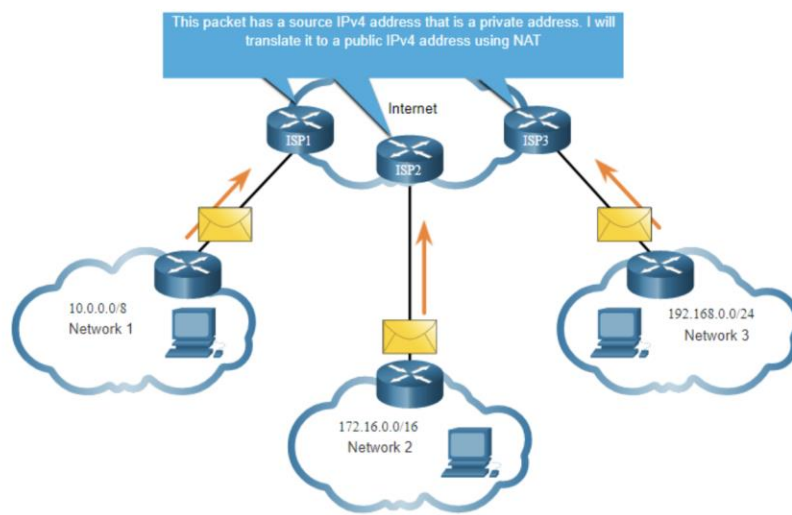
- As defined in RFC 1918, public IPv4 addresses are globally routed between internet service provider (ISP) routers.
- Private IPv4 addresses are not unique and can be used internally within any network.
- Private addresses are common blocks of addresses used by most organizations to assign IPv4 addresses to internal hosts.
- However, private addresses are not globally routable.

Network Address and Prefix	RFC 1918 Private Address Range
10.0.0.0/8	10.0.0.0 - 10.255.255.255
172.16.0.0/12	172.16.0.0 - 172.31.255.255
192.168.0.0/16	192.168.0.0 - 192.168.255.255

○

- **Routing to the Internet**

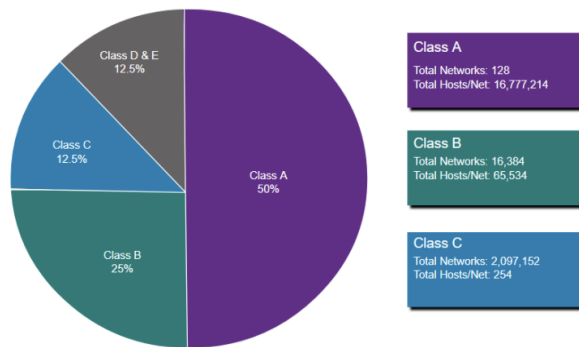
- Network Address Translation (NAT) translates private IPv4 addresses to public IPv4 addresses.
- NAT is typically enabled on the edge router connecting to the internet.
- It translates the internal private address to a public global IP address.



-
- Special Use IPv4 Addresses
 - Loopback addresses
 - 127.0.0.0 /8 (127.0.0.1 to 127.255.255.254)
 - Commonly identified as only 127.0.0.1
 - Used on a host to test if TCP/IP is operational.
 - Link-Local addresses
 - 169.254.0.0 /16 (169.254.0.1 to 169.254.255.254)
 - Commonly known as the Automatic Private IP Addressing (APIPA) addresses or self assigned addresses.
 - Used by Windows DHCP clients to self-configure when no DHCP servers are available.

```
C:\Users\NetAcad> ping 127.0.0.1
Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

-
- Legacy Classful Addressing
 - RFC 790 (1981) allocated IPv4 addresses in classes
 - Class A (0.0.0.0/8 to 127.0.0.0/8)
 - Class B (128.0.0.0 /16 – 191.255.0.0 /16)
 - Class C (192.0.0.0 /24 – 223.255.255.0 /24)
 - Class D (224.0.0.0 to 239.0.0.0)
 - Class E (240.0.0.0 – 255.0.0.0)
 - Classful addressing wasted many IPv4 addresses.
 - Classful address allocation was replaced with classless addressing which ignores the rules of classes (A, B, C).



• Assignment of IP Addresses

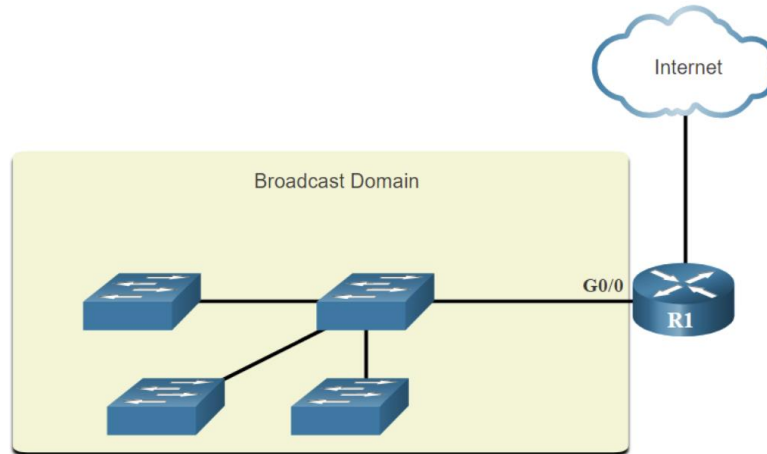
- The Internet Assigned Numbers Authority (IANA) manages and allocates blocks of IPv4 and IPv6 addresses to five Regional Internet Registries (RIRs).
- RIRs are responsible for allocating IP addresses to ISPs who provide IPv4 address blocks to smaller ISPs and organizations.



Network Segmentation

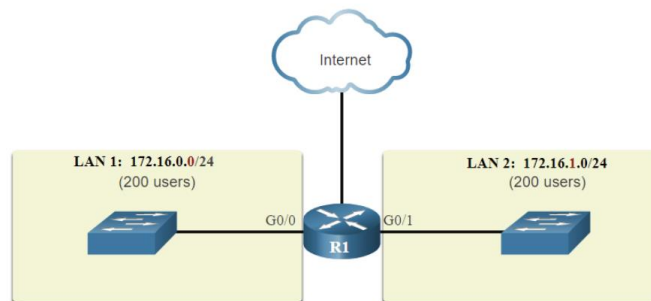
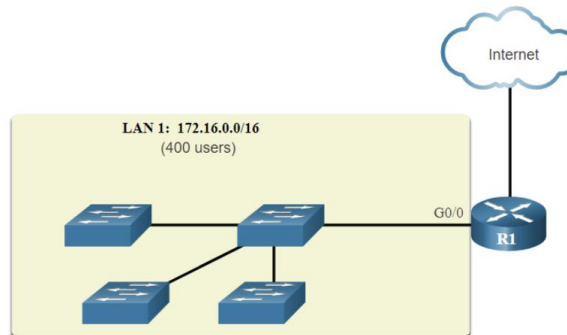
• Broadcast Domains and Segmentation

- Many protocols use broadcasts or multicasts (e.g., ARP use broadcasts to locate other devices, hosts send DHCP discover broadcasts to locate a DHCP server.)
- Switches propagate broadcasts out all interfaces except the interface on which it was received.
- The only device that stops broadcasts is a router.
- Routers do not propagate broadcasts.
- Each router interface connects to a broadcast domain and broadcasts are only propagated within that specific broadcast domain.



- Problems with Large Broadcast Domains

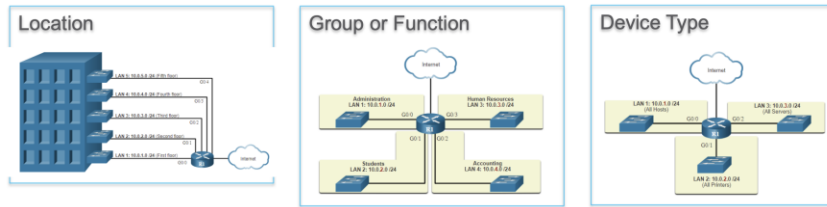
- A problem with a large broadcast domain is that these hosts can generate excessive broadcasts and negatively affect the network.
- The solution is to reduce the size of the network to create smaller broadcast domains in a process called subnetting.
- Dividing the network address 172.16.0.0 /16 into two subnets of 200 users each: 172.16.0.0 /24 and 172.16.1.0 /24.
- Broadcasts are only propagated within the smaller broadcast domains.



- Reasons for Segmenting Networks

- Subnetting reduces overall network traffic and improves network performance.
- It can be used to implement security policies between subnets.
- Subnetting reduces the number of devices affected by abnormal broadcast traffic.

- Subnets are used for a variety of reasons including by:



Subnet an IPv4 Network

Subnet on an Octet Boundary

- Networks are most easily subnetted at the octet boundary of /8, /16, and /24.
- Notice that using longer prefix lengths decreases the number of hosts per subnet.

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of hosts
/8	255.0.0.0	nnnnnnnn . hhhhhhhh . hhhhhhhh . hhhhhhhh 11111111 . 00000000 . 00000000 . 00000000	16,777,214
/16	255.255.0.0	nnnnnnnn . nnnnnnnn . hhhhhhhh . hhhhhhhh 11111111 . 11111111 . 00000000 . 00000000	65,534
/24	255.255.255.0	nnnnnnnn . nnnnnnnn . nnnnnnnn . hhhhhhhh 11111111 . 11111111 . 11111111 . 00000000	254

- In the first table 10.0.0.0/8 is subnetted using /16 and in the second table, a /24 mask.

Subnet Address (256 Possible Subnets)	Host Range (65,534 possible hosts per subnet)	Broadcast
10.0.0.0/16	10.0.0.1 - 10.0.255.254	10.0.255.255
10.1.0.0/16	10.1.0.1 - 10.1.255.254	10.1.255.255
10.2.0.0/16	10.2.0.1 - 10.2.255.254	10.2.255.255
10.3.0.0/16	10.3.0.1 - 10.3.255.254	10.3.255.255
10.4.0.0/16	10.4.0.1 - 10.4.255.254	10.4.255.255
10.5.0.0/16	10.5.0.1 - 10.5.255.254	10.5.255.255
10.6.0.0/16	10.6.0.1 - 10.6.255.254	10.6.255.255
10.7.0.0/16	10.7.0.1 - 10.7.255.254	10.7.255.255
...	...	...
10.255.0.0/16	10.255.0.1 - 10.255.255.254	10.255.255.255

Subnet Address (65,536 Possible Subnets)	Host Range (254 possible hosts per subnet)	Broadcast
10.0.0.0/24	10.0.0.1 - 10.0.0.254	10.0.0.255
10.0.1.0/24	10.0.1.1 - 10.0.1.254	10.0.1.255
10.0.2.0/24	10.0.2.1 - 10.0.2.254	10.0.2.255
...	...	...
10.0.255.0/24	10.0.255.1 - 10.0.255.254	10.0.255.255
10.1.0.0/24	10.1.0.1 - 10.1.0.254	10.1.0.255
10.1.1.0/24	10.1.1.1 - 10.1.1.254	10.1.1.255
10.1.2.0/24	10.1.2.1 - 10.1.2.254	10.1.2.255
...	...	...
10.100.0.0/24	10.100.0.1 - 10.100.0.254	10.100.0.255
...	...	...
10.255.255.0/24	10.255.255.1 - 10.255.255.254	10.255.255.255

Subnet within an Octet Boundary

- Refer to the table to see six ways to subnet a /24 network.

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnnnnnnh 11111111 . 11111111 . 11111111 . 10000000	2	126
/26	255.255.255.192	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnnnnnhh 11111111 . 11111111 . 11111111 . 11000000	4	62
/27	255.255.255.224	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnnnnhhh 11111111 . 11111111 . 11111111 . 11100000	8	30
/28	255.255.255.240	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnnnnhhh 11111111 . 11111111 . 11111111 . 11110000	16	14
/29	255.255.255.248	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnnnnhhh 11111111 . 11111111 . 11111111 . 11111000	32	6
/30	255.255.255.252	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnnnnhhh 11111111 . 11111111 . 11111111 . 11111100	64	2

Subnet a Slash 16 and a Slash 8 Prefix

- Create Subnets with a Slash 16 prefix
 - The table highlights all the possible scenarios for subnetting a /16 prefix.

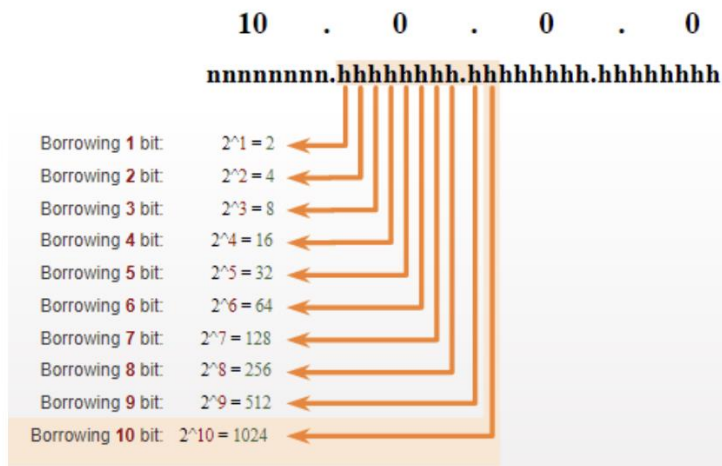
Prefix Length	Subnet Mask	Network Address (n = network, h = host)	# of subnets	# of hosts
/17	255.255.128.0	nnnnnnnn.nnnnnnnn.nnnnnnnn.hhhhhhhh 11111111.11111111.10000000.00000000	2	32766
/18	255.255.192.0	nnnnnnnn.nnnnnnnn.nnnnnnnn.hhhhhhhh 11111111.11111111.11000000.00000000	4	16382
/19	255.255.224.0	nnnnnnnn.nnnnnnnn.nnnnnnnn.hhhhhhhh 11111111.11111111.11100000.00000000	8	8190
/20	255.255.240.0	nnnnnnnn.nnnnnnnn.nnnnnnnn.hhhhhhhh 11111111.11111111.11110000.00000000	16	4094
/21	255.255.248.0	nnnnnnnn.nnnnnnnn.nnnnnnnn.hhhhhhhh 11111111.11111111.11111000.00000000	32	2046
/22	255.255.252.0	nnnnnnnn.nnnnnnnn.nnnnnnnn.hhhhhhhh 11111111.11111111.11111100.00000000	64	1022
/23	255.255.254.0	nnnnnnnn.nnnnnnnn.nnnnnnnn.hhhhhhhh 11111111.11111111.11111110.00000000	128	510
/24	255.255.255.0	nnnnnnnn.nnnnnnnn.nnnnnnnn.hhhhhhhh 11111111.11111111.11111111.00000000	256	254
/25	255.255.255.128	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnnnn 11111111.11111111.11111111.10000000	512	126
/26	255.255.255.192	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnnnn 11111111.11111111.11111111.11000000	1024	62
/27	255.255.255.224	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnnnn 11111111.11111111.11111111.11100000	2048	30
/28	255.255.255.240	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnnnn 11111111.11111111.11111111.11110000	4096	14
/29	255.255.255.248	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnnnn 11111111.11111111.11111111.11111000	8192	6
/30	255.255.255.252	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnnnn 11111111.11111111.11111111.11111100	16384	2

- Create 100 Subnets with a Slash 16 prefix
 - Consider a large enterprise that requires at least 100 subnets and has chosen the private address 172.16.0.0/16 as its internal network address.
 - The figure displays the number of subnets that can be created when borrowing bits from the third octet and the fourth octet.
 - Notice there are now up to 14 host bits that can be borrowed (i.e., last two bits cannot be borrowed).
 - To satisfy the requirement of 100 subnets for the
 - enterprise, 7 bits (i.e., $2^7 = 128$ subnets) would need to be
 - borrowed (for a total of 128 subnets).



-
- Create 1000 Subnets with a Slash 8 prefix

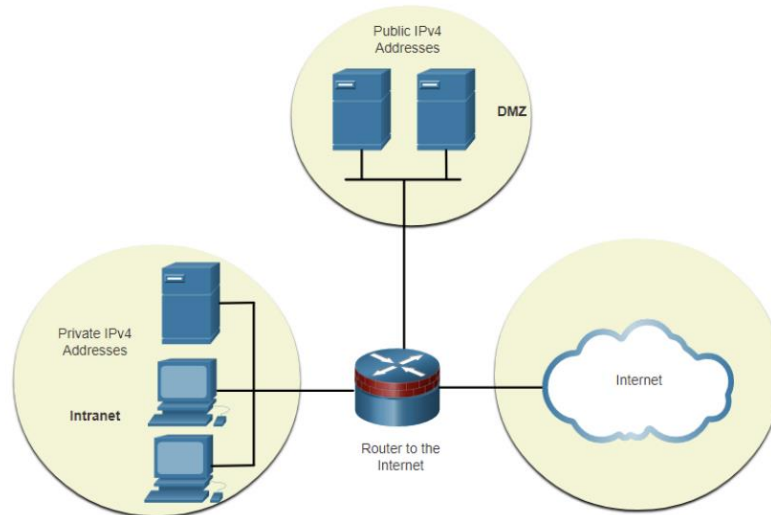
- Consider a small ISP that requires 1000 subnets for its clients using network address 10.0.0.0/8 which means there are 8 bits in the network portion and 24 host bits available to borrow toward subnetting.
 - The figure displays the number of subnets that can be created when borrowing bits from the second and third.
 - Notice there are now up to 22 host bits that can be borrowed (i.e., last two bits cannot be borrowed).
- To satisfy the requirement of 1000 subnets for the enterprise, 10 bits (i.e., $2^{10}=1024$ subnets) would need to be borrowed (for a total of 128 subnets)



Subnet to Meet Requirements

- Subnet Private versus Public IPv4 Address Space

- Enterprise networks will have an:
 - Intranet - A company's internal network typically using private IPv4 addresses.
 - DMZ – A companies internet facing servers. Devices in the DMZ use public IPv4 addresses.
 - A company could use the 10.0.0.0/8 and subnet on the /16 or /24 network boundary.
 - The DMZ devices would have to be configured with public IP addresses.



-
- Minimize Unused Host IPv4 Addresses and Maximize Subnets

There are two considerations when planning subnets:

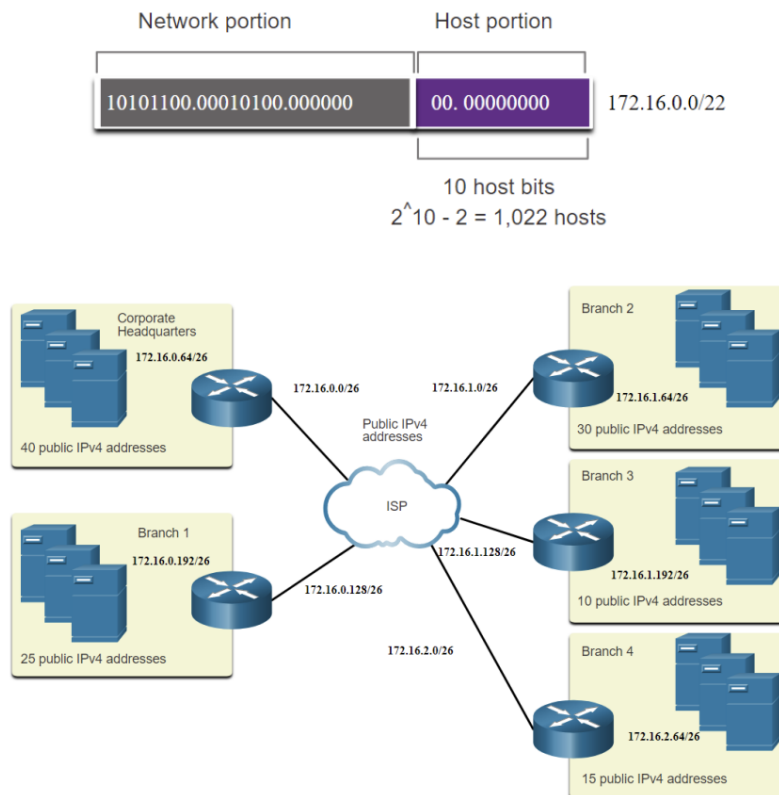
- The number of host addresses required for each network
- The number of individual subnets needed

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	nnnnnnnn.nnnnnnnn.nnnnnnnn.nhhhhhhh 1111111.1111111.1111111.10000000	2	126
/26	255.255.255.192	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnhhhhh 1111111.1111111.1111111.11000000	4	62
/27	255.255.255.224	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnhhh 1111111.1111111.1111111.11100000	8	30
/28	255.255.255.240	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnhhh 1111111.1111111.1111111.11110000	16	14
/29	255.255.255.248	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnhhh 1111111.1111111.1111111.11111000	32	6
/30	255.255.255.252	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnhhh 1111111.1111111.1111111.11111100	64	2

-
- Example: Efficient IPv4 Subnetting

- In this example, corporate headquarters has been allocated a public network address of 172.16.0.0/22 (10 host bits) by its ISP providing 1,022 host addresses.
- There are five sites and therefore five internet connections which means the organization requires 10 subnets with the largest subnet requires 40 addresses.

- It allocated 10 subnets with a /26 (i.e., 255.255.255.192) subnet mask.



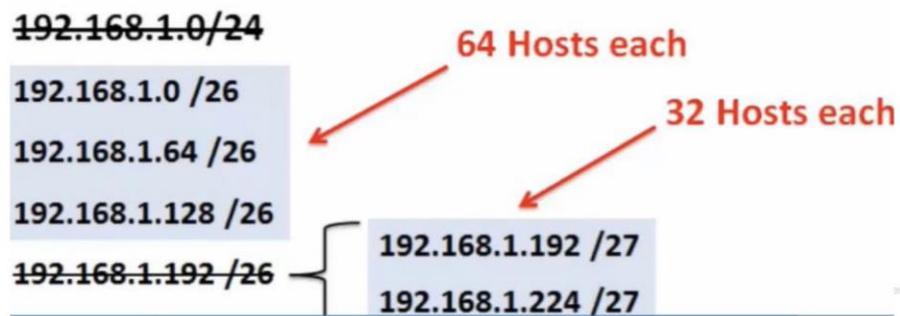
○

VLSM

- VLSM Basics - Video

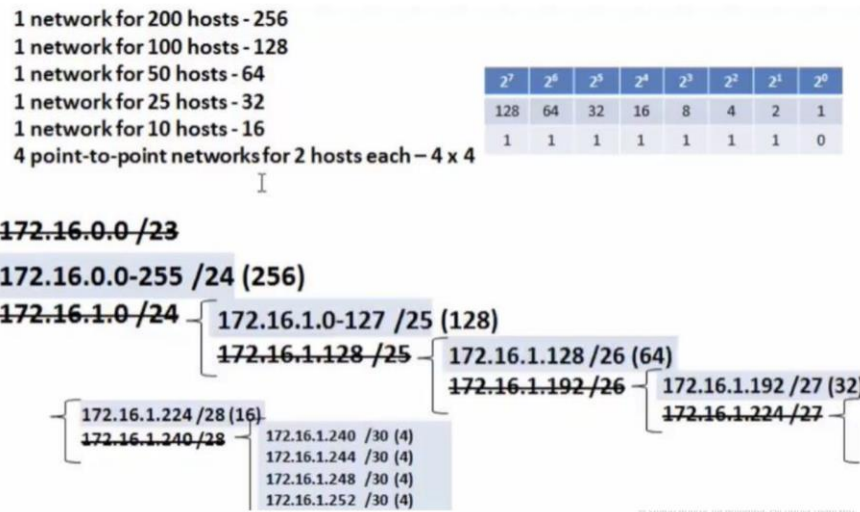
Basic VLSM

1. Subnets do not have to be equal sizes, as long as their address ranges do not overlap
2. When creating subnets it is easier to work from larger to smaller



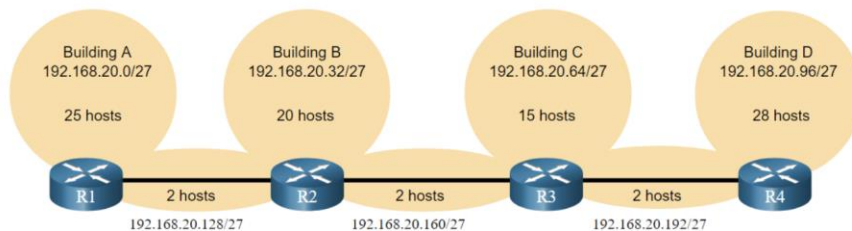
○

- VLSM Example - Video



- IPv4 Address Conservation

- Given the topology, 7 subnets are required (i.e, four LANs and three WAN links)
 - and the largest number of host is in Building D with 28 hosts.
 - A /27 mask would provide 8 subnets of 30 host IP addresses and therefore support this topology.



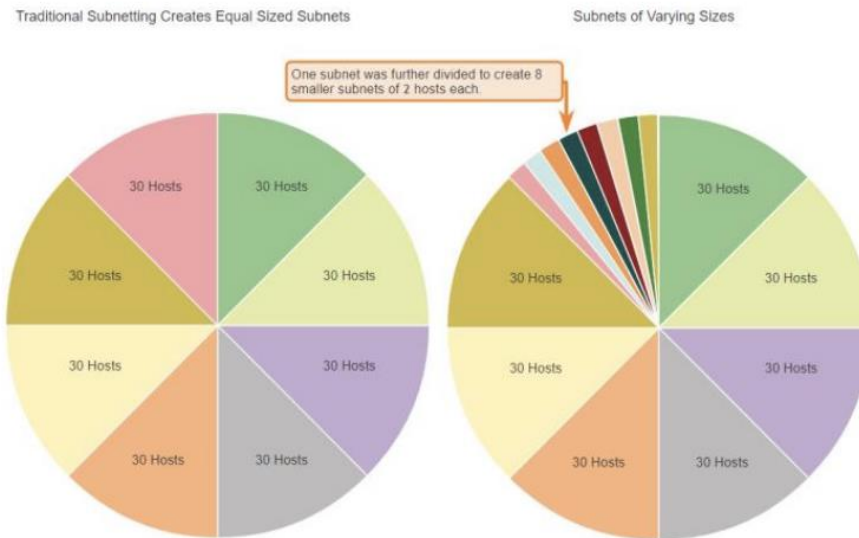
- However, the point-to-point WAN links only require two addresses and therefore waste 28 addresses each for a total of 84 unused addresses.
- Applying a traditional subnetting scheme to this scenario is not very efficient and is wasteful.
- VLSM was developed to avoid wasting addresses by enabling us to subnet a subnet.

Host portion
 $2^5 - 2 = 30$ host IP addresses per subnet
 $30 - 2 = 28$
 Each WAN subnet wastes 28 addresses
 $28 \times 3 = 84$
 84 addresses are unused

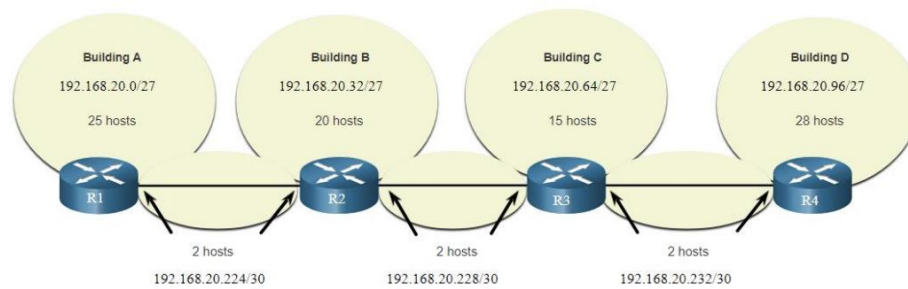
- VLSM

- The left side displays the traditional subnetting scheme (i.e., the same subnet mask) while the right side illustrates how VLSM can be used to subnet a subnet and divided the last subnet into eight /30 subnets.
- When using VLSM, always begin by satisfying the host requirements of the largest subnet and continue subnetting until the host requirements of the smallest subnet are satisfied.

- The resulting topology with VLSM applied.



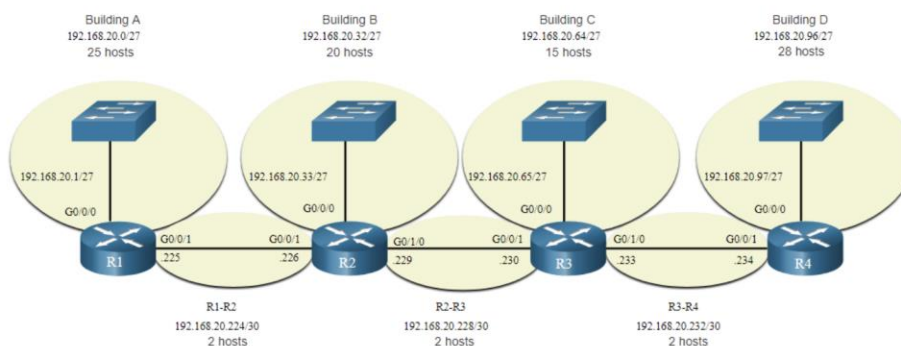
○



○

● VLSM Topology Address Assignment

- Using VLSM subnets, the LAN and inter-router networks can be addressed without unnecessary waste as shown in the logical topology diagram.



○

Structured Design

● IPv4 Network Address Planning

- IP network planning is crucial to develop a scalable solution to an enterprise network.

- To develop an IPv4 network wide addressing scheme, you need to know how many subnets are needed, how many hosts a particular subnet requires, what devices are part of the subnet, which parts of your network use private addresses, and which use public, and many other determining factors.
 - Examine the needs of an organization's network usage and how the subnets will be structured.
 - Perform a network requirement study by looking at the entire network to determining how each area will be segmented.
 - Determine how many subnets are needed and how many hosts per subnet.
 - Determine DHCP address pools and Layer 2 VLAN pools.
- **Device Address Assignment**
 - Within a network, there are different types of devices that require addresses:
 - End user clients
 - Most use DHCP to reduce errors and burden on network support staff. IPv6 clients can obtain address information using DHCPv6 or SLAAC.
 - Servers and peripherals
 - These should have a predictable static IP address.
 - Servers that are accessible from the internet
 - Servers must have a public IPv4 address, most often accessed using NAT.
 - Intermediary devices
 - Devices are assigned addresses for network management, monitoring, and security.
 - Gateway
 - Routers and firewall devices are gateway for the hosts in that network.
 - When developing an IP addressing scheme, it is generally recommended that you have a set pattern of how addresses are allocated to each type of device.

IPv6 Addressing

IPv4 Issues

- **Need for IPv6**
 - IPv4 is running out of addresses. IPv6 is the successor to IPv4. IPv6 has a much larger 128-bit address space.
 - The development of IPv6 also included fixes for IPv4 limitations and other enhancements.
 - With an increasing internet population, a limited IPv4 address space, issues with NAT and the IoT, the time has come to begin the transition to IPv6.

- **IPv4 and IPv6 Coexistence**

- Both IPv4 and IPv6 will coexist in the near future and the transition will take several years.
- The IETF has created various protocols and tools to help network administrators migrate their networks to IPv6. These migration techniques can be divided into three categories:
 - Dual stack
 - The devices run both IPv4 and IPv6 protocol stacks simultaneously.
 - Tunneling
 - A method of transporting an IPv6 packet over an IPv4 network. The IPv6 packet is encapsulated inside an IPv4 packet.
 - Translation
 - Network Address Translation 64 (NAT64) allows IPv6-enabled devices to communicate with IPv4-enabled devices using a translation technique similar to NAT for IPv4.
- Note: Tunneling and translation are for transitioning to native IPv6 and should only be used where needed. The goal should be native IPv6 communications from source to destination.

IPv6 Address Representation

- **IPv6 Addressing Formats**

- IPv6 addresses are 128 bits in length and written in hexadecimal.
- IPv6 addresses are not case-sensitive and can be written in either lowercase or uppercase.
- The preferred format for writing an IPv6 address is x:x:x:x:x:x, with each “x” consisting of four hexadecimal values.
- In IPv6, a hextet is the unofficial term used to refer to a segment of 16 bits, or four hexadecimal values.
- Examples of IPv6 addresses in the preferred format:
2001:0db8:0000:1111:0000:0000:0000:0200
2001:0db8:0000:00a3:abcd:0000:0000:1234

- **Rule 1 – Omit Leading Zero**

- The first rule to help reduce the notation of IPv6 addresses is to omit any leading 0s (zeros).
- Examples:
 - 01ab can be represented as 1ab
 - 09f0 can be represented as 9f0
 - 0a00 can be represented as a00
 - 00ab can be represented as ab
- Note: This rule only applies to leading 0s, NOT to trailing 0s, otherwise the address would be ambiguous.

Type	Format
Preferred	2001 : 0db8 : 0000 : 1111 : 0000 : 0000 : 0000 : 0200
No leading zeros	2001 : db8 : 0 : 1111 : 0 : 0 : 0 : 200

-
- Rule 2 – Double Colon

- A double colon (::) can replace any single, contiguous string of one or more 16-bit hexets consisting of all zeros.
- Example:
 - 2001:db8:cafe:1:0:0:0:1 (leading 0s omitted) could be represented as 2001:db8:cafe:1::1
- Note: The double colon (::) can only be used once within an address, otherwise there would be more than one possible resulting address.

Type	Format
Preferred	2001 : 0db8 : 0000 : 1111 : 0000 : 0000 : 0000 : 0200
Compressed	2001:db8:0:1111::200

IPv6 Address Types

- Unicast, Multicast, Anycast

- There are three broad categories of IPv6 addresses:
 - Unicast
 - Unicast uniquely identifies an interface on an IPv6-enabled device.
 - Multicast
 - Multicast is used to send a single IPv6 packet to multiple destinations.
 - Anycast
 - This is any IPv6 unicast address that can be assigned to multiple devices.
- A packet sent to an anycast address is routed to the nearest device having that address.
- Note: Unlike IPv4, IPv6 does not have a broadcast address. However, there is an IPv6 all-nodes multicast address that essentially gives the same result.

- IPv6 Prefix Length

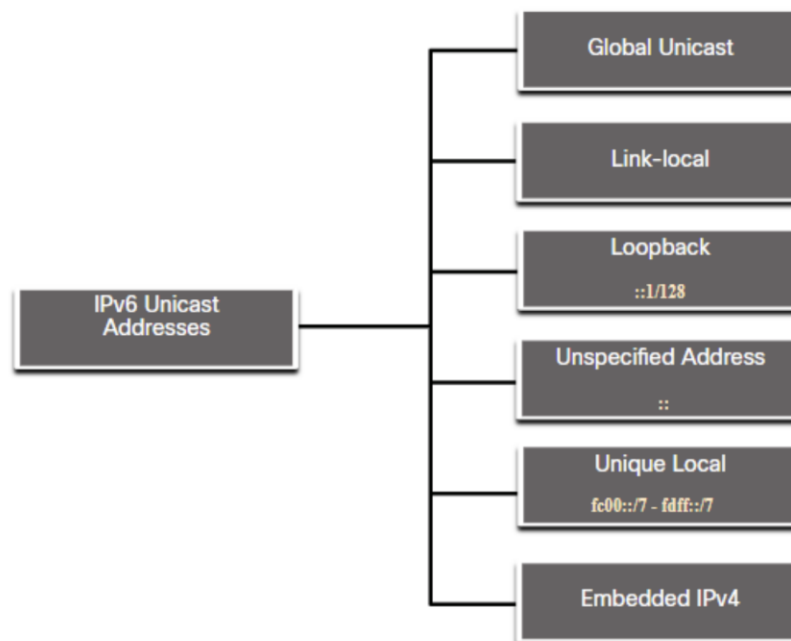
- Prefix length is represented in slash notation and is used to indicate the network portion of an IPv6 address.
- The IPv6 prefix length can range from 0 to 128. The recommended IPv6 prefix length for LANs and most other types of networks is /64.



-
- Note: It is strongly recommended to use a 64-bit Interface ID for most networks. This is because stateless address autoconfiguration (SLAAC) uses 64 bits for the Interface ID. It also makes subnetting easier to create and manage.

● Types of IPv6 Unicast Addresses

- Unlike IPv4 devices that have only a single address, IPv6 addresses typically have two unicast addresses:
 - Global Unicast Address (GUA)
 - This is similar to a public IPv4 address. These are globally unique, internet-routable addresses.
 - Link-local Address (LLA)
 - Required for every IPv6-enabled device and used to communicate with other devices on the same local link. LLAs are not routable and are confined to a single link.



○

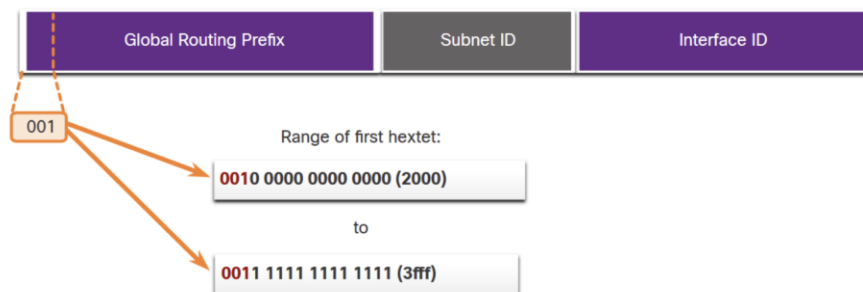
● A Note About the Unique Local Address

- The IPv6 unique local addresses (range fc00::/7 to fdff::/7) have some similarity to RFC 1918 private addresses for IPv4, but there are significant differences:
 - Unique local addresses are used for local addressing within a site or between a limited number of sites.

- Unique local addresses can be used for devices that will never need to access another network.
 - Unique local addresses are not globally routed or translated to a global IPv6 address.
- Note: Many sites use the private nature of RFC 1918 addresses to attempt to secure or hide their network from potential security risks. This was never the intended use of ULAs.

● IPv6 GUA

- IPv6 global unicast addresses (GUAs) are globally unique and routable on the IPv6 internet.
 - Currently, only GUAs with the first three bits of 001 or 2000::/3 are being assigned.
 - Currently available GUAs begins with a decimal 2 or a 3 (This is only 1/8th of the total available IPv6 address space).



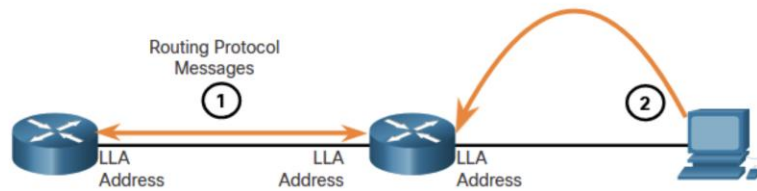
● IPv6 GUA Structure

- Global Routing Prefix:
 - The global routing prefix is the prefix, or network, portion of the address that is assigned by the provider, such as an ISP, to a customer or site. The global routing prefix will vary depending on ISP policies.
- Subnet ID:
 - The Subnet ID field is the area between the Global Routing Prefix and the Interface ID. The Subnet ID is used by an organization to identify subnets within its site.
- Interface ID:
 - The IPv6 interface ID is equivalent to the host portion of an IPv4 address. It is strongly recommended that in most cases /64 subnets should be used, which creates a 64-bit interface ID.
- Note: IPv6 allows the all-0s and all-1s host addresses can be assigned to a device. The all-0s address is reserved as a Subnet-Router anycast address, and should be assigned only to routers.

● IPv6 LLA

- An IPv6 link-local address (LLA) enables a device to communicate with other IPv6-enabled devices on the same link and only on that link (subnet).
 - Packets with a source or destination LLA cannot be routed.
 - Every IPv6-enabled network interface must have an LLA.

- If an LLA is not configured manually on an interface, the device will automatically create one.
- IPv6 LLAs are in the fe80::/10 range.



1. Routers use the LLA of neighbor routers to send routing updates.
2. Hosts use the LLA of a local router as the default-gateway.

GUA and LLA Static Configuration

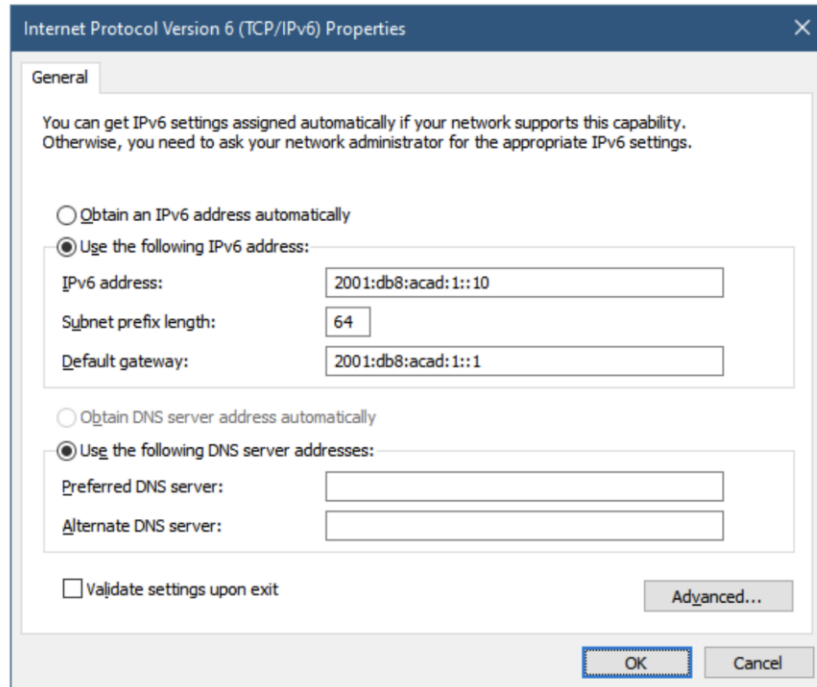
• Static GUA Configuration on a Router

- Most IPv6 configuration and verification commands in the Cisco IOS are similar to their IPv4 counterparts. In many cases, the only difference is the use of ipv6 in place of ip within the commands.
 - The command to configure an IPv6 GUA on an interface is: `ipv6 address ipv6-address/prefix-length`.
 - The example shows commands to configure a GUA on the G0/0/0 interface on R1:

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
```

• Static GUA Configuration on a Windows Host

- Manually configuring the IPv6 address on a host is similar to configuring an IPv4 address.
- The GUA or LLA of the router interface can be used as the default gateway. Best practice is to use the LLA.
- Note: When DHCPv6 or SLAAC is used, the LLA of the router will automatically be specified as the default gateway address.



- Static GUA Configuration of a Link-Local Unicast Address

- Configuring the LLA manually lets you create an address that is recognizable and easier to remember.
 - LLAs can be configured manually using the ipv6 address ipv6-link-local-address link-local command.
 - The example shows commands to configure a LLA on the G0/0/0 interface on R1

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ipv6 address fe80::1:1 link-local
R1(config-if)# no shutdown
R1(config-if)# exit
```

- Note: The same LLA can be configured on each link as long as it is unique on that link. Common practice is to create a different LLA on each interface of the router to make it easy to identify the router and the specific interface.

Dynamic Addressing for IPv6 GUAs

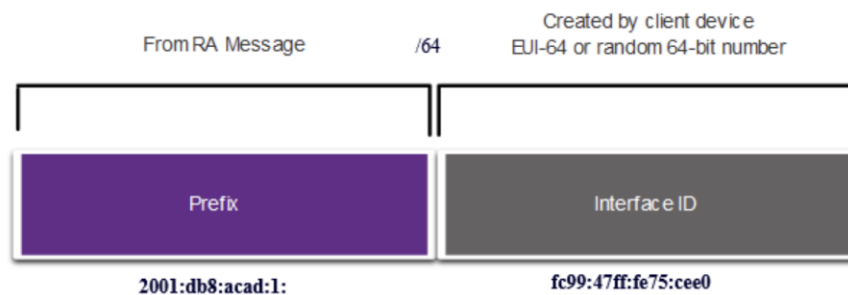
- RS and RA Messages

- Devices obtain GUA addresses dynamically through Internet Control Message Protocol version 6 (ICMPv6) messages.
 - Router Solicitation (RS) messages are sent by host devices to discover IPv6 routers
 - Router Advertisement (RA) messages are sent by routers to inform hosts on how to obtain an IPv6 GUA and provide useful network information such as:
 - Network prefix and prefix length
 - Default gateway address

- DNS addresses and domain name
- The RA can provide three methods for configuring an IPv6 GUA :
 - SLAAC
 - SLAAC with stateless DHCPv6 server
 - Stateful DHCPv6 (no SLAAC)

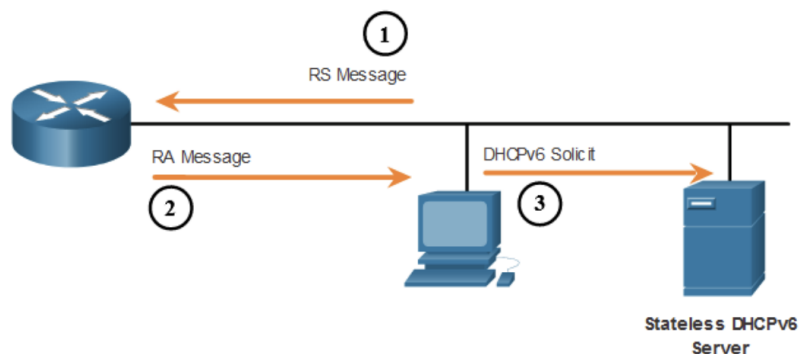
• Method 1: SLAAC

- SLAAC allows a device to configure a GUA without the services of DHCPv6.
- Devices obtain the necessary information to configure a GUA from the ICMPv6 RA messages of the local router.
- The prefix is provided by the RA and the device uses either the EUI-64 or random generation method to create an interface ID.



• Method 2: SLAAC and Stateless DHCP

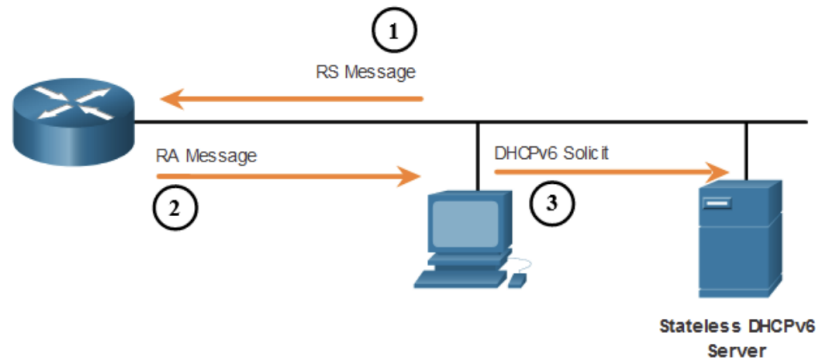
- An RA can instruct a device to use both SLAAC and stateless DHCPv6.
- The RA message suggests devices use the following:
 - SLAAC to create its own IPv6 GUA
 - The router LLA, which is the RA source IPv6 address, as the default gateway address
 - A stateless DHCPv6 server to obtain other information such as a DNS server address and a domain name



• Method 3: Stateful DHCPv6

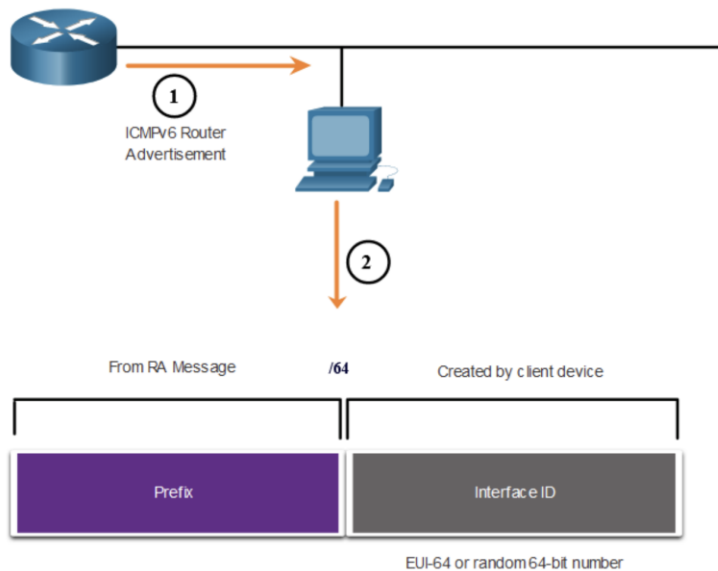
- An RA can instruct a device to use stateful DHCPv6 only.
- Stateful DHCPv6 is similar to DHCP for IPv4. A device can automatically receive a GUA,
- prefix length, and the addresses of DNS servers from a stateful DHCPv6 server.
- The RA message suggests devices use the following:

- The router LLA, which is the RA source IPv6 address, for the default gateway address.
- A stateful DHCPv6 server to obtain a GUA, DNS server address, domain name and other necessary information.



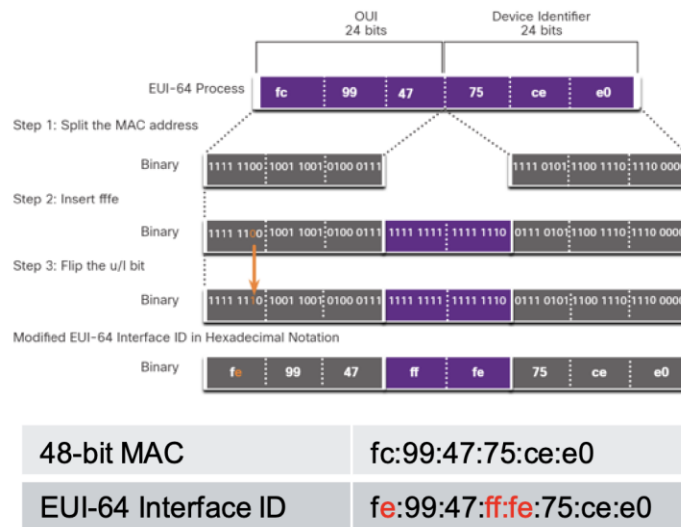
• EUI-64 Process vs. Randomly Generated

- When the RA message is either SLAAC or SLAAC with stateless DHCPv6, the client must generate its own interface ID.
- The interface ID can be created using the EUI-64 process or a randomly generated 64-bit number.



• EUI-64 Process

- The IEEE defined the Extended Unique Identifier (EUI) or modified EUI-64 process which performs the following:
 - A 16 bit value of fffe (in hexadecimal) is inserted into the middle of the 48-bit Ethernet MAC address of the client.
 - The 7th bit of the client MAC address is reversed from binary 0 to 1.
 - Example:



- Randomly Generated Interface IDs

- Depending upon the operating system, a device may use a randomly generated interface ID instead of using the MAC address and the EUI-64 process.
- Beginning with Windows Vista, Windows uses a randomly generated interface ID instead of one created with EUI-64.

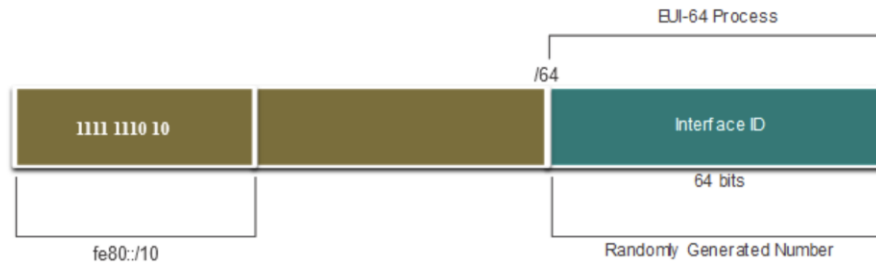
```
C:\> ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . : 
IPv6 Address. . . . . : 2001:db8:acad:1:50a5:8a35:a5bb:66e1
Link-local IPv6 Address . . . . . : fe80::50a5:8a35:a5bb:66e1
Default Gateway . . . . . : fe80::1
C:\>
```

- Note: To ensure the uniqueness of any IPv6 unicast address, the client may use a process known as Duplicate Address Detection (DAD). This is similar to an ARP request for its own address. If there is no reply, then the address is unique.

Dynamic Addressing for IPv6 LLAs

- Dynamic LLAs

- All IPv6 interfaces must have an IPv6 LLA.
- Like IPv6 GUAs, LLAs can be configured dynamically.
- The figure shows the LLA is dynamically created using the fe80::/10 prefix and the interface ID using the EUI-64 process, or a randomly generated 64-bit number.



- Dynamic LLAs on Windows

- Operating systems, such as Windows, will typically use the same method for both a SLAAC-created GUA and a dynamically assigned LLA.

EUI-64 Generated Interface ID:

```
C:\> ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2001:db8:acad:1:fc99:47ff:fe75:cee0
Link-local IPv6 Address . . . . . : fe80::fc99:47ff:fe75:cee0
Default Gateway . . . . . : fe80::1
C:\>
```

Random 64-bit Generated Interface ID:

```
C:\> ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2001:db8:acad:1:50a5:8a35:a5bb:66e1
Link-local IPv6 Address . . . . . : fe80::50a5:8a35:a5bb:66e1
Default Gateway . . . . . : fe80::1
C:\>
```

- Dynamic LLAs on Cisco Routers

- Cisco routers automatically create an IPv6 LLA whenever a GUA is assigned to the interface. By default, Cisco IOS routers use EUI-64 to generate the interface ID for all LLAs on IPv6 interfaces.
- Here is an example of a LLA dynamically configured on the G0/0/0 interface of R1:

```
R1# show interface gigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Hardware is ISR4221-2x1GE, address is 7079.b392.3640 (bia 7079.b392.3640)
(Output omitted)
R1# show ipv6 interface brief
GigabitEthernet0/0/0 [up/up]
FE80::7279:B3FF:FE92:3640
2001:DB8:ACAD:1::1
```

- Verify IPv6 Address Configuration

- Cisco routers automatically create an IPv6 LLA whenever a GUA is assigned to the interface. By default, Cisco IOS routers use EUI-64 to generate the interface ID for all LLAs on IPv6 interfaces.
- Here is an example of a LLA dynamically configured on the G0/0/0 interface of R1:

```

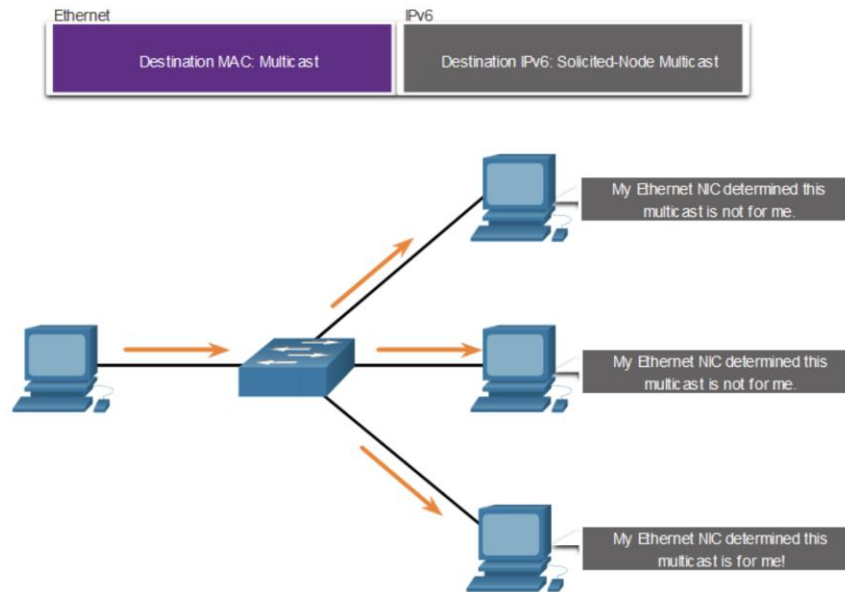
R1# show interface gigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Hardware is ISR4221-2x1GE, address is 7079.b392.3640 (bia 7079.b392.3640)
(Output omitted)
R1# show ipv6 interface brief
GigabitEthernet0/0/0 [up/up]
FE80::7279:B3FF:FE92:3640
2001:DB8:ACAD:1::1

```

○

IPv6 Multicast Addresses

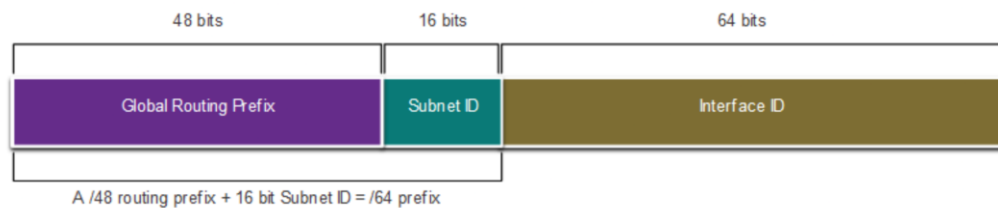
- Assigned IPv6 Multicast Addresses
 - IPv6 multicast addresses have the prefix ff00::/8. There are two types of IPv6 multicast addresses:
 - Well-Known multicast addresses
 - Solicited node multicast addresses
 - Note: Multicast addresses can only be destination addresses and not source addresses.
- Well-Known IPv6 Multicast Addresses
 - Well-known IPv6 multicast addresses are assigned and are reserved for predefined groups of devices.
 - There are two common IPv6 Assigned multicast groups:
 - ff02::1 All-nodes multicast group - This is a multicast group that all IPv6-enabled devices join. A packet sent to this group is received and processed by all IPv6 interfaces on the link or network.
 - ff02::2 All-routers multicast group - This is a multicast group that all IPv6 routers join. A router becomes a member of this group when it is enabled as an IPv6 router with the ipv6 unicast-routing global configuration command.
- Solicited-Node IPv6 Multicast
 - A solicited-node multicast address is similar to the all-nodes multicast address.
 - A solicited-node multicast address is mapped to a special Ethernet multicast address.
 - The Ethernet NIC can filter the frame by examining the destination MAC address without sending it to the IPv6 process to see if the device is the intended target of the IPv6 packet.



○

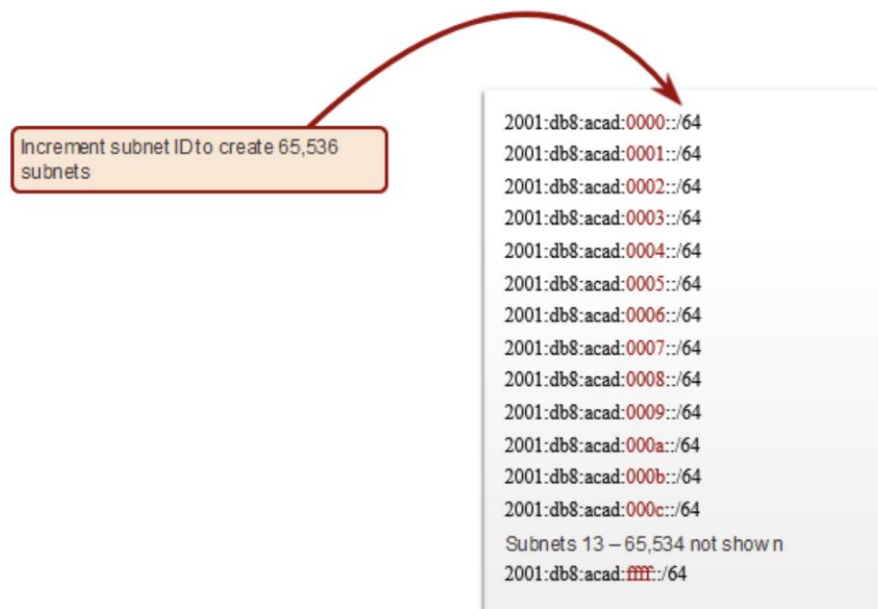
Subnet an IPv6 Network

- Subnet Using the Subnet ID
 - IPv6 was designed with subnetting in mind.
 - A separate subnet ID field in the IPv6 GUA is used to create subnets.
 - The subnet ID field is the area between the Global Routing Prefix and the interface ID.

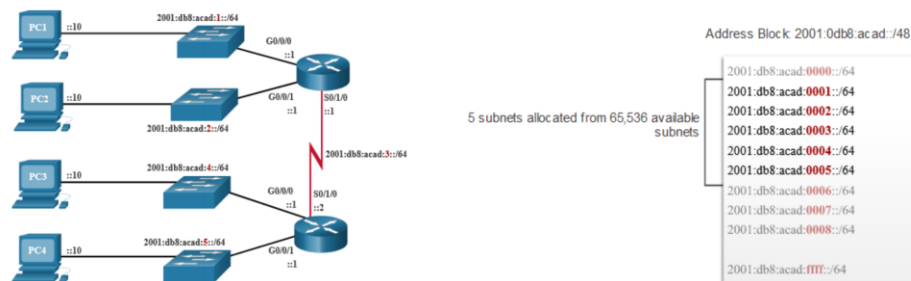


○

- IPv6 Subnetting Example
 - Given the 2001:db8:acad::/48 global routing prefix with a 16 bit subnet ID.
 - Allows 65,536 /64 subnets
 - The global routing prefix is the same for all subnets.
 - Only the subnet ID hexet is incremented in hexadecimal for each subnet.



-
- IPv6 Subnet Allocation
 - The example topology requires five subnets, one for each LAN as well as for the serial link between R1 and R2.
 - The five IPv6 subnets were allocated, with the subnet ID field 0001 through 0005. Each /64 subnet will provide more addresses than will ever be needed.



-
- Router Configured with IPv6 Subnets
 - The example shows that each of the router interfaces on R1 has been configured to be on a different IPv6 subnet.

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitethernet 0/0/1
R1(config-if)# ipv6 address 2001:db8:acad:2::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface serial 0/1/0
R1(config-if)# ipv6 address 2001:db8:acad:3::1/64
R1(config-if)# no shutdown
```

○

ICMP

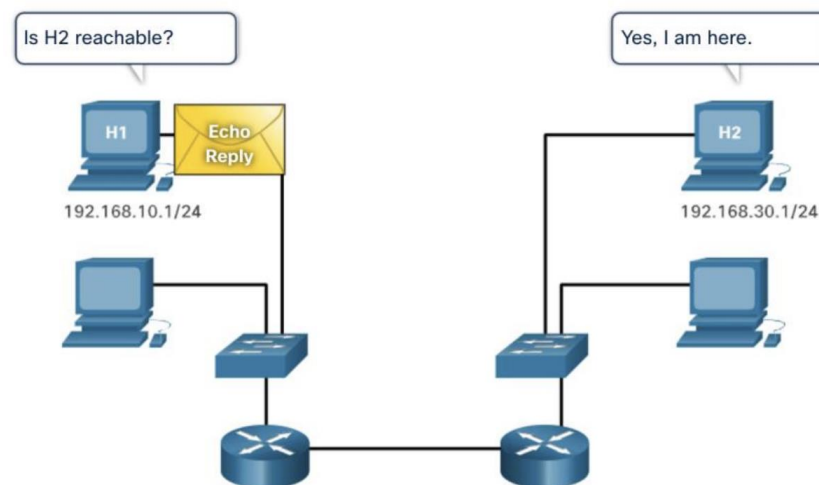
ICMP Messages

- ICMPv4 and ICMPv6 Messages

- Internet Control Message Protocol (ICMP) provides feedback about issues related to the processing of IP packets under certain conditions.
- ICMPv4 is the messaging protocol for IPv4. ICMPv6 is the messaging protocol for IPv6 and includes additional functionality.
- The ICMP messages common to both ICMPv4 and ICMPv6 include:
 - Host reachability
 - Destination or Service Unreachable
 - Time exceeded
- Note: ICMPv4 messages are not required and are often not allowed within a network for security reasons.

- Host Reachability

- ICMP Echo Message can be used to test the reachability of a host on an IP network.
- In the example:
 - The local host sends an ICMP Echo Request to a host.
 - If the host is available, the destination host responds with an Echo Reply.



- Destination or Service Unreachable

- An ICMP Destination Unreachable message can be used to notify the source that a destination or service is unreachable.
- The ICMP message will include a code indicating why the packet could not be delivered.
- A few Destination Unreachable codes for ICMPv4 are as follows:
 - 0 - Net unreachable
 - 1 - Host unreachable
 - 2 - Protocol unreachable

- 3 - Port unreachable
- A few Destination Unreachable codes for ICMPv6 are as follows:
 - 0 - No route to destination
 - 1 - Communication with the destination is administratively prohibited (e.g., firewall)
 - 2 - Beyond scope of the source address
 - 3 - Address unreachable
 - 4 - Port unreachable
- Note: ICMPv6 has similar but slightly different codes for Destination Unreachable messages.

● Time Exceeded

- When the Time to Live (TTL) field in a packet is decremented to 0, an ICMPv4 Time Exceeded message will be sent to the source host.
- ICMPv6 also sends a Time Exceeded message. Instead of the IPv4 TTL field, ICMPv6 uses the IPv6 Hop Limit field to determine if the packet has expired

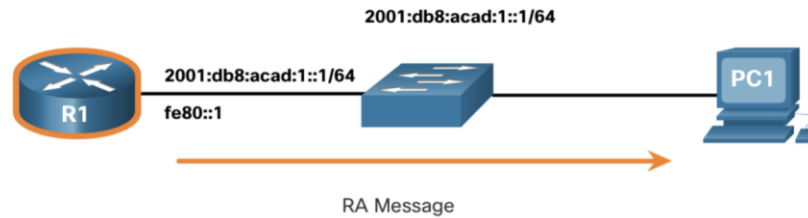
```
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 192.168.1.1: TTL expired in transit.
Reply from 192.168.1.1: TTL expired in transit.
Reply from 192.168.1.1: TTL expired in transit.
Reply from 192.168.1.1: TTL expired in transit.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

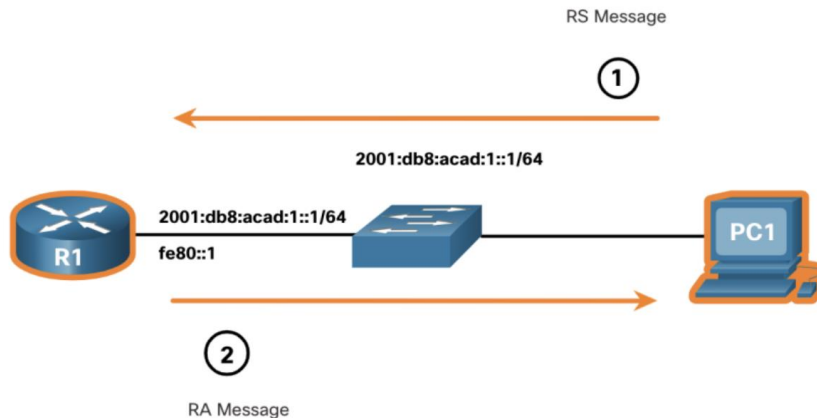
-
- Note: Time Exceeded messages are used by the traceroute tool.

● ICMPv6 Messages

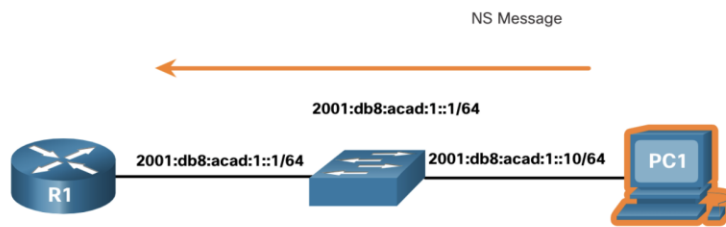
- ICMPv6 has new features and improved functionality not found in ICMPv4, including four new protocols as part of the Neighbor Discovery Protocol (ND or NDP).
- Messaging between an IPv6 router and an IPv6 device, including dynamic address allocation are as follows:
 - Router Solicitation (RS) message
 - Router Advertisement (RA) message
- Messaging between IPv6 devices, including duplicate address detection and address resolution are as follows:
 - Neighbor Solicitation (NS) message
 - Neighbor Advertisement (NA) message
- Note: ICMPv6 ND also includes the redirect message, which has a similar function to the redirect message used in ICMPv4.
- RA messages are sent by IPv6-enabled routers every 200 seconds to provide addressing information to IPv6-enabled hosts.
- RA message can include addressing information for the host such as the prefix, prefix length, DNS address, and domain name.
- A host using Stateless Address Autoconfiguration (SLAAC) will set its default gateway to the link-local address of the router that sent the RA.



-
- An IPv6-enabled router will also send out an RA message in response to an RS message.
- In the figure, PC1 sends a RS message to determine how to receive its IPv6 address information dynamically.
- R1 replies to the RS with an RA message.
- PC1 sends an RS message, "Hi, I just booted up. Is there an IPv6 router on the network? I need to know how to get my IPv6 address information dynamically."
- R1 replies with an RA message. "Hi all IPv6- enabled devices. I'm R1 and you can use SLAAC to create an IPv6 global unicast address. The prefix is `2001:db8:acad:1::1/64`. By the way, use my link-local address `fe80::1` as your default gateway."

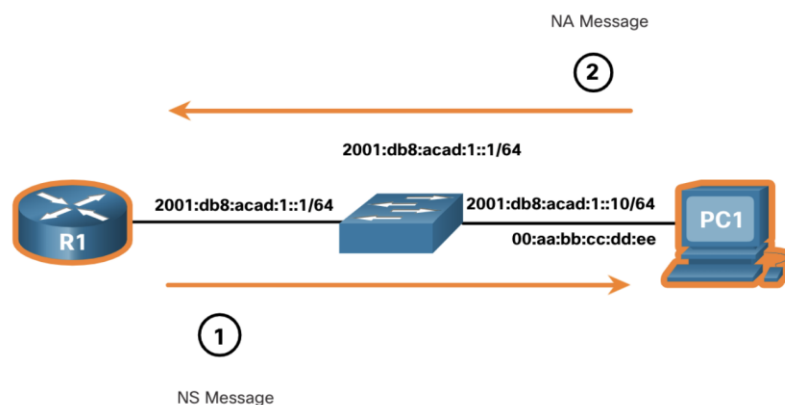


-
- A device assigned a global IPv6 unicast or link-local unicast address, may perform duplicate address detection (DAD) to ensure that the IPv6 address is unique.
- To check the uniqueness of an address, the device will send an NS message with its own IPv6 address as the targeted IPv6 address.
- If another device on the network has this address, it will respond with an NA message notifying to the sending device that the address is in use.



Note: DAD is not required, but RFC 4861 recommends that DAD is performed on unicast addresses.

-
- To determine the MAC address for the destination, the device will send an NS message to the solicited node address.
- The message will include the known (targeted) IPv6 address. The device that has the targeted IPv6 address will respond with an NA message containing its Ethernet MAC address.
- In the figure, R1 sends a NS message to 2001:db8:acad:1::10 asking for its MAC address.



Ping and Traceroute Tests

● Ping – Test Connectivity

- The ping command is an IPv4 and IPv6 testing utility that uses ICMP echo request and echo reply messages to test connectivity between hosts and provides a summary that includes the success rate and average round-trip time to the destination.
- If a reply is not received within the timeout, ping provides a message indicating that a response was not received.

- It is common for the first ping to timeout if address resolution (ARP or ND) needs to be performed before sending the ICMP Echo Request.

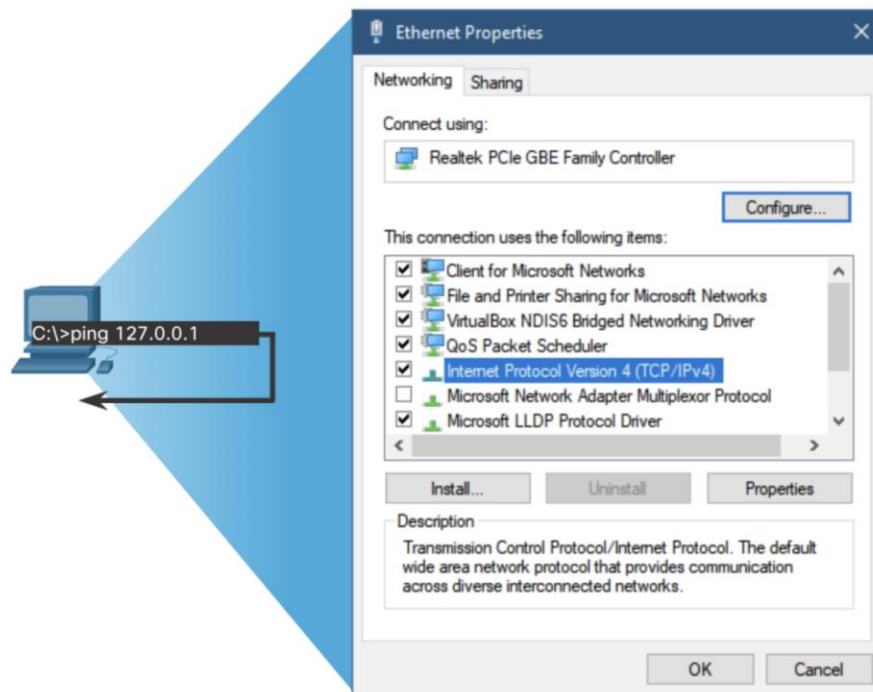
```
S1#ping 192.168.20.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms
```

```
R1#ping 2001:db8:acad:1::2

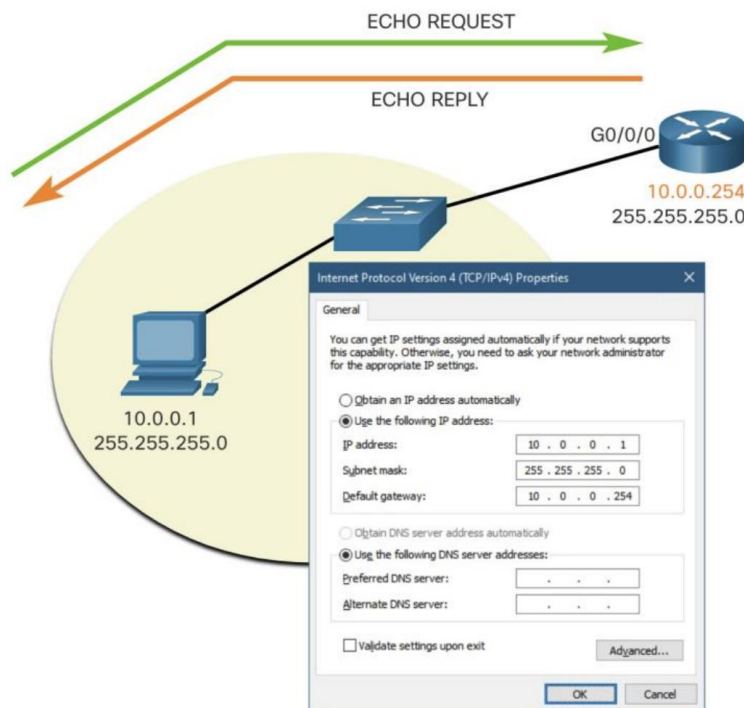
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:db8:acad:1::2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

-
- Ping the Loopback
 - Ping can be used to test the internal configuration of IPv4 or IPv6 on the local host. To do this, ping the local loopback address of 127.0.0.1 for IPv4 (::1 for IPv6).
 - A response from 127.0.0.1 for IPv4, or ::1 for IPv6, indicates that IP is properly installed on the host.
 - An error message indicates that TCP/IP is not operational on the host.



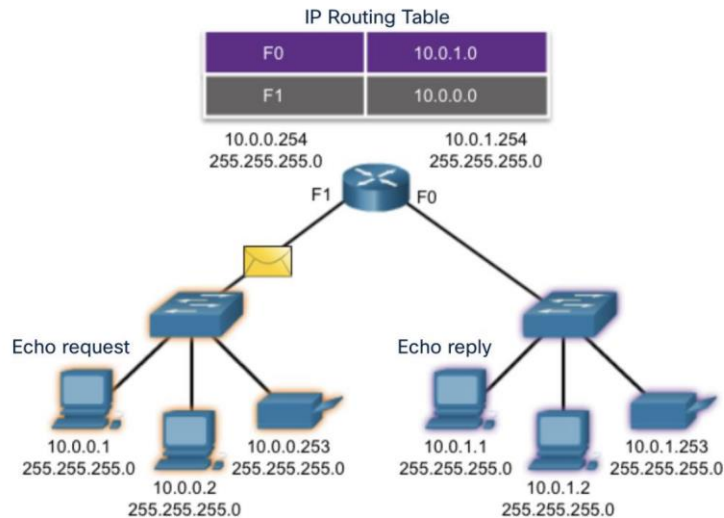
-
- Ping the Default Gateway
 - The ping command can be used to test the ability of a host to communicate on the local network. The default gateway address is most often used because the router is normally always operational.

- A successful ping to the default gateway indicates that the host and the router interface serving as the default gateway are both operational on the local network.
- If the default gateway address does not respond, a ping can be sent to the IP address of another host on the local network that is known to be operational.



● Ping a Remote Host

- Ping can also be used to test the ability of a local host to communicate across an internetwork.
- A local host can ping a host on a remote network. A successful ping across the internetwork confirms communication on the local network.
- Note: Many network administrators limit or prohibit the entry of ICMP messages therefore, the lack of a ping response could be due to security restrictions.



- Traceroute – Test the Path

- Traceroute (tracert) is a utility that is used to test the path between two hosts and provide a list of hops that were successfully reached along that path.
- Traceroute provides round-trip time for each hop along the path and indicates if a hop fails to respond. An asterisk (*) is used to indicate a lost or unreplied packet.
- This information can be used to locate a problematic router in the path or may indicate that the router is configured not to reply

```

R1#traceroute 192.168.40.2
Type escape sequence to abort.
Tracing the route to 192.168.40.2

 1  192.168.10.2    1 msec    0 msec    0 msec
 2  192.168.20.2   2 msec    1 msec    0 msec
 3  192.168.30.2   1 msec    0 msec    0 msec
 4  192.168.40.2   0 msec    0 msec    0 msec
  
```

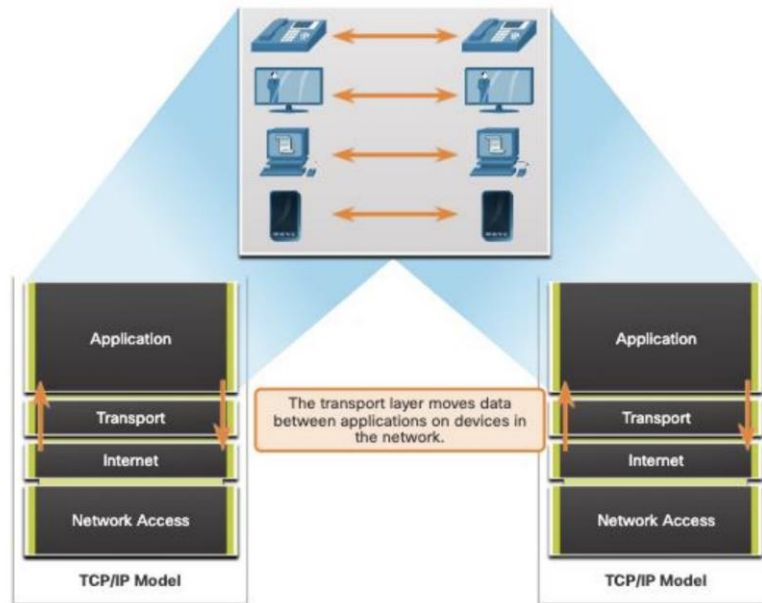
- Note: Traceroute makes use of a function of the TTL field in IPv4 and the Hop Limit field in IPv6 in the Layer 3 headers, along with the ICMP Time Exceeded message.
- The first message sent from traceroute will have a TTL field value of 1. This causes the TTL to time out at the first router. This router then responds with a ICMPv4 Time Exceeded message.
- Traceroute then progressively increments the TTL field (2, 3, 4...) for each sequence of messages. This provides the trace with the address of each hop as the packets time out further down the path.
- The TTL field continues to be increased until the destination is reached, or it is incremented to a predefined maximum

Transport Layer

Transportation of Data

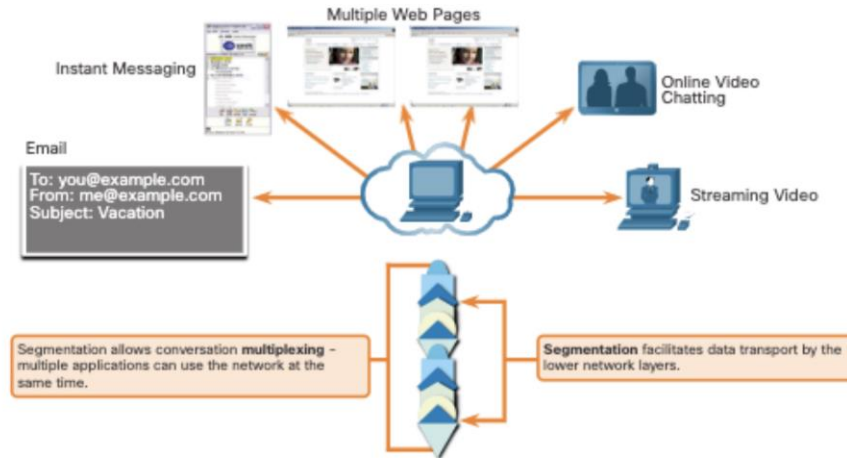
- Role of the Transport Layer

- The transport layer is:
 - responsible for logical communications between applications running on different hosts.
 - The link between the application layer and the lower layers that are responsible for network transmission.



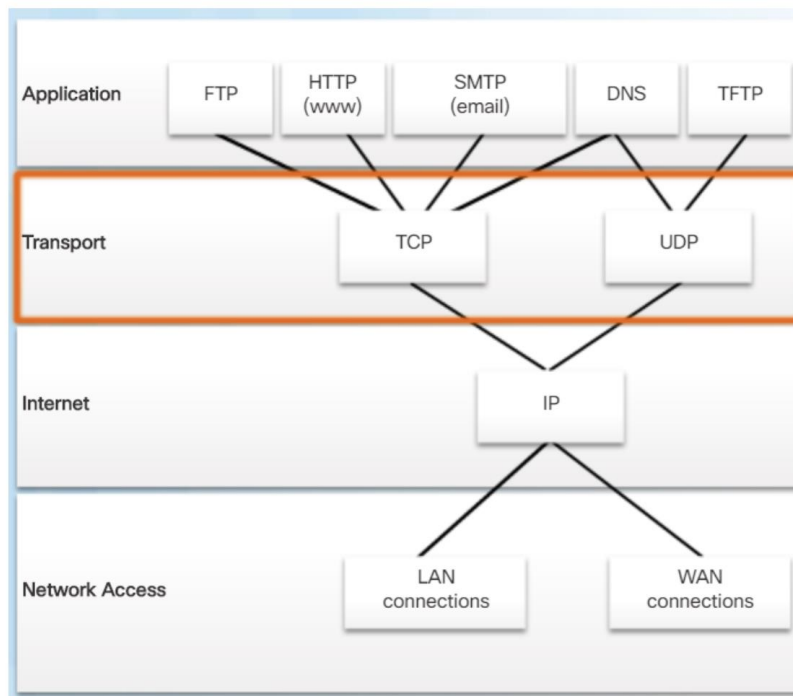
- - Transport Layer Responsibilities

- The transport layer has the following responsibilities:
 - Tracking individual conversations
 - Segmenting data and reassembling segments
 - Adds header information
 - Identify, separate, and manage multiple conversations
 - Uses segmentation and multiplexing to enable different communication conversations to be interleaved on the same network



• Transport Layer Protocols

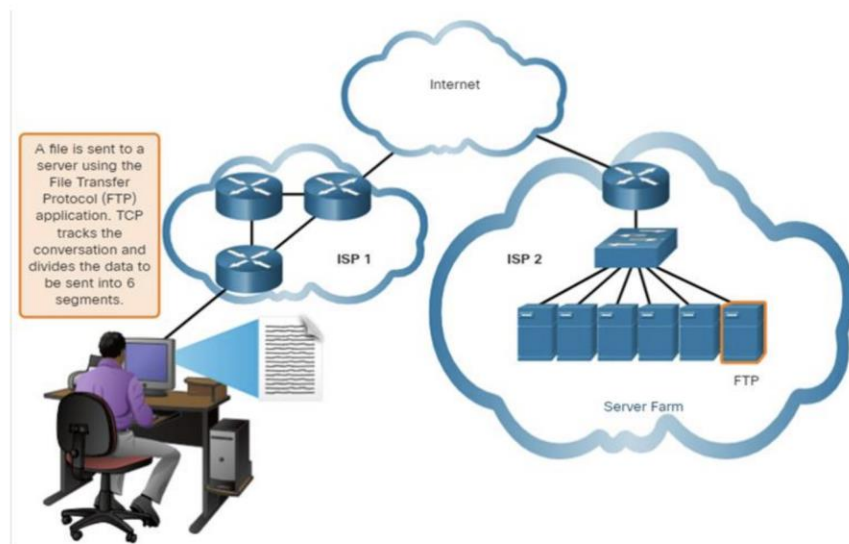
- IP does not specify how the delivery or transportation of the packets takes place.
- Transport layer protocols specify how to transfer messages between hosts, and are responsible for managing reliability requirements of a conversation.
- The transport layer includes the TCP and UDP protocols.



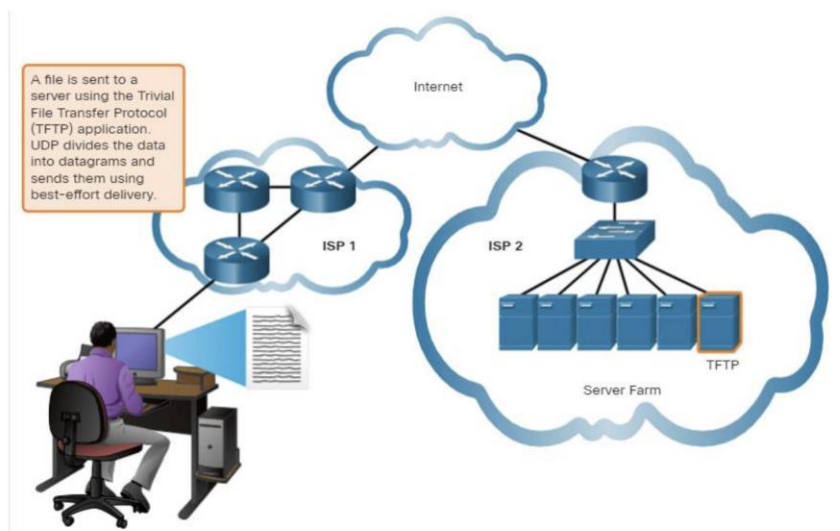
• Transmission Control Protocol

- TCP provides reliability and flow control. TCP basic operations:
 - Number and track data segments transmitted to a specific host from a specific application
 - Acknowledge received data
 - Retransmit any unacknowledged data after a certain amount of time

- Sequence data that might arrive in wrong order
- Send data at an efficient rate that is acceptable by the receiver

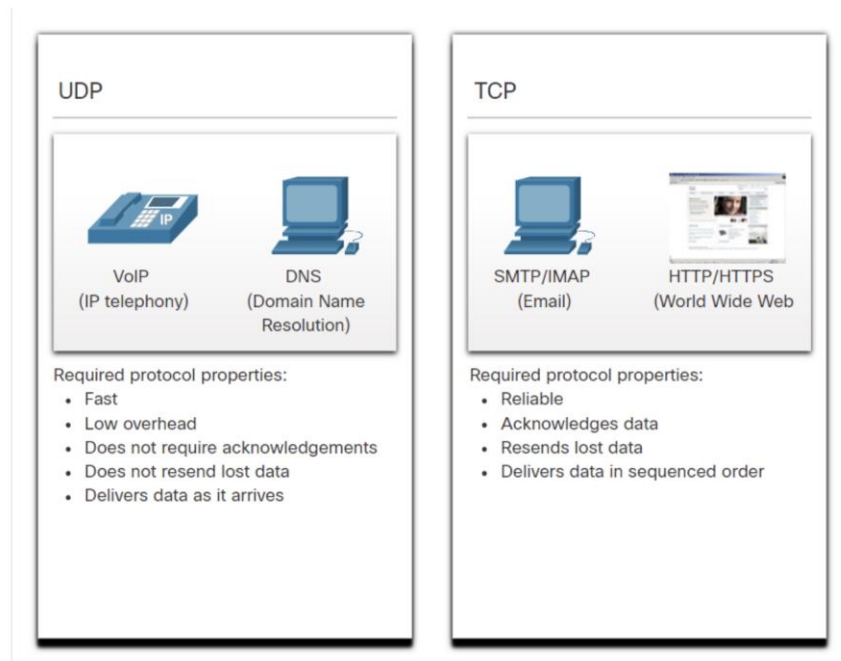


-
- User Datagram Protocol (UDP)
 - UDP provides the basic functions for delivering datagrams between the appropriate applications, with very little overhead and data checking.
 - UDP is a connectionless protocol.
 - UDP is known as a best-effort delivery protocol because there is no acknowledgment that the data is received at the destination.



-
- The Right Transport Layer Protocol for the Right Application
 - UDP is also used by request-and-reply applications where the data is minimal, and retransmission can be done quickly.

- If it is important that all the data arrives and that it can be processed in its proper sequence, TCP is used as the transport protocol.



○

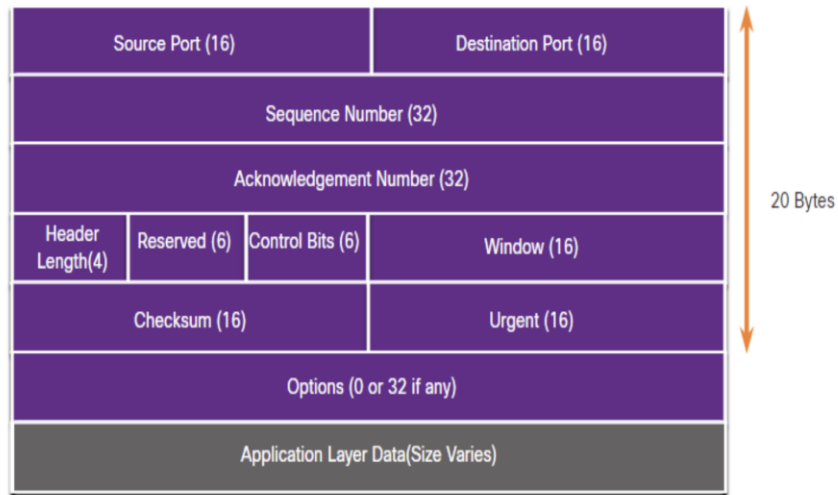
TCP Overview

● TCP Features

- Establishes a Session - TCP is a connection-oriented protocol that negotiates and establishes a permanent connection (or session) between source and destination devices prior to forwarding any traffic.
- Ensures Reliable Delivery - For many reasons, it is possible for a segment to become corrupted or lost completely, as it is transmitted over the network. TCP ensures that each segment that is sent by the source arrives at the destination.
- Provides Same-Order Delivery - Because networks may provide multiple routes that can have different transmission rates, data can arrive in the wrong order.
- Supports Flow Control - Network hosts have limited resources (i.e., memory and processing power). When TCP is aware that these resources are overtaxed, it can request that the sending application reduce the rate of data flow.

● TCP Header

- TCP is a stateful protocol which means it keeps track of the state of the communication session.
- TCP records which information it has sent, and which information has been acknowledged.

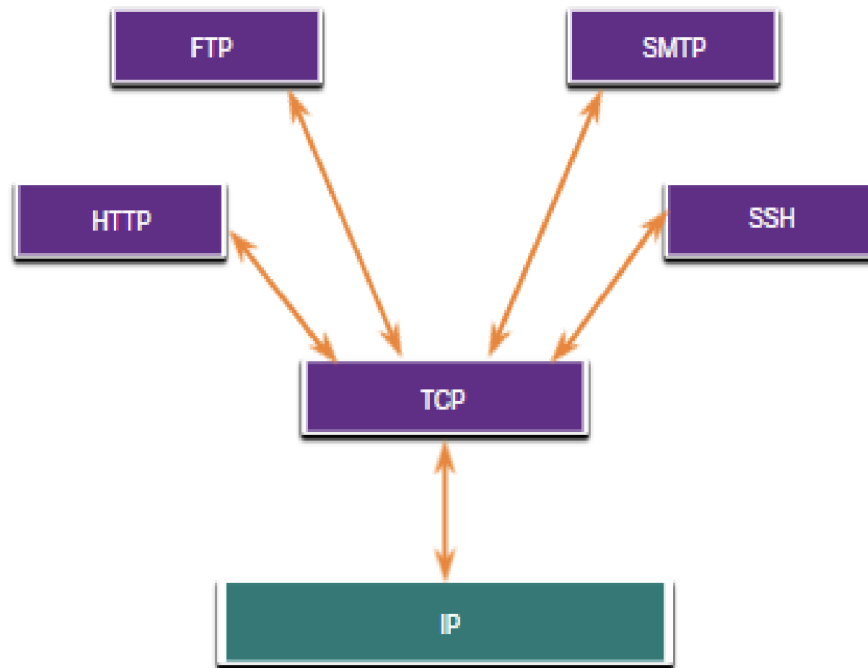


- - TCP Header Fields

TCP Header Field	Description
Source Port	A 16-bit field used to identify the source application by port number.
Destination Port	A 16-bit field used to identify the destination application by port number.
Sequence Number	A 32-bit field used for data reassembly purposes.
Acknowledgment Number	A 32-bit field used to indicate that data has been received and the next byte expected from the source.
Header Length	A 4-bit field known as "data offset" that indicates the length of the TCP segment header.
Reserved	A 6-bit field that is reserved for future use.
Control bits	A 6-bit field used that includes bit codes, or flags, which indicate the purpose and function of the TCP segment.
Window size	A 16-bit field used to indicate the number of bytes that can be accepted at one time.
Checksum	A 16-bit field used for error checking of the segment header and data.
Urgent	A 16-bit field used to indicate if the contained data is urgent.

- - Applications that use TCP

- TCP handles all tasks associated with dividing the data stream into segments, providing reliability, controlling data flow, and reordering segments.



○

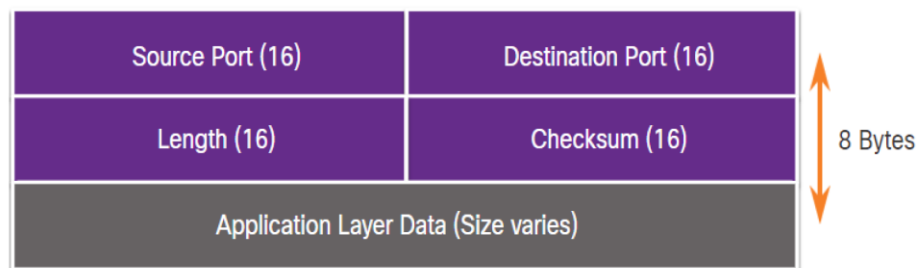
UDP Overview

- UDP Features

- UDP features include the following:
 - Data is reconstructed in the order that it is received.
 - Any segments that are lost are not resent.
 - There is no session establishment.
 - The sender is not informed about resource availability.

- UDP Header

- The UDP header is far simpler than the TCP header because it only has four fields and requires 8 bytes (i.e. 64 bits).



○

- UDP Header Fields

- The table identifies and describes the four fields in a UDP header.

UDP Header Field	Description
Source Port	A 16-bit field used to identify the source application by port number.
Destination Port	A 16-bit field used to identify the destination application by port number.
Length	A 16-bit field that indicates the length of the UDP datagram header.
Checksum	A 16-bit field used for error checking of the datagram header and data.